

Examples:

For examples relating to Facts 1, 2, and 5, see Chapter 4.

1. Let

$$A = \begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & 1 \\ -1 & 2 & 2 \end{bmatrix} \quad \text{and} \quad x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, y = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}.$$

Then, as noted by Fact 7,

$$\det \begin{bmatrix} A & x \\ y^T & -1 \end{bmatrix} = \begin{bmatrix} 1 & 3 & -1 & 1 \\ 0 & 1 & 1 & 1 \\ -1 & 2 & 2 & 1 \\ 2 & 1 & -1 & -1 \end{bmatrix} = -\det(A + xy^T) = \begin{bmatrix} 3 & 4 & -2 \\ 2 & 2 & 0 \\ 1 & 3 & 1 \end{bmatrix} = 10.$$

Next, letting $c = [121]^T$ and $d = [0 - 1 - 1]^T$, by Fact 6, we have

$$\det(A + cd^T) = \det(A)(1 + d^T A^{-1}c) = (-4) \cdot (-1) = 4.$$

2. To illustrate Facts 8 and 9, let

$$A = \left[\begin{array}{cc|c} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{array} \right] = \begin{bmatrix} X & Y \\ Y^* & Z \end{bmatrix}.$$

Note that A is positive definite and so Hadamard's inequality says that

$$\det A \leq 3 \cdot 5 \cdot 3 = 45;$$

in fact, $\det A = 36$. Fischer's inequality gives a smaller upper bound for the determinant:

$$\det A \leq \det X \det Z = 13 \cdot 3 = 39.$$

3. Consider the matrix

$$B = \begin{bmatrix} 1 & 2 & -2 \\ 4 & -1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

The first inequality about general matrices in Fact 8 applied to B gives

$$|\det B| \leq \sqrt{9 \cdot 18 \cdot 2} = 18.$$

As the rows of B are mutually orthogonal, we have that $|\det B| = 18$; in fact, $\det B = -18$.

14.5 Rank and Nullity Equalities and Inequalities

Let A be a matrix over a field F . Here we present relations among the fundamental subspaces of A and their dimensions. As general references consult, e.g., [HJ85] and [Mey00, Sec. 4.2, 4.4, 4.5] (even though the matrices discussed there are complex, most of the proofs remain valid for any field). Additional material on rank and nullity can also be found in Section 2.4.

Facts:

- Let $A \in F^{m \times n}$. Then $\text{rank}(A) = \dim \text{range } A = \dim \text{range } A^T$.
If $F = \mathbb{C}$, then $\text{rank}(A) = \dim \text{range } A^* = \dim \overline{\text{range } A}$.

2. If $A \in \mathbb{C}^{m \times n}$, then $\text{range } A = (\ker A^*)^\perp$ and $\text{range } A^* = (\ker A)^\perp$.
3. If $A \in F^{m \times n}$ and $\text{rank}(A) = k$, then there exist $X \in F^{m \times k}$ and $Y \in F^{k \times n}$ such that $A = XY$.
4. Let $A, B \in F^{m \times n}$. Then $\text{rank}(A) = \text{rank}(B)$ if and only if there exist invertible matrices $X \in F^{m \times m}$ and $Y \in F^{n \times n}$ such that $B = XAY$.
5. (Dimension Theorem) Let $A \in F^{m \times n}$. Then

$$\text{rank}(A) + \text{null}(A) = n \quad \text{and} \quad \text{rank}(A) + \text{null}(A^T) = m.$$

If $F = \mathbb{C}$, then $\text{rank}(A) + \text{null}(A^*) = m$.

6. Let $A, B \in F^{m \times n}$. Then

$$\text{rank}(A) - \text{rank}(B) \leq \text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B).$$

7. Let $A \in F^{m \times n}$, $B \in F^{m \times k}$, and $C = [A|B] \in F^{m \times (n+k)}$. Then

- $\text{rank}(C) = \text{rank}(A) + \text{rank}(B) - \dim(\text{range } A \cap \text{range } B)$.
- $\text{null}(C) = \text{null}(A) + \text{null}(B) + \dim(\text{range } A \cap \text{range } B)$.

8. Let $A \in F^{m \times n}$ and $B \in F^{n \times k}$. Then

- $\text{rank}(AB) = \text{rank}(B) - \dim(\ker A \cap \text{range } B)$.
- If $F = \mathbb{C}$, then $\text{rank}(AB) = \text{rank}(A) - \dim(\ker B^* \cap \text{range } A^*)$.
- Multiplication of a matrix from the left or right by an invertible matrix leaves the rank unchanged.
- $\text{null}(AB) = \text{null}(B) + \dim(\ker A \cap \text{range } B)$.
- $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$.
- $\text{rank}(AB) \geq \text{rank}(A) + \text{rank}(B) - n$.

9. (Sylvester's law of nullity) Let $A, B \in \mathbb{C}^{n \times n}$. Then

$$\begin{aligned} \max\{\text{null}(A), \text{null}(B)\} &\leq \text{null}(AB) \\ &\leq \text{null}(A) + \text{null}(B). \end{aligned}$$

The above fact is valid *only* for square matrices.

10. (Frobenius inequality) Let $A \in F^{m \times n}$, $B \in F^{n \times k}$, and $C \in F^{k \times p}$. Then

$$\text{rank}(AB) + \text{rank}(BC) \leq \text{rank}(B) + \text{rank}(ABC).$$

11. Let $A \in \mathbb{C}^{m \times n}$. Then

$$\text{rank}(A^*A) = \text{rank}(A) = \text{rank}(AA^*).$$

In fact,

$$\text{range}(A^*A) = \text{range } A^* \quad \text{and} \quad \text{range } A = \text{range}(AA^*),$$

as well as

$$\ker(A^*A) = \ker A \quad \text{and} \quad \ker(AA^*) = \ker A^*.$$

12. Let $A \in F^{m \times n}$ and $B \in F^{k \times p}$. The rank of their direct sum is

$$\text{rank}(A \oplus B) = \text{rank} \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} = \text{rank}(A) + \text{rank}(B).$$

13. Let $A = [a_{ij}] \in F^{m \times n}$ and $B \in F^{k \times p}$. The rank of the Kronecker product $A \otimes B = [a_{ij}B] \in F^{mk \times np}$ is

$$\text{rank}(A \otimes B) = \text{rank}(A)\text{rank}(B).$$

14. Let $A = [a_{ij}] \in F^{m \times n}$ and $B = [b_{ij}] \in F^{m \times n}$. The rank of the Hadamard product $A \circ B = [a_{ij}b_{ij}] \in F^{m \times n}$ satisfies

$$\text{rank}(A \circ B) \leq \text{rank}(A)\text{rank}(B).$$

Examples:

1. Consider the matrices

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 3 & -2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix}, \quad \text{and} \quad C = \begin{bmatrix} 1 & 2 & -1 & 1 \\ 1 & 2 & -1 & 1 \\ 2 & 4 & -2 & 2 \end{bmatrix}.$$

We have that

$$\begin{aligned} \text{rank}(A) = 2, \quad \text{rank}(B) = 3, \quad \text{rank}(C) = 1, \quad \text{rank}(A + B) = 3, \\ \text{rank}(AB) = 2, \quad \text{rank}(BC) = 1, \quad \text{rank}(ABC) = 1. \end{aligned}$$

- As a consequence of Fact 5, we have

$$\begin{aligned} \text{null}(A) = 3 - 2 = 1, \quad \text{null}(B) = 3 - 3 = 0, \quad \text{null}(C) = 4 - 1 = 3, \\ \text{null}(A + B) = 3 - 3 = 0, \quad \text{null}(AB) = 3 - 2 = 1, \\ \text{null}(BC) = 3 - 1 = 2, \quad \text{null}(ABC) = 4 - 1 = 3. \end{aligned}$$

- Fact 6 states that

$$-1 = 2 - 3 = \text{rank}(A) - \text{rank}(B) \leq \text{rank}(A + B) = 0 \leq \text{rank}(A) + \text{rank}(B) = 5.$$

- Since $\text{range } A \cap \text{range } B = \text{range } A$, Fact 7 states that

$$\begin{aligned} \text{rank}([A|B]) &= \text{rank}(A) + \text{rank}(B) - \dim(\text{range } A \cap \text{range } B) = 2 + 3 - 2 = 3, \\ \text{null}([A|B]) &= \text{null}(A) + \text{null}(B) + \dim(\text{range } A \cap \text{range } B) = 1 + 0 + 2 = 3. \end{aligned}$$

- Since $\ker A \cap \text{range } B = \ker A$, Fact 8 states that

$$\begin{aligned} 2 &= \text{rank}(AB) = \text{rank}(B) - \dim(\ker A \cap \text{range } B) = 3 - 1 = 2. \\ 2 &= \text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\} = 2. \\ 2 &= \text{rank}(AB) \geq \text{rank}(A) + \text{rank}(B) - n = 2 + 3 - 3 = 2. \end{aligned}$$

- Fact 9 states that

$$\begin{aligned} 1 &= \max\{\text{null}(A), \text{null}(B)\} \leq \text{null}(AB) = 1 \\ &\leq \text{Null}(A) + \text{null}(B) = 1. \end{aligned}$$

Fact 9 can fail for nonsquare matrices. For example, if

$$D = \begin{bmatrix} 1 & 1 \end{bmatrix},$$

then

$$1 = \max\{\text{null}(D), \text{null}(D^T)\} \not\leq \text{null}(DD^T) = 0.$$

- Fact 10 states that

$$3 = \text{rank}(AB) + \text{rank}(BC) \leq \text{rank}(B) + \text{rank}(ABC) = 4.$$

14.6 Useful Identities for the Inverse

This section presents facts and formulas related to inversion of matrices.

Facts:

1. [Oue81, (1.9)], [HJ85, p. 18] Recall that $A/A[\alpha]$ denotes the Schur complement of the principal submatrix $A[\alpha]$ in A . (See Section 4.2 and Section 10.3.) If $A \in F^{n \times n}$ is partitioned in blocks as

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix},$$

where A_{11} and A_{22} are square matrices, then, provided that A , A_{11} , and A_{22} are invertible, we have that the Schur complements A/A_{11} and A/A_{22} are invertible and

$$A^{-1} = \begin{bmatrix} (A/A_{22})^{-1} & -A_{11}^{-1}A_{12}(A/A_{11})^{-1} \\ -(A/A_{11})^{-1}A_{21}A_{11}^{-1} & (A/A_{11})^{-1} \end{bmatrix}.$$

More generally, given an invertible $A \in F^{n \times n}$ and $\alpha \subseteq \{1, 2, \dots, n\}$ such that $A[\alpha]$ and $A(\alpha)$ are invertible, A^{-1} is obtained from A by replacing

- $A[\alpha]$ by $(A/A(\alpha))^{-1}$,
 - $A[\alpha, \alpha^c]$ by $-A[\alpha]^{-1}A[\alpha, \alpha^c](A/A[\alpha])^{-1}$,
 - $A[\alpha^c, \alpha]$ by $-(A/A[\alpha])^{-1}A[\alpha^c, \alpha]A[\alpha]^{-1}$, and
 - $A(\alpha)$ by $(A/A[\alpha])^{-1}$.
2. [HJ85, pp. 18–19] Let $A \in F^{n \times n}$, $X \in F^{n \times r}$, $R \in F^{r \times r}$, and $Y \in F^{r \times n}$. Let $B = A + XRY$. Suppose that A , B , and R are invertible. Then

$$B^{-1} = (A + XRY)^{-1} = A^{-1} - A^{-1}X(R^{-1} + YA^{-1}X)^{-1}YA^{-1}.$$

3. (Sherman–Morrison) Let $A \in F^{n \times n}$, $x, y \in F^n$. Let $B = A + xy^T$. Suppose that A and B are invertible. Then, if $y^T A^{-1}x \neq -1$,

$$B^{-1} = (A + xy^T)^{-1} = A^{-1} - \frac{1}{1 + y^T A^{-1}x} A^{-1}xy^T A^{-1}.$$

In particular, if $y^T x \neq -1$, then

$$(I + xy^T)^{-1} = I - \frac{1}{1 + y^T x} xy^T.$$

4. Let $A \in F^{n \times n}$. Then the adjugate of A (see Section 4.2) satisfies

$$(\text{adj } A)A = A(\text{adj } A) = (\det A)I.$$

If A is invertible, then

$$A^{-1} = \frac{1}{\det A} \text{adj } A.$$

5. Let $A \in F^{n \times n}$ be invertible and let its characteristic polynomial be $p_A(x) = x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1x + a_0$. Then,

$$A^{-1} = \frac{(-1)^{n+1}}{\det A} (A^{n+1} + a_1A^n + a_2A^{n-1} + \cdots + a_{n-1}A).$$

6. [Mey00, Sec. 7.10, p. 618] Let $A \in \mathbb{C}^{n \times n}$. The following statements are equivalent.

- The **Neumann series**, $I + A + A^2 + \cdots$, converges.
- $(I - A)^{-1}$ exists and $(I - A)^{-1} = \sum_{k=0}^{\infty} A^k$.
- $\rho(A) < 1$.
- $\lim_{k \rightarrow \infty} A^k = 0$.

Examples:

1. Consider the partitioned matrix

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \left[\begin{array}{cc|c} 1 & 3 & -1 \\ 0 & 2 & 1 \\ \hline -1 & -1 & 1 \end{array} \right].$$

Since

$$(A/A_{22})^{-1} = \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} -1.5 & 1 \\ 0.5 & 0 \end{bmatrix} \quad \text{and} \quad (A/A_{11})^{-1} = (-1)^{-1} = -1,$$

by Fact 1, we have

$$A^{-1} = \begin{bmatrix} (A/A_{22})^{-1} & -A_{11}^{-1}A_{12}(A/A_{11})^{-1} \\ -(A/A_{11})^{-1}A_{21}A_{11}^{-1} & (A/A_{11})^{-1} \end{bmatrix} = \left[\begin{array}{cc|c} -1.5 & 1 & -2.5 \\ 0.5 & 0 & 0.5 \\ \hline -1 & 1 & -1 \end{array} \right].$$

2. To illustrate Fact 3, consider the invertible matrix

$$A = \begin{bmatrix} 1 & i & -1 \\ 1 & 0 & 1 \\ -2i & 1 & -2 \end{bmatrix}$$

and the vectors $x = y = [1 \ 1 \ 1]^T$. We have that

$$A^{-1} = \begin{bmatrix} 0.5i & 1 + 0.5i & 0.5 \\ -1 - i & -1 + i & i \\ -0.5i & -0.5i & -0.5 \end{bmatrix}.$$

Adding xy^T to A amounts to adding 1 to each entry of A ; since

$$1 + y^T A^{-1} x = i \neq 0,$$

the resulting matrix is invertible and its inverse is given by

$$\begin{aligned}(A + xy^T)^{-1} &= A^{-1} - \frac{1}{i} A^{-1} xy^T A^{-1} \\ &= \begin{bmatrix} 2.5 & -0.5 - 0.5i & -1 - i \\ -2 + 2i & 1 & 2 \\ -1.5 - i & 0.5 + 0.5i & i \end{bmatrix}.\end{aligned}$$

3. Consider the matrix

$$A = \begin{bmatrix} -1 & 1 & -1 \\ 1 & -1 & 3 \\ 1 & -1 & 2 \end{bmatrix}.$$

Since $A^3 = 0$, A is a nilpotent matrix and, thus, all its eigenvalues equal 0. That is, $\rho(A) = 0 < 1$. As a consequence of Fact 6, $I - A$ is invertible and

$$(I - A)^{-1} = I + A + A^2 = \begin{bmatrix} 1 & 0 & 1 \\ 2 & -1 & 5 \\ 1 & -1 & 3 \end{bmatrix}.$$

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15

Matrix Perturbation Theory

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There is a vast amount of material in matrix (operator) perturbation theory. Related books that are worth mentioning are [SS90], [Par98], [Bha96], [Bau85], and [Kat70]. In this chapter, we attempt to include the most fundamental results up to date, except those for linear systems and least squares problems for which the reader is referred to Section 38.1 and Section 39.6.

Throughout this chapter, $\|\cdot\|_{\text{ui}}$ denotes a general unitarily invariant norm. Two commonly used ones are the spectral norm $\|\cdot\|_2$ and the Frobenius norm $\|\cdot\|_F$.

15.1 Eigenvalue Problems

The reader is referred to Sections 4.3, 14.1, and 14.2 for more information on eigenvalues and their locations.

Definitions:

Let $A \in \mathbb{C}^{n \times n}$. A scalar–vector pair $(\lambda, \mathbf{x}) \in \mathbb{C} \times \mathbb{C}^n$ is an **eigenpair** of A if $\mathbf{x} \neq 0$ and $A\mathbf{x} = \lambda\mathbf{x}$. A vector–scalar–vector triplet $(\mathbf{y}, \lambda, \mathbf{x}) \in \mathbb{C}^n \times \mathbb{C} \times \mathbb{C}^n$ is an **eigentriplet** if $\mathbf{x} \neq 0$, $\mathbf{y} \neq 0$, and $A\mathbf{x} = \lambda\mathbf{x}$, $\mathbf{y}^*A = \lambda\mathbf{y}^*$. The quantity

$$\text{cond}(\lambda) = \frac{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2}{|\mathbf{y}^*\mathbf{x}|}$$

is the **individual condition number** for λ , where $(\mathbf{y}, \lambda, \mathbf{x}) \in \mathbb{C}^n \times \mathbb{C} \times \mathbb{C}^n$ is an eigentriplet.

Let $\sigma(A) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$, the multiset of A 's eigenvalues, and set

$$\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n), \quad \Lambda_\tau = \text{diag}(\lambda_{\tau(1)}, \lambda_{\tau(2)}, \dots, \lambda_{\tau(n)}),$$

where τ is a **permutation** of $\{1, 2, \dots, n\}$. For real Λ , i.e., all λ_j 's are real,

$$\Lambda^\uparrow = \text{diag}(\lambda_1^\uparrow, \lambda_2^\uparrow, \dots, \lambda_n^\uparrow).$$

$\Lambda^\uparrow$ is in fact a Λ_τ for which the permutation τ makes $\lambda_{\tau(j)} = \lambda_j^\uparrow$ for all j .

Given two square matrices A_1 and A_2 , the **separation** $\text{sep}(A_1, A_2)$ between A_1 and A_2 is defined as [SS90, p. 231]

$$\text{sep}(A_1, A_2) = \inf_{\|X\|_2=1} \|XA_1 - A_2X\|_2.$$

A is perturbed to $\tilde{A} = A + \Delta A$. The same notation is adopted for $\tilde{A}$, except all symbols with tildes.

Let $X, Y \in \mathbb{C}^{n \times k}$ with $\text{rank}(X) = \text{rank}(Y) = k$. The **canonical angles** between their column spaces are $\theta_i = \arccos \sigma_i$, where $\{\sigma_i\}_{i=1}^k$ are the singular values of $(Y^*Y)^{-1/2}Y^*X(X^*X)^{-1/2}$. Define the **canonical angle matrix** between X and Y as

$$\Theta(X, Y) = \text{diag}(\theta_1, \theta_2, \dots, \theta_k).$$

For $k = 1$, i.e., $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$ (both nonzero), we use $\angle(\mathbf{x}, \mathbf{y})$, instead, to denote the canonical angle between the two vectors.

Facts:

- [SS90, p. 168] (**Elsner**) $\max_i \min_j |\tilde{\lambda}_i - \lambda_j| \leq (\|A\|_2 + \|\tilde{A}\|_2)^{1-1/n} \|\Delta A\|_2^{1/n}$.
- [SS90, p. 170] (**Elsner**) There exists a permutation τ of $\{1, 2, \dots, n\}$ such that

$$\|\Lambda - \tilde{\Lambda}_\tau\|_2 \leq 2 \left\lfloor \frac{n}{2} \right\rfloor (\|A\|_2 + \|\tilde{A}\|_2)^{1-1/n} \|\Delta A\|_2^{1/n}.$$

- [SS90, p. 183] Let $(\mathbf{y}, \mu, \mathbf{x})$ be an eigentriplet of A . ΔA changes μ to $\mu + \Delta\mu$ with

$$\Delta\mu = \frac{\mathbf{y}^*(\Delta A)\mathbf{x}}{\mathbf{y}^*\mathbf{x}} + O(\|\Delta A\|_2^2),$$

and $|\Delta\mu| \leq \text{cond}(\mu)\|\Delta A\|_2 + O(\|\Delta A\|_2^2)$.

- [SS90, p. 205] If A and $A + \Delta A$ are Hermitian, then

$$\|\Lambda^\uparrow - \tilde{\Lambda}^\uparrow\|_{\text{UI}} \leq \|\Delta A\|_{\text{UI}}.$$

- [Bha96, p. 165] (**Hoffman–Wielandt**) If A and $A + \Delta A$ are normal, then there exists a permutation τ of $\{1, 2, \dots, n\}$ such that $\|\Lambda - \tilde{\Lambda}_\tau\|_{\text{F}} \leq \|\Delta A\|_{\text{F}}$.
- [Sun96] If A is normal, then there exists a permutation τ of $\{1, 2, \dots, n\}$ such that $\|\Lambda - \tilde{\Lambda}_\tau\|_{\text{F}} \leq \sqrt{n}\|\Delta A\|_{\text{F}}$.
- [SS90, p. 192] (**Bauer–Fike**) If A is diagonalizable and $A = X\Lambda X^{-1}$ is its eigendecomposition, then

$$\max_i \min_j |\tilde{\lambda}_i - \lambda_j| \leq \|X^{-1}(\Delta A)X\|_p \leq \kappa_p(X)\|\Delta A\|_p.$$

- [BKL97] Suppose both A and $\tilde{A}$ are diagonalizable and have eigendecompositions $A = X\Lambda X^{-1}$ and $\tilde{A} = \tilde{X}\tilde{\Lambda}\tilde{X}^{-1}$.

(a) There exists a permutation τ of $\{1, 2, \dots, n\}$ such that

$$\|\Lambda - \tilde{\Lambda}_\tau\|_{\text{F}} \leq \sqrt{\kappa_2(X)\kappa_2(\tilde{X})}\|\Delta A\|_{\text{F}}.$$

(b) $\|\Lambda^\uparrow - \tilde{\Lambda}^\uparrow\|_{\text{UI}} \leq \sqrt{\kappa_2(X)\kappa_2(\tilde{X})}\|\Delta A\|_{\text{UI}}$ for real Λ and $\tilde{\Lambda}$.

9. [KPJ82] Let residuals $\mathbf{r} = A\tilde{\mathbf{x}} - \tilde{\mu}\tilde{\mathbf{x}}$ and $\mathbf{s}^* = \tilde{\mathbf{y}}^*A - \tilde{\mu}\tilde{\mathbf{y}}^*$, where $\|\tilde{\mathbf{x}}\|_2 = \|\tilde{\mathbf{y}}\|_2 = 1$, and let $\varepsilon = \max\{\|\mathbf{r}\|_2, \|\mathbf{s}\|_2\}$. The smallest error matrix ΔA in the 2-norm, for which $(\tilde{\mathbf{y}}, \tilde{\mu}, \tilde{\mathbf{x}})$ is an exact eigentriple of $\tilde{A} = A + \Delta A$, satisfies $\|\Delta A\|_2 = \varepsilon$, and $|\tilde{\mu} - \mu| \leq \text{cond}(\tilde{\mu})\varepsilon + O(\varepsilon^2)$ for some $\mu \in \sigma(A)$.
10. [KPJ82], [DK70], [Par98, pp. 73, 244] Suppose A is Hermitian, and let residual $\mathbf{r} = A\tilde{\mathbf{x}} - \tilde{\mu}\tilde{\mathbf{x}}$ and $\|\tilde{\mathbf{x}}\|_2 = 1$.
- (a) The smallest Hermitian error matrix ΔA (in the 2-norm), for which $(\tilde{\mu}, \tilde{\mathbf{x}})$ is an exact eigenpair of $\tilde{A} = A + \Delta A$, satisfies $\|\Delta A\|_2 = \|\mathbf{r}\|_2$.
 - (b) $|\tilde{\mu} - \mu| \leq \|\mathbf{r}\|_2$ for some eigenvalue μ of A .
 - (c) Let μ be the closest eigenvalue in $\sigma(A)$ to $\tilde{\mu}$ and $\mathbf{x}$ be its associated eigenvector with $\|\mathbf{x}\|_2 = 1$, and let $\eta = \min_{\mu \neq \lambda \in \sigma(A)} |\tilde{\mu} - \lambda|$. If $\eta > 0$, then

$$|\tilde{\mu} - \mu| \leq \frac{\|\mathbf{r}\|_2^2}{\eta}, \quad \sin \angle(\tilde{\mathbf{x}}, \mathbf{x}) \leq \frac{\|\mathbf{r}\|_2}{\eta}.$$

11. Let A be Hermitian, $X \in \mathbb{C}^{n \times k}$ have full column rank, and $M \in \mathbb{C}^{k \times k}$ be Hermitian having eigenvalues $\mu_1 \leq \mu_2 \leq \dots \leq \mu_k$. Set

$$R = AX - XM.$$

There exist k eigenvalues $\lambda_{i_1} \leq \lambda_{i_2} \leq \dots \leq \lambda_{i_k}$ of A such that the following inequalities hold. Note that subset $\{\lambda_{i_j}\}_{j=1}^k$ may be different at different occurrences.

- (a) [Par98, pp. 253–260], [SS90, Remark 4.16, p. 207] (**Kahan–Cao–Xie–Li**)

$$\begin{aligned} \max_{1 \leq j \leq k} |\mu_j - \lambda_{i_j}| &\leq \frac{\|R\|_2}{\sigma_{\min}(X)}, \\ \sqrt{\sum_{j=1}^k (\mu_j - \lambda_{i_j})^2} &\leq \frac{\|R\|_F}{\sigma_{\min}(X)}. \end{aligned}$$

- (b) [SS90, pp. 254–257], [Sun91] If $X^*X = I$ and $M = X^*AX$, and if all but k of A 's eigenvalues differ from every one of M 's by at least $\eta > 0$ and $\varepsilon_F = \|R\|_F/\eta < 1$, then

$$\sqrt{\sum_{j=1}^k (\mu_k - \lambda_{i_j})^2} \leq \frac{\|R\|_F^2}{\eta \sqrt{1 - \varepsilon_F^2}}.$$

- (c) [SS90, pp. 254–257], [Sun91] If $X^*X = I$ and $M = X^*AX$, and there is a number $\eta > 0$ such that either all but k of A 's eigenvalues lie outside the open interval $(\mu_1 - \eta, \mu_k + \eta)$ or all but k of A 's eigenvalues lie inside the closed interval $[\mu_\ell + \eta, \mu_{\ell+1} - \eta]$ for some $1 \leq \ell \leq k - 1$, and $\varepsilon = \|R\|_2/\eta < 1$, then

$$\max_{1 \leq j \leq k} |\mu_j - \lambda_{i_j}| \leq \frac{\|R\|_2^2}{\eta \sqrt{1 - \varepsilon^2}}.$$

12. [DK70] Let A be Hermitian and have decomposition

$$\begin{bmatrix} X_1^* \\ X_2^* \end{bmatrix} A \begin{bmatrix} X_1 & X_2 \end{bmatrix} = \begin{bmatrix} A_1 & \\ & A_2 \end{bmatrix},$$

where $\begin{bmatrix} X_1 & X_2 \end{bmatrix}$ is unitary and $X_1 \in \mathbb{C}^{n \times k}$. Let $Q \in \mathbb{C}^{n \times k}$ have orthonormal columns and for a $k \times k$ Hermitian matrix M set

$$R = AQ - QM.$$

Let $\eta = \min |\mu - \nu|$ over all $\mu \in \sigma(M)$ and $\nu \in \sigma(A_2)$. If $\eta > 0$, then $\|\sin \Theta(X_1, Q)\|_F \leq \frac{\|R\|_F}{\eta}$.

13. [LL05] Let

$$A = \begin{bmatrix} M & E^* \\ E & H \end{bmatrix}, \quad \tilde{A} = \begin{bmatrix} M & 0 \\ 0 & H \end{bmatrix}$$

be Hermitian, and set $\eta = \min |\mu - \nu|$ over all $\mu \in \sigma(M)$ and $\nu \in \sigma(H)$. Then

$$\max_{1 \leq j \leq n} |\lambda_j^\uparrow - \tilde{\lambda}_j^\uparrow| \leq \frac{2\|E\|_2^2}{\eta + \sqrt{\eta^2 + 4\|E\|_2^2}}.$$

14. [SS90, p. 230] Let $\begin{bmatrix} X_1 & Y_2 \end{bmatrix}$ be unitary and $X_1 \in \mathbb{C}^{n \times k}$, and let

$$\begin{bmatrix} X_1^* \\ Y_2^* \end{bmatrix} A \begin{bmatrix} X_1 & Y_2 \end{bmatrix} = \begin{bmatrix} A_1 & G \\ E & A_2 \end{bmatrix}.$$

Assume that $\sigma(A_1) \cap \sigma(A_2) = \emptyset$, and set $\eta = \text{sep}(A_1, A_2)$. If $\|G\|_2 \|E\|_2 < \eta^2/4$, then there is a unique $W \in \mathbb{C}^{(n-k) \times k}$, satisfying $\|W\|_2 \leq 2\|E\|_2/\eta$, such that $\begin{bmatrix} \tilde{X}_1 & \tilde{Y}_2 \end{bmatrix}$ is unitary and

$$\begin{bmatrix} \tilde{X}_1^* \\ \tilde{Y}_2^* \end{bmatrix} A \begin{bmatrix} \tilde{X}_1 & \tilde{Y}_2 \end{bmatrix} = \begin{bmatrix} \tilde{A}_1 & \tilde{G} \\ 0 & \tilde{A}_2 \end{bmatrix},$$

where

$$\begin{aligned} \tilde{X}_1 &= (X_1 + Y_2 W)(I + W^* W)^{-1/2}, \\ \tilde{Y}_2 &= (Y_2 - X_1 W^*)(I + W W^*)^{-1/2}, \\ \tilde{A}_1 &= (I + W^* W)^{1/2} (A_1 + G W) (I + W^* W)^{-1/2}, \\ \tilde{A}_2 &= (I + W W^*)^{-1/2} (A_2 - W G) (I + W W^*)^{1/2}. \end{aligned}$$

Thus, $\|\tan \Theta(X_1, \tilde{X}_1)\|_2 < \frac{2\|E\|_2}{\eta}$.

Examples:

1. Bounds on $\|\Lambda - \tilde{\Lambda}_\tau\|_{\text{UI}}$ are, in fact, bounds on $\lambda_j - \lambda_{\tau(j)}$ in disguise, only more convenient and concise. For example, for $\|\cdot\|_{\text{UI}} = \|\cdot\|_2$ (spectral norm), $\|\Lambda - \tilde{\Lambda}_\tau\|_2 = \max_j |\lambda_j - \lambda_{\tau(j)}|$, and for $\|\cdot\|_{\text{UI}} = \|\cdot\|_F$ (Frobenius norm), $\|\Lambda - \tilde{\Lambda}_\tau\|_F = \left[\sum_{j=1}^n |\lambda_j - \lambda_{\tau(j)}|^2 \right]^{1/2}$.
2. Let $A, \tilde{A} \in \mathbb{C}^{n \times n}$ as follows, where $\varepsilon > 0$.

$$A = \begin{bmatrix} \mu & 1 & & & \\ & \mu & \ddots & & \\ & & \ddots & 1 & \\ & & & & \mu \end{bmatrix}, \quad \tilde{A} = \begin{bmatrix} \mu & 1 & & & \\ & \mu & \ddots & & \\ & & \ddots & 1 & \\ \varepsilon & & & & \mu \end{bmatrix}.$$

It can be seen that $\sigma(A) = \{\mu, \dots, \mu\}$ (repeated n times) and the characteristic polynomial $\det(tI - \tilde{A}) = (t - \mu)^n - \varepsilon$, which gives $\sigma(\tilde{A}) = \{\mu + \varepsilon^{1/n} e^{2ij\pi/n}, 0 \leq j \leq n-1\}$. Thus,

$|\tilde{\lambda} - \mu| = \varepsilon^{1/n} = \|\Delta A\|_2^{1/n}$. This shows that the fractional power $\|\Delta A\|_2^{1/n}$ in Facts 1 and 2 cannot be removed in general.

3. Consider

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 4.001 \end{bmatrix} \quad \text{is perturbed by} \quad \Delta A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0.001 & 0 & 0 \end{bmatrix}.$$

A 's eigenvalues are easily read off, and

$$\lambda_1 = 1, \mathbf{x}_1 = [1, 0, 0]^T, \mathbf{y}_1 = [0.8285, -0.5523, 0.0920]^T,$$

$$\lambda_2 = 4, \mathbf{x}_2 = [0.5547, 0.8321, 0]^T, \mathbf{y}_2 = [0, 0.0002, -1.0000]^T,$$

$$\lambda_3 = 4.001, \mathbf{x}_3 = [0.5547, 0.8321, 0.0002]^T, \mathbf{y}_3 = [0, 0, 1]^T.$$

On the other hand, $\tilde{A}$'s eigenvalues computed by MATLAB's `eig` are $\tilde{\lambda}_1 = 1.0001$, $\tilde{\lambda}_2 = 3.9427$, $\tilde{\lambda}_3 = 4.0582$. The following table gives $|\tilde{\lambda}_j - \lambda_j|$ with upper bounds up to the 1st order by Fact 3.

j	$\text{cond}(\lambda_j)$	$\text{cond}(\lambda_j)\ \Delta A\ _2$	$ \tilde{\lambda}_j - \lambda_j $
1	1.2070	0.0012	0.0001
2	$6.0 \cdot 10^3$	6.0	0.057
3	$6.0 \cdot 10^3$	6.0	0.057

We see that $\text{cond}(\lambda_j)\|\Delta A\|_2$ gives a fairly good error bound for $j = 1$, but dramatically worse for $j = 2, 3$. There are two reasons for this: One is in the choice of ΔA and the other is that ΔA 's order of magnitude is too big for the first order bound $\text{cond}(\lambda_j)\|\Delta A\|_2$ to be effective for $j = 2, 3$. Note that ΔA has the same order of magnitude as the difference between λ_2 and λ_3 and that is too big usually. For better understanding of this first order error bound, the reader may play with this example with $\Delta A = \varepsilon \frac{\mathbf{y}_j \mathbf{x}_j^*}{\|\mathbf{y}_j\|_2 \|\mathbf{x}_j^*\|_2}$ for various tiny parameters ε .

4. Let $\Sigma = \text{diag}(c_1, c_2, \dots, c_k)$ and $\Gamma = \text{diag}(s_1, s_2, \dots, s_k)$, where $c_j, s_j \geq 0$ and $c_j^2 + s_j^2 = 1$ for all j . The canonical angles between

$$X = Q \begin{bmatrix} I_k \\ 0 \\ 0 \end{bmatrix} V^*, \quad Y = Q \begin{bmatrix} \Sigma \\ \Gamma \\ 0 \end{bmatrix} U^*$$

are $\theta_j = \arccos c_j$, $j = 1, 2, \dots, k$, where Q, U, V are unitary. On the other hand, every pair of $X, Y \in \mathbb{C}^{n \times k}$ with $2k \leq n$ and $X^*X = Y^*Y = I_k$, having canonical angles $\arccos c_j$, can be represented this way [SS90, p. 40].

5. Fact 13 is most useful when $\|E\|_2$ is tiny and the computation of A 's eigenvalues is then decoupled into two smaller ones. In eigenvalue computations, we often seek unitary $[X_1 \ X_2]$ such that

$$\begin{bmatrix} X_1^* \\ X_2^* \end{bmatrix} A [X_1 \ X_2] = \begin{bmatrix} M & E^* \\ E & H \end{bmatrix}, \quad \begin{bmatrix} X_1^* \\ X_2^* \end{bmatrix} \tilde{A} [X_1 \ X_2] = \begin{bmatrix} M & 0 \\ 0 & H \end{bmatrix},$$

and $\|E\|_2$ is tiny. Since a unitarily similarity transformation does not alter eigenvalues, Fact 13 still applies.

6. [LL05] Consider the 2×2 Hermitian matrix

$$A = \begin{bmatrix} \alpha & \varepsilon \\ \varepsilon & \beta \end{bmatrix},$$

where $\alpha > \beta$ and $\varepsilon > 0$. It has two eigenvalues

$$\lambda_{\pm} = \frac{\alpha + \beta \pm \sqrt{(\alpha - \beta)^2 + 4\varepsilon^2}}{2},$$

and

$$0 < \left\{ \begin{array}{c} \lambda_+ - \alpha \\ \beta - \lambda_- \end{array} \right\} = \frac{2\varepsilon^2}{(\alpha - \beta) + \sqrt{(\alpha - \beta)^2 + 4\varepsilon^2}}.$$

The inequalities in Fact 13 become equalities for the example.

15.2 Singular Value Problems

The reader is referred to Section 5.6, Chapters 17 and 45 for more information on singular value decompositions.

Definitions:

$B \in \mathbb{C}^{m \times n}$ has a (first standard form) SVD $B = U\Sigma V^*$, where $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$ are unitary, and $\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots) \in \mathbb{R}^{m \times n}$ is leading diagonal (σ_j starts in the top-left corner) with all $\sigma_j \geq 0$.

Let $\text{sv}(B) = \{\sigma_1, \sigma_2, \dots, \sigma_{\min\{m,n\}}\}$, the set of B 's singular values, and $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$, and let $\text{sv}_{\text{ext}}(B) = \text{sv}(B)$ unless $m > n$ for which $\text{sv}_{\text{ext}}(B) = \text{sv}(B) \cup \{0, \dots, 0\}$ (additional $m - n$ zeros).

A vector–scalar–vector triplet $(\mathbf{u}, \sigma, \mathbf{v}) \in \mathbb{C}^m \times \mathbb{R} \times \mathbb{C}^n$ is a **singular-triplet** if $\mathbf{u} \neq 0$, $\mathbf{v} \neq 0$, $\sigma \geq 0$, and $B\mathbf{v} = \sigma\mathbf{u}$, $B^*\mathbf{u} = \sigma\mathbf{v}$.

B is perturbed to $\tilde{B} = B + \Delta B$. The same notation is adopted for $\tilde{B}$, except all symbols with tildes.

Facts:

1. [SS90, p. 204] (Mirsky) $\|\Sigma - \tilde{\Sigma}\|_{\text{UI}} \leq \|\Delta B\|_{\text{UI}}$.
2. Let residuals $\mathbf{r} = B\tilde{\mathbf{v}} - \tilde{\mu}\tilde{\mathbf{u}}$ and $\mathbf{s} = B^*\tilde{\mathbf{u}} - \tilde{\mu}\tilde{\mathbf{v}}$, and $\|\tilde{\mathbf{v}}\|_2 = \|\tilde{\mathbf{u}}\|_2 = 1$.
 - (a) [Sun98] The smallest error matrix ΔB (in the 2-norm), for which $(\tilde{\mathbf{u}}, \tilde{\mu}, \tilde{\mathbf{v}})$ is an exact singular-triplet of $\tilde{B} = B + \Delta B$, satisfies $\|\Delta B\|_2 = \varepsilon$, where $\varepsilon = \max\{\|\mathbf{r}\|_2, \|\mathbf{s}\|_2\}$.
 - (b) $|\tilde{\mu} - \mu| \leq \varepsilon$ for some singular value μ of B .
 - (c) Let μ be the closest singular value in $\text{sv}_{\text{ext}}(B)$ to $\tilde{\mu}$ and $(\mathbf{u}, \sigma, \mathbf{v})$ be the associated singular-triplet with $\|\mathbf{u}\|_2 = \|\mathbf{v}\|_2 = 1$, and let $\eta = \min |\tilde{\mu} - \sigma|$ over all $\sigma \in \text{sv}_{\text{ext}}(B)$ and $\sigma \neq \mu$. If $\eta > 0$, then $|\tilde{\mu} - \mu| \leq \varepsilon^2/\eta$, and [SS90, p. 260]

$$\sqrt{\sin^2 \angle(\tilde{\mathbf{u}}, \mathbf{u}) + \sin^2 \angle(\tilde{\mathbf{v}}, \mathbf{v})} \leq \frac{\sqrt{\|\mathbf{r}\|_2^2 + \|\mathbf{s}\|_2^2}}{\eta}.$$

3. [LL05] Let

$$B = \begin{bmatrix} B_1 & F \\ E & B_2 \end{bmatrix} \in \mathbb{C}^{m \times n}, \quad \tilde{B} = \begin{bmatrix} B_1 & 0 \\ 0 & B_2 \end{bmatrix},$$

where $B_1 \in \mathbb{C}^{k \times k}$, and set $\eta = \min |\mu - \nu|$ over all $\mu \in \text{sv}(B_1)$ and $\nu \in \text{sv}_{\text{ext}}(B_2)$, and $\varepsilon = \max\{\|E\|_2, \|F\|_2\}$. Then

$$\max_j |\sigma_j - \tilde{\sigma}_j| \leq \frac{2\varepsilon^2}{\eta + \sqrt{\eta^2 + 4\varepsilon^2}}.$$

4. [SS90, p. 260] (**Wedin**) Let $B, \tilde{B} \in \mathbb{C}^{m \times n}$ ($m \geq n$) have decompositions

$$\begin{bmatrix} U_1^* \\ U_2^* \end{bmatrix} B [V_1 \ V_2] = \begin{bmatrix} B_1 & 0 \\ 0 & B_2 \end{bmatrix}, \quad \begin{bmatrix} \tilde{U}_1^* \\ \tilde{U}_2^* \end{bmatrix} \tilde{B} [\tilde{V}_1 \ \tilde{V}_2] = \begin{bmatrix} \tilde{B}_1 & 0 \\ 0 & \tilde{B}_2 \end{bmatrix},$$

where $[U_1 \ U_2], [V_1 \ V_2], [\tilde{U}_1 \ \tilde{U}_2]$, and $[\tilde{V}_1 \ \tilde{V}_2]$ are unitary, and $U_1, \tilde{U}_1 \in \mathbb{C}^{m \times k}$, $V_1, \tilde{V}_1 \in \mathbb{C}^{n \times k}$. Set

$$R = B\tilde{V}_1 - \tilde{U}_1\tilde{B}_1, \quad S = B^*\tilde{U}_1 - \tilde{V}_1\tilde{B}_1.$$

If $\text{sv}(\tilde{B}_1) \cap \text{sv}_{\text{ext}}(B_2) = \emptyset$, then

$$\sqrt{\|\sin \Theta(U_1, \tilde{U}_1)\|_{\text{F}}^2 + \|\sin \Theta(V_1, \tilde{V}_1)\|_{\text{F}}^2} \leq \frac{\sqrt{\|R\|_{\text{F}}^2 + \|S\|_{\text{F}}^2}}{\eta},$$

where $\eta = \min |\tilde{\mu} - \nu|$ over all $\tilde{\mu} \in \text{sv}(\tilde{B}_1)$ and $\nu \in \text{sv}_{\text{ext}}(B_2)$.

Examples:

1. Let

$$B = \begin{bmatrix} 3 \cdot 10^{-3} & 1 \\ 2 & 4 \cdot 10^{-3} \end{bmatrix}, \quad \tilde{B} = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} = [\mathbf{e}_2 \ \mathbf{e}_1] \begin{bmatrix} 2 & \\ & 1 \end{bmatrix} \begin{bmatrix} \mathbf{e}_1^T \\ \mathbf{e}_2^T \end{bmatrix}.$$

Then $\sigma_1 = 2.000012$, $\sigma_2 = 0.999988$, and $\tilde{\sigma}_1 = 2$, $\tilde{\sigma}_2 = 1$. Fact 1 gives

$$\max_{1 \leq j \leq 2} |\sigma_j - \tilde{\sigma}_j| \leq 4 \cdot 10^{-3}, \quad \sqrt{\sum_{j=1}^2 |\sigma_j - \tilde{\sigma}_j|^2} \leq 5 \cdot 10^{-3}.$$

2. Let B be as in the previous example, and let $\tilde{\mathbf{v}} = \mathbf{e}_1, \tilde{\mathbf{u}} = \mathbf{e}_2, \tilde{\mu} = 2$. Then $\mathbf{r} = B\tilde{\mathbf{v}} - \tilde{\mu}\tilde{\mathbf{u}} = 3 \cdot 10^{-3}\mathbf{e}_1$ and $\mathbf{s} = B^*\tilde{\mathbf{u}} - \tilde{\mu}\tilde{\mathbf{v}} = 4 \cdot 10^{-3}\mathbf{e}_2$. Fact 2 applies. Note that, without calculating $\text{sv}(B)$, one may bound η needed for Fact 2(c) from below as follows. Since B has two singular values that are near 1 and $\tilde{\mu} = 2$, respectively, with errors no bigger than $4 \cdot 10^{-3}$, then $\eta \geq 2 - (1 + 4 \cdot 10^{-3}) = 1 - 4 \cdot 10^{-3}$.
3. Let B and $\tilde{B}$ be as in Example 1. Fact 3 gives $\max_{1 \leq j \leq 2} |\sigma_j - \tilde{\sigma}_j| \leq 1.6 \cdot 10^{-5}$, a much better bound than by Fact 1.
4. Let B and $\tilde{B}$ be as in Example 1. Note $\tilde{B}$'s SVD there. Apply Fact 4 with $k = 1$ to give a similar bound as by Fact 2(c).
5. Since unitary transformations do not change singular values, Fact 3 applies to $B, \tilde{B} \in \mathbb{C}^{m \times n}$ having decompositions

$$\begin{bmatrix} U_1^* \\ U_2^* \end{bmatrix} B [V_1 \ V_2] = \begin{bmatrix} B_1 & F \\ E & B_2 \end{bmatrix}, \quad \begin{bmatrix} \tilde{U}_1^* \\ \tilde{U}_2^* \end{bmatrix} \tilde{B} [\tilde{V}_1 \ \tilde{V}_2] = \begin{bmatrix} \tilde{B}_1 & 0 \\ 0 & \tilde{B}_2 \end{bmatrix},$$

where $[U_1 \ U_2]$ and $[V_1 \ V_2]$ are unitary and $U_1 \in \mathbb{C}^{m \times k}$, $V_1 \in \mathbb{C}^{n \times k}$.

15.3 Polar Decomposition

The reader is referred to Chapter 17.1 for definition and for more information on polar decompositions.

Definitions:

$B \in \mathbb{F}^{m \times n}$ is perturbed to $\tilde{B} = B + \Delta B$, and their polar decompositions are

$$B = QH, \quad \tilde{B} = \tilde{Q}\tilde{H} = (Q + \Delta Q)(H + \Delta H),$$

where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. ΔB is restricted to $\mathbb{F}$ for $B \in \mathbb{F}$.

Denote the singular values of B and $\tilde{B}$ as $\sigma_1 \geq \sigma_2 \geq \dots$ and $\tilde{\sigma}_1 \geq \tilde{\sigma}_2 \geq \dots$, respectively. The **condition numbers for the polar factors** in the Frobenius norm are defined as

$$\text{cond}_{\mathbb{F}}(X) = \lim_{\delta \rightarrow 0} \sup_{\|\Delta B\|_{\mathbb{F}} \leq \delta} \frac{\|\Delta X\|_{\mathbb{F}}}{\delta}, \quad \text{for } X = H \text{ or } Q.$$

B is **multiplicatively perturbed** to $\tilde{B}$ if $\tilde{B} = D_{\text{L}}^* B D_{\text{R}}$ for some $D_{\text{L}} \in \mathbb{F}^{m \times m}$ and $D_{\text{R}} \in \mathbb{F}^{n \times n}$.

B is said to be **graded** if it can be scaled as $B = GS$ such that G is “well-behaved” (i.e., $\kappa_2(G)$ is of modest magnitude), where S is a scaling matrix, often diagonal but not required so for the facts below. Interesting cases are when $\kappa_2(G) \ll \kappa_2(B)$.

Facts:

1. [CG00] The condition numbers $\text{cond}_{\mathbb{F}}(Q)$ and $\text{cond}_{\mathbb{F}}(H)$ are tabulated as follows, where $\kappa_2(B) = \sigma_1/\sigma_n$.

		$\mathbb{R}$	$\mathbb{C}$
Factor Q	$m = n$	$2/(\sigma_{n-1} + \sigma_n)$	$1/\sigma_n$
	$m > n$	$1/\sigma_n$	$1/\sigma_n$
Factor H	$m \geq n$	$\frac{\sqrt{2(1 + \kappa_2(B)^2)}}{1 + \kappa_2(B)}$	

2. [Kit86] $\|\Delta H\|_{\mathbb{F}} \leq \sqrt{2}\|\Delta B\|_{\mathbb{F}}$.
3. [Li95] If $m = n$ and $\text{rank}(B) = n$, then

$$\|\Delta Q\|_{\text{UI}} \leq \frac{2}{\sigma_n + \tilde{\sigma}_n} \|\Delta B\|_{\text{UI}}.$$

4. [Li95], [LS02] If $\text{rank}(B) = n$, then

$$\begin{aligned} \|\Delta Q\|_{\text{UI}} &\leq \left(\frac{2}{\sigma_n + \tilde{\sigma}_n} + \frac{1}{\max\{\sigma_n, \tilde{\sigma}_n\}} \right) \|\Delta B\|_{\text{UI}}, \\ \|\Delta Q\|_{\mathbb{F}} &\leq \frac{2}{\sigma_n + \tilde{\sigma}_n} \|\Delta B\|_{\mathbb{F}}. \end{aligned}$$

5. [Mat93] If $B \in \mathbb{R}^{n \times n}$, $\text{rank}(B) = n$, and $\|\Delta B\|_2 < \sigma_n$, then

$$\|\Delta Q\|_{\text{UI}} \leq -\frac{2\|\Delta B\|_{\text{UI}}}{\|\Delta B\|_2} \ln \left(1 - \frac{\|\Delta B\|_2}{\sigma_n + \sigma_{n-1}} \right),$$

where $\|\cdot\|_2$ is the **Ky Fan 2-norm**, i.e., the sum of the first two largest singular values. (See Chapter 17.3.)

6. [LS02] If $B \in \mathbb{R}^{n \times n}$, $\text{rank}(B) = n$, and $\|\Delta B\|_2 < \sigma_n + \tilde{\sigma}_n$, then

$$\|\Delta Q\|_{\mathbb{F}} \leq \frac{4}{\sigma_{n-1} + \sigma_n + \tilde{\sigma}_{n-1} + \tilde{\sigma}_n} \|\Delta B\|_{\mathbb{F}}.$$

7. [Li97] Let B and $\tilde{B} = D_{\text{L}}^* B D_{\text{R}}$ having full column rank. Then

$$\|\Delta Q\|_{\mathbb{F}} \leq \sqrt{\|I - D_{\text{L}}^{-1}\|_{\mathbb{F}}^2 + \|D_{\text{L}} - I\|_{\mathbb{F}}^2} + \sqrt{\|I - D_{\text{R}}^{-1}\|_{\mathbb{F}}^2 + \|D_{\text{R}} - I\|_{\mathbb{F}}^2}.$$

8. [Li97], [Li05] Let $B = GS$ and $\tilde{B} = \tilde{G}S$ and assume that G and B have full column rank. If $\|\Delta G\|_2 \|G^\dagger\|_2 < 1$, then

$$\begin{aligned}\|\Delta Q\|_F &\leq \gamma \|G^\dagger\|_2 \|\Delta G\|_F, \\ \|(\Delta H)S^{-1}\|_F &\leq \left(\gamma \|G^\dagger\|_2 \|G\|_2 + 1\right) \|\Delta G\|_F,\end{aligned}$$

$$\text{where } \gamma = \sqrt{1 + (1 - \|G^\dagger\|_2 \|\Delta G\|_2)^{-2}}.$$

Examples:

1. Take both B and $\tilde{B}$ to have orthonormal columns to see that some of the inequalities above on ΔQ are attainable.
2. Let

$$B = \frac{1}{\sqrt{2}} \begin{bmatrix} 2.01 & 502 \\ -1.99 & -498 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 10^{-2} & 2 \\ 2 & 5 \cdot 10^2 \end{bmatrix}$$

and

$$\tilde{B} = \begin{bmatrix} 1.4213 & 3.5497 \cdot 10^2 \\ -1.4071 & -3.5214 \cdot 10^2 \end{bmatrix}$$

obtained by rounding each entry of B to have five significant decimal digits. $B = QH$ can be read off above and $\tilde{B} = \tilde{Q}\tilde{H}$ can be computed by $\tilde{Q} = \tilde{U}\tilde{V}^*$ and $\tilde{H} = \tilde{V}\tilde{\Sigma}\tilde{V}^*$, where $\tilde{B}$'s SVD is $\tilde{U}\tilde{\Sigma}\tilde{V}^*$. One has

$$\text{sv}(B) = \{5.00 \cdot 10^2, 2.00 \cdot 10^{-3}\}, \text{sv}(\tilde{B}) = \{5.00 \cdot 10^2, 2.04 \cdot 10^{-3}\}$$

and

$\ \Delta B\ _2$	$\ \Delta B\ _F$	$\ \Delta Q\ _2$	$\ \Delta Q\ _F$	$\ \Delta H\ _2$	$\ \Delta H\ _F$
$3 \cdot 10^{-3}$	$3 \cdot 10^{-3}$	$2 \cdot 10^{-6}$	$3 \cdot 10^{-6}$	$2 \cdot 10^{-3}$	$2 \cdot 10^{-3}$

Fact 2 gives $\|\Delta H\|_F \leq 3 \cdot 10^{-3}$ and Fact 6 gives $\|\Delta Q\|_F \leq 10^{-5}$.

3. [Li97] and [Li05] have examples on the use of inequalities in Facts 7 and 8.

15.4 Generalized Eigenvalue Problems

The reader is referred to Section 43.1 for more information on generalized eigenvalue problems.

Definitions:

Let $A, B \in \mathbb{C}^{m \times n}$. A **matrix pencil** is a family of matrices $A - \lambda B$, parameterized by a (complex) number λ . The associated **generalized eigenvalue problem** is to find the nontrivial solutions of the equations

$$A\mathbf{x} = \lambda B\mathbf{x} \quad \text{and/or} \quad \mathbf{y}^* A = \lambda \mathbf{y}^* B,$$

where $\mathbf{x} \in \mathbb{C}^n$, $\mathbf{y} \in \mathbb{C}^m$, and $\lambda \in \mathbb{C}$.

$A - \lambda B$ is **regular** if $m = n$ and $\det(A - \lambda B) \neq 0$ for some $\lambda \in \mathbb{C}$.

In what follows, all pencils in question are assumed regular.

An eigenvalue λ is conveniently represented by a nonzero number pair, so-called a **generalized eigenvalue** $\langle \alpha, \beta \rangle$, interpreted as $\lambda = \alpha/\beta$. $\beta = 0$ corresponds to eigenvalue infinity.

A **generalized eigenpair** of $A - \lambda B$ refers to $(\langle \alpha, \beta \rangle, \mathbf{x})$ such that $\beta A \mathbf{x} = \alpha B \mathbf{x}$, where $\mathbf{x} \neq 0$ and $|\alpha|^2 + |\beta|^2 > 0$. A **generalized eigentriplet** of $A - \lambda B$ refers to $(\mathbf{y}, \langle \alpha, \beta \rangle, \mathbf{x})$ such that $\beta A \mathbf{x} = \alpha B \mathbf{x}$ and $\beta \mathbf{y}^* A = \alpha \mathbf{y}^* B$, where $\mathbf{x} \neq 0$, $\mathbf{y} \neq 0$, and $|\alpha|^2 + |\beta|^2 > 0$. The quantity

$$\text{cond}(\langle \alpha, \beta \rangle) = \frac{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2}{\sqrt{|\mathbf{y}^* A \mathbf{x}|^2 + |\mathbf{y}^* B \mathbf{x}|^2}}$$

is the **individual condition number** for the generalized eigenvalue $\langle \alpha, \beta \rangle$, where $(\mathbf{y}, \langle \alpha, \beta \rangle, \mathbf{x})$ is a generalized eigentriplet of $A - \lambda B$.

$A - \lambda B$ is perturbed to $\tilde{A} - \lambda \tilde{B} = (A + \Delta A) - \lambda(B + \Delta B)$.

Let $\sigma(A, B) = \{\langle \alpha_1, \beta_1 \rangle, \langle \alpha_2, \beta_2 \rangle, \dots, \langle \alpha_n, \beta_n \rangle\}$ be the set of the generalized eigenvalues of $A - \lambda B$, and set $Z = [A, B] \in \mathbb{C}^{2n \times n}$.

$A - \lambda B$ is **diagonalizable** if it is equivalent to a diagonal pencil, i.e., there are nonsingular $X, Y \in \mathbb{C}^{n \times n}$ such that $Y^* A X = \Lambda$, $Y^* B X = \Omega$, where $\Lambda = \text{diag}(\alpha_1, \alpha_2, \dots, \alpha_n)$ and $\Omega = \text{diag}(\beta_1, \beta_2, \dots, \beta_n)$.

$A - \lambda B$ is a **definite pencil** if both A and B are Hermitian and

$$\gamma(A, B) = \min_{\mathbf{x} \in \mathbb{C}^n, \|\mathbf{x}\|_2=1} |\mathbf{x}^* A \mathbf{x} + i \mathbf{x}^* B \mathbf{x}| > 0.$$

The same notation is adopted for $\tilde{A} - \lambda \tilde{B}$, except all symbols with tildes.

The **chordal distance** between two nonzero pairs $\langle \alpha, \beta \rangle$ and $\langle \tilde{\alpha}, \tilde{\beta} \rangle$ is

$$\chi(\langle \alpha, \beta \rangle, \langle \tilde{\alpha}, \tilde{\beta} \rangle) = \frac{|\tilde{\beta} \alpha - \tilde{\alpha} \beta|}{\sqrt{|\alpha|^2 + |\beta|^2} \sqrt{|\tilde{\alpha}|^2 + |\tilde{\beta}|^2}}.$$

Facts:

- [SS90, p. 293] Let $(\mathbf{y}, \langle \alpha, \beta \rangle, \mathbf{x})$ be a generalized eigentriplet of $A - \lambda B$. $[\Delta A, \Delta B]$ changes $\langle \alpha, \beta \rangle = \langle \mathbf{y}^* A \mathbf{x}, \mathbf{y}^* B \mathbf{x} \rangle$ to

$$\langle \tilde{\alpha}, \tilde{\beta} \rangle = \langle \alpha, \beta \rangle + \langle \mathbf{y}^* (\Delta A) \mathbf{x}, \mathbf{y}^* (\Delta B) \mathbf{x} \rangle + O(\varepsilon^2),$$

where $\varepsilon = \|[\Delta A, \Delta B]\|_2$, and $\chi(\langle \alpha, \beta \rangle, \langle \tilde{\alpha}, \tilde{\beta} \rangle) \leq \text{cond}(\langle \alpha, \beta \rangle) \varepsilon + O(\varepsilon^2)$.

- [SS90, p. 301], [Li88] If $A - \lambda B$ is diagonalizable, then

$$\max_i \min_j \chi(\langle \alpha_i, \beta_i \rangle, \langle \tilde{\alpha}_j, \tilde{\beta}_j \rangle) \leq \kappa_2(X) \|\sin \Theta(Z^*, \tilde{Z}^*)\|_2.$$

- [Li94, Lemma 3.3] (**Sun**)

$$\|\sin \Theta(Z^*, \tilde{Z}^*)\|_{\text{UI}} \leq \frac{\|Z - \tilde{Z}\|_{\text{UI}}}{\max\{\sigma_{\min}(Z), \sigma_{\min}(\tilde{Z})\}},$$

where $\sigma_{\min}(Z)$ is Z 's smallest singular value.

- The quantity $\gamma(A, B)$ is the minimum distance of the numerical range $W(A + iB)$ to the origin for definite pencil $A - \lambda B$.
- [SS90, p. 316] Suppose $A - \lambda B$ is a definite pencil. If $\tilde{A}$ and $\tilde{B}$ are Hermitian and $\|[\Delta A, \Delta B]\|_2 < \gamma(A, B)$, then $\tilde{A} - \lambda \tilde{B}$ is also a definite pencil and there exists a permutation τ of $\{1, 2, \dots, n\}$ such that

$$\max_{1 \leq j \leq n} \chi(\langle \alpha_j, \beta_j \rangle, \langle \tilde{\alpha}_{\tau(j)}, \tilde{\beta}_{\tau(j)} \rangle) \leq \frac{\|[\Delta A, \Delta B]\|_2}{\gamma(A, B)}.$$

- [SS90, p. 318] Definite pencil $A - \lambda B$ is always diagonalizable: $X^* A X = \Lambda$ and $X^* B X = \Omega$, and with real spectra. Facts 7 and 10 apply.

7. [Li03] Suppose $A - \lambda B$ and $\tilde{A} - \lambda \tilde{B}$ are diagonalizable with real spectra, i.e.,

$$Y^* A X = \Lambda, Y^* B X = \Omega \quad \text{and} \quad \tilde{Y}^* \tilde{A} \tilde{X} = \tilde{\Lambda}, \tilde{Y}^* \tilde{B} \tilde{X} = \tilde{\Omega},$$

and all $\langle \alpha_j, \beta_j \rangle$ and all $\langle \tilde{\alpha}_j, \tilde{\beta}_j \rangle$ are real. Then the follow statements hold, where

$$\Xi = \text{diag}(\chi(\langle \alpha_1, \beta_1 \rangle, \langle \tilde{\alpha}_{\tau(1)}, \tilde{\beta}_{\tau(1)} \rangle), \dots, \chi(\langle \alpha_n, \beta_n \rangle, \langle \tilde{\alpha}_{\tau(n)}, \tilde{\beta}_{\tau(n)} \rangle))$$

for some permutation τ of $\{1, 2, \dots, n\}$ (possibly depending on the norm being used). In all cases, the constant factor $\pi/2$ can be replaced by 1 for the 2-norm and the Frobenius norm.

$$(a) \quad \|\Xi\|_{\text{UI}} \leq \frac{\pi}{2} \sqrt{\kappa_2(X) \kappa_2(\tilde{X})} \|\sin \Theta(Z^*, \tilde{Z}^*)\|_{\text{UI}}.$$

- (b) If all $|\alpha_j|^2 + |\beta_j|^2 = |\tilde{\alpha}_j|^2 + |\tilde{\beta}_j|^2 = 1$ in their eigendecompositions, then

$$\|\Xi\|_{\text{UI}} \leq \frac{\pi}{2} \sqrt{\|X\|_2 \|Y^*\|_2 \|\tilde{X}\|_2 \|\tilde{Y}^*\|_2} \|\Delta A, \Delta B\|_{\text{UI}}.$$

8. Let residuals $\mathbf{r} = \tilde{\beta} A \tilde{\mathbf{x}} - \tilde{\alpha} B \tilde{\mathbf{x}}$ and $\mathbf{s}^* = \tilde{\beta} \tilde{\mathbf{y}}^* A - \tilde{\alpha} \tilde{\mathbf{y}}^* B$, where $\|\tilde{\mathbf{x}}\|_2 = \|\tilde{\mathbf{y}}\|_2 = 1$. The smallest error matrix $[\Delta A, \Delta B]$ in the 2-norm, for which $(\tilde{\mathbf{y}}, \langle \tilde{\alpha}, \tilde{\beta} \rangle, \tilde{\mathbf{x}})$ is an exact generalized eigentriplet of $\tilde{A} - \lambda \tilde{B}$, satisfies $\|[\Delta A, \Delta B]\|_2 = \varepsilon$, where $\varepsilon = \max\{\|\mathbf{r}\|_2, \|\mathbf{s}\|_2\}$, and $\chi(\langle \alpha, \beta \rangle, \langle \tilde{\alpha}, \tilde{\beta} \rangle) \leq \text{cond}(\langle \tilde{\alpha}, \tilde{\beta} \rangle) \varepsilon + O(\varepsilon^2)$ for some $\langle \alpha, \beta \rangle \in \sigma(A, B)$.

9. [BDD00, p. 128] Suppose A and B are Hermitian and B is positive definite, and let residual $\mathbf{r} = A \tilde{\mathbf{x}} - \tilde{\mu} B \tilde{\mathbf{x}}$ and $\|\tilde{\mathbf{x}}\|_2 = 1$.

- (a) For some eigenvalue μ of $A - \lambda B$,

$$|\tilde{\mu} - \mu| \leq \frac{\|\mathbf{r}\|_{B^{-1}}}{\|\tilde{\mathbf{x}}\|_B} \leq \|B^{-1}\|_2 \|\mathbf{r}\|_2,$$

where $\|\mathbf{z}\|_M = \sqrt{\mathbf{z}^* M \mathbf{z}}$.

- (b) Let μ be the closest eigenvalue to $\tilde{\mu}$ among all eigenvalues of $A - \lambda B$ and $\mathbf{x}$ its associated eigenvector with $\|\mathbf{x}\|_2 = 1$, and let $\eta = \min |\tilde{\mu} - \nu|$ over all other eigenvalues $\nu \neq \mu$ of $A - \lambda B$. If $\eta > 0$, then

$$|\tilde{\mu} - \mu| \leq \frac{1}{\eta} \cdot \left(\frac{\|\mathbf{r}\|_{B^{-1}}}{\|\tilde{\mathbf{x}}\|_B} \right)^2 \leq \|B^{-1}\|_2^2 \frac{\|\mathbf{r}\|_2^2}{\eta},$$

$$\sin \angle(\tilde{\mathbf{x}}, \mathbf{x}) \leq \|B^{-1}\|_2 \sqrt{2\kappa_2(B)} \frac{\|\mathbf{r}\|_2}{\eta}.$$

10. [Li94] Suppose $A - \lambda B$ and $\tilde{A} - \lambda \tilde{B}$ are diagonalizable and have eigendecompositions

$$\begin{bmatrix} Y_1^* \\ Y_2^* \end{bmatrix} A [X_1, X_2] = \begin{bmatrix} \Lambda_1 & \\ & \Lambda_2 \end{bmatrix}, \begin{bmatrix} Y_1^* \\ Y_2^* \end{bmatrix} B [X_1, X_2] = \begin{bmatrix} \Omega_1 & \\ & \Omega_2 \end{bmatrix},$$

$$X^{-1} = [W_1, W_2]^*,$$

and the same for $\tilde{A} - \lambda \tilde{B}$ except all symbols with tildes, where $X_1, Y_1, W_1 \in \mathbb{C}^{n \times k}$, $\Lambda_1, \Omega_1 \in \mathbb{C}^{k \times k}$. Suppose $|\alpha_j|^2 + |\beta_j|^2 = |\tilde{\alpha}_j|^2 + |\tilde{\beta}_j|^2 = 1$ for $1 \leq j \leq n$ in the eigendecompositions, and set $\eta = \min \chi(\langle \alpha, \beta \rangle, \langle \tilde{\alpha}, \tilde{\beta} \rangle)$ taken over all $\langle \alpha, \beta \rangle \in \sigma(\Lambda_1, \Omega_1)$ and $\langle \tilde{\alpha}, \tilde{\beta} \rangle \in \sigma(\tilde{\Lambda}_2, \tilde{\Omega}_2)$. If $\eta > 0$, then

$$\left\| \sin \Theta(X_1, \tilde{X}_1) \right\|_{\text{F}} \leq \frac{\|X_1^\dagger\|_2 \|\tilde{W}_2^\dagger\|_2}{\eta} \left\| \tilde{Y}_2^* (\tilde{Z} - Z) \begin{bmatrix} X_1 & \\ & X_1 \end{bmatrix} \right\|_{\text{F}}.$$

15.5 Generalized Singular Value Problems

Definitions:

Let $A \in \mathbb{C}^{m \times n}$ and $B \in \mathbb{C}^{\ell \times n}$. A matrix pair $\{A, B\}$ is an (m, ℓ, n) -**Grassmann matrix pair** if $\text{rank} \begin{pmatrix} A \\ B \end{pmatrix} = n$.

In what follows, all matrix pairs are (m, ℓ, n) -Grassmann matrix pairs.

A pair $\langle \alpha, \beta \rangle$ is a **generalized singular value** of $\{A, B\}$ if

$$\det(\beta^2 A^* A - \alpha^2 B^* B) = 0, \langle \alpha, \beta \rangle \neq \langle 0, 0 \rangle, \alpha, \beta \geq 0,$$

i.e., $\langle \alpha, \beta \rangle = \langle \sqrt{\mu}, \sqrt{\nu} \rangle$ for some generalized eigenvalue $\langle \mu, \nu \rangle$ of matrix pencil $A^* A - \lambda B^* B$.

Generalized Singular Value Decomposition (GSVD) of $\{A, B\}$:

$$U^* A X = \Sigma_A, \quad V^* B X = \Sigma_B,$$

where $U \in \mathbb{C}^{m \times m}$, $V \in \mathbb{C}^{\ell \times \ell}$ are unitary, $X \in \mathbb{C}^{n \times n}$ is nonsingular, $\Sigma_A = \text{diag}(\alpha_1, \alpha_2, \dots)$ is **leading diagonal** (α_j starts in the top left corner), and $\Sigma_B = \text{diag}(\dots, \beta_{n-1}, \beta_n)$ is **trailing diagonal** (β_j ends in the bottom-right corner), $\alpha_j, \beta_j \geq 0$ and $\alpha_j^2 + \beta_j^2 = 1$ for $1 \leq j \leq n$. (Set some $\alpha_j = 0$ and/or some $\beta_j = 0$, if necessary.)

$\{A, B\}$ is perturbed to $\{\tilde{A}, \tilde{B}\} = \{A + \Delta A, B + \Delta B\}$.

Let $\text{sv}(A, B) = \{\langle \alpha_1, \beta_1 \rangle, \langle \alpha_2, \beta_2 \rangle, \dots, \langle \alpha_n, \beta_n \rangle\}$ be the set of the generalized singular values of $\{A, B\}$,

and set $Z = \begin{bmatrix} A \\ B \end{bmatrix} \in \mathbb{C}^{(m+\ell) \times n}$.

The same notation is adopted for $\{\tilde{A}, \tilde{B}\}$, except all symbols with tildes.

Facts:

1. If $\{A, B\}$ is an (m, ℓ, n) -Grassmann matrix pair, then $A^* A - \lambda B^* B$ is a definite matrix pencil.
2. [Van76] The GSVD of an (m, ℓ, n) -Grassmann matrix pair $\{A, B\}$ exists.
3. [Li93] There exist permutations τ and ω of $\{1, 2, \dots, n\}$ such that

$$\begin{aligned} \max_{1 \leq j \leq n} \chi(\langle \alpha_i, \beta_i \rangle, \langle \tilde{\alpha}_{\tau(i)}, \tilde{\beta}_{\tau(i)} \rangle) &\leq \|\sin \Theta(Z, \tilde{Z})\|_2, \\ \sqrt{\sum_{j=1}^n \left[\chi(\langle \alpha_i, \beta_i \rangle, \langle \tilde{\alpha}_{\omega(i)}, \tilde{\beta}_{\omega(i)} \rangle) \right]^2} &\leq \|\sin \Theta(Z, \tilde{Z})\|_F. \end{aligned}$$

4. [Li94, Lemma 3.3] (**Sun**)

$$\|\sin \Theta(Z, \tilde{Z})\|_{\text{UI}} \leq \frac{\|Z - \tilde{Z}\|_{\text{UI}}}{\max\{\sigma_{\min}(Z), \sigma_{\min}(\tilde{Z})\}},$$

where $\sigma_{\min}(Z)$ is Z 's smallest singular value.

5. [Pai84] If $\alpha_i^2 + \beta_i^2 = \tilde{\alpha}_i^2 + \tilde{\beta}_i^2 = 1$ for $i = 1, 2, \dots, n$, then there exists a permutation ϖ of $\{1, 2, \dots, n\}$ such that

$$\sqrt{\sum_{j=1}^n \left[(\alpha_j - \tilde{\alpha}_{\varpi(j)})^2 + (\beta_j - \tilde{\beta}_{\varpi(j)})^2 \right]} \leq \min_{Q \text{ unitary}} \|Z_0 - \tilde{Z}_0 Q\|_F,$$

where $Z_0 = Z(Z^* Z)^{-1/2}$ and $\tilde{Z}_0 = \tilde{Z}(\tilde{Z}^* \tilde{Z})^{-1/2}$.

6. [Li93], [Sun83] Perturbation bounds on generalized singular subspaces (those spanned by one or a few columns of U , V , and X in GSVD) are also available, but it is quite complicated.

15.6 Relative Perturbation Theory for Eigenvalue Problems

Definitions:

Let scalar $\tilde{\alpha}$ be an approximation to α , and $1 \leq p \leq \infty$. Define **relative distances** between α and $\tilde{\alpha}$ as follows. For $|\alpha|^2 + |\tilde{\alpha}|^2 \neq 0$,

$$d(\alpha, \tilde{\alpha}) = \left| \frac{\tilde{\alpha}}{\alpha} - 1 \right| = \frac{|\tilde{\alpha} - \alpha|}{|\alpha|}, \quad (\text{classical measure})$$

$$\varrho_p(\alpha, \tilde{\alpha}) = \frac{|\tilde{\alpha} - \alpha|}{\sqrt[p]{|\alpha|^p + |\tilde{\alpha}|^p}}, \quad ([\text{Li98}])$$

$$\zeta(\alpha, \tilde{\alpha}) = \frac{|\tilde{\alpha} - \alpha|}{\sqrt{|\alpha\tilde{\alpha}|}}, \quad ([\text{BD90}], [\text{DV92}])$$

$$\varsigma(\alpha, \tilde{\alpha}) = |\ln(\tilde{\alpha}/\alpha)|, \quad \text{for } \tilde{\alpha}\alpha > 0, \quad ([\text{LM99a}], [\text{Li99b}])$$

and $d(0, 0) = \varrho_p(0, 0) = \zeta(0, 0) = \varsigma(0, 0) = 0$.

$A \in \mathbb{C}^{n \times n}$ is **multiplicatively perturbed** to $\tilde{A}$ if $\tilde{A} = D_L^* A D_R$ for some $D_L, D_R \in \mathbb{C}^{n \times n}$.

Denote $\sigma(A) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$ and $\sigma(\tilde{A}) = \{\tilde{\lambda}_1, \tilde{\lambda}_2, \dots, \tilde{\lambda}_n\}$.

$A \in \mathbb{C}^{n \times n}$ is said to be **graded** if it can be scaled as $A = S^* H S$ such that H is “well-behaved” (i.e., $\kappa_2(H)$ is of modest magnitude), where S is a scaling matrix, often diagonal but not required so for the facts below. Interesting cases are when $\kappa_2(H) \ll \kappa_2(A)$.

Facts:

1. [Bar00] $\varrho_p(\cdot, \cdot)$ is a metric on $\mathbb{C}$ for $1 \leq p \leq \infty$.
2. Let $A, \tilde{A} = D^* A D \in \mathbb{C}^{n \times n}$ be Hermitian, where D is nonsingular.

(a) [HJ85, p. 224] (**Ostrowski**) There exists t_j , satisfying

$$\lambda_{\min}(D^* D) \leq t_j \leq \lambda_{\max}(D^* D),$$

such that $\tilde{\lambda}_j^\uparrow = t_j \lambda_j^\uparrow$ for $j = 1, 2, \dots, n$ and, thus,

$$\max_{1 \leq j \leq n} d(\lambda_j^\uparrow, \tilde{\lambda}_j^\uparrow) \leq \|I - D^* D\|_2.$$

(b) [LM99], [Li98]

$$\|\text{diag}(\varsigma(\lambda_1^\uparrow, \tilde{\lambda}_1^\uparrow), \dots, \varsigma(\lambda_n^\uparrow, \tilde{\lambda}_n^\uparrow))\|_{\text{UI}} \leq \|\ln(D^* D)\|_{\text{UI}},$$

$$\|\text{diag}(\zeta(\lambda_1^\uparrow, \tilde{\lambda}_1^\uparrow), \dots, \zeta(\lambda_n^\uparrow, \tilde{\lambda}_n^\uparrow))\|_{\text{UI}} \leq \|D^* - D^{-1}\|_{\text{UI}}.$$

3. [Li98], [LM99] Let $A = S^* H S$ be a positive semidefinite Hermitian matrix, perturbed to $\tilde{A} = S^*(H + \Delta H)S$. Suppose H is positive definite and $\|H^{-1/2}(\Delta H)H^{-1/2}\|_2 < 1$, and set

$$M = H^{1/2} S S^* H^{1/2}, \quad \widetilde{M} = D M D,$$

where $D = [I + H^{-1/2}(\Delta H)H^{-1/2}]^{1/2} = D^*$. Then $\sigma(A) = \sigma(M)$ and $\sigma(\tilde{A}) = \sigma(\widetilde{M})$, and the inequalities in Fact 2 above hold with D here. Note that

$$\begin{aligned}\|D - D^{-1}\|_{\text{UI}} &\leq \frac{\|H^{-1/2}(\Delta H)H^{-1/2}\|_{\text{UI}}}{\sqrt{1 - \|H^{-1/2}(\Delta H)H^{-1/2}\|_2}} \\ &\leq \frac{\|H^{-1}\|_2}{\sqrt{1 - \|H^{-1}\|_2\|\Delta H\|_2}} \|\Delta H\|_{\text{UI}}.\end{aligned}$$

4. [BD90], [VS93] Suppose A and $\tilde{A}$ are Hermitian, and let $|A| = (A^2)^{1/2}$ be the positive semidefinite square root of A^2 . If there exists $0 \leq \delta < 1$ such that

$$|\mathbf{x}^*(\Delta A)\mathbf{x}| \leq \delta \mathbf{x}^*|A|\mathbf{x} \quad \text{for all } \mathbf{x} \in \mathbb{C}^n,$$

then either $\tilde{\lambda}_j^\uparrow = \lambda_j^\uparrow = 0$ or $1 - \delta \leq \tilde{\lambda}_j^\uparrow/\lambda_j^\uparrow \leq 1 + \delta$.

5. [Li99a] Let Hermitian A , $\tilde{A} = D^*AD$ have decompositions

$$\begin{bmatrix} X_1^* \\ X_2^* \end{bmatrix} A \begin{bmatrix} X_1 & X_2 \end{bmatrix} = \begin{bmatrix} A_1 & \\ & A_2 \end{bmatrix}, \quad \begin{bmatrix} \tilde{X}_1^* \\ \tilde{X}_2^* \end{bmatrix} \tilde{A} \begin{bmatrix} \tilde{X}_1 & \tilde{X}_2 \end{bmatrix} = \begin{bmatrix} \tilde{A}_1 & \\ & \tilde{A}_2 \end{bmatrix},$$

where $\begin{bmatrix} X_1 & X_2 \end{bmatrix}$ and $\begin{bmatrix} \tilde{X}_1 & \tilde{X}_2 \end{bmatrix}$ are unitary and $X_1, \tilde{X}_1 \in \mathbb{C}^{n \times k}$. If $\eta_2 = \min_{\mu \in \sigma(A_1), \tilde{\mu} \in \sigma(\tilde{A}_2)} \varrho_2(\mu, \tilde{\mu}) > 0$, then

$$\|\sin \Theta(X_1, \tilde{X}_1)\|_{\text{F}} \leq \frac{\sqrt{\|(I - D^{-1})X_1\|_{\text{F}}^2 + \|(I - D^*)X_1\|_{\text{F}}^2}}{\eta_2}.$$

6. [Li99a] Let $A = S^*HS$ be a positive semidefinite Hermitian matrix, perturbed to $\tilde{A} = S^*(H + \Delta H)S$, having decompositions, in notation, the same as in Fact 5. Let $D = [I + H^{-1/2}(\Delta H)H^{-1/2}]^{1/2}$. Assume H is positive definite and $\|H^{-1/2}(\Delta H)H^{-1/2}\|_2 < 1$. If $\eta_\zeta = \min_{\mu \in \sigma(A_1), \tilde{\mu} \in \sigma(\tilde{A}_2)} \zeta(\mu, \tilde{\mu}) > 0$, then

$$\|\sin \Theta(X_1, \tilde{X}_1)\|_{\text{F}} \leq \frac{\|D - D^{-1}\|_{\text{F}}}{\eta_\zeta}.$$

Examples:

1. [DK90], [EI95] Let A be a real symmetric tridiagonal matrix with zero diagonal and off-diagonal entries $b_1, b_2, \dots, b_{n-1}$. Suppose $\tilde{A}$ is identical to A except for its off-diagonal entries which change to $\beta_1 b_1, \beta_2 b_2, \dots, \beta_{n-1} b_{n-1}$, where all β_i are real and supposedly close to 1. Then $\tilde{A} = DAD$, where $D = \text{diag}(d_1, d_2, \dots, d_n)$ with

$$d_{2k} = \frac{\beta_1 \beta_3 \cdots \beta_{2k-1}}{\beta_2 \beta_4 \cdots \beta_{2k-2}}, \quad d_{2k+1} = \frac{\beta_2 \beta_4 \cdots \beta_{2k}}{\beta_1 \beta_3 \cdots \beta_{2k-1}}.$$

Let $\beta = \prod_{j=1}^{n-1} \max\{\beta_j, 1/\beta_j\}$. Then $\beta^{-1}I \leq D^2 \leq \beta I$, and Fact 2 and Fact 5 apply. Now if all $1 - \varepsilon \leq \beta_j \leq 1 + \varepsilon$, then $(1 - \varepsilon)^{n-1} \leq \beta^{-1} \leq \beta \leq (1 + \varepsilon)^{n-1}$.

2. Let $A = SHS$ with $S = \text{diag}(1, 10, 10^2, 10^3)$, and

$$A = \begin{bmatrix} 1 & & & \\ & 1 & & \\ 1 & 10^2 & 10^2 & \\ & 10^2 & 10^4 & 10^4 \\ & & 10^4 & 10^6 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 10^{-1} & & \\ 10^{-1} & 1 & 10^{-1} & \\ & 10^{-1} & 1 & 10^{-1} \\ & & 10^{-1} & 1 \end{bmatrix}.$$

Suppose that each entry A_{ij} of A is perturbed to $A_{ij}(1 + \delta_{ij})$ with $|\delta_{ij}| \leq \varepsilon$. Then $|(\Delta H)_{ij}| \leq \varepsilon |H_{ij}|$ and thus $\|\Delta H\|_2 \leq 1.2\varepsilon$. Since $\|H^{-1}\|_2 \leq 10/8$, Fact 3 implies

$$\zeta(\lambda_j^\uparrow, \tilde{\lambda}_j^\uparrow) \leq 1.5\varepsilon/\sqrt{1 - 1.5\varepsilon} \approx 1.5\varepsilon.$$

15.7 Relative Perturbation Theory for Singular Value Problems

Definitions:

$B \in \mathbb{C}^{m \times n}$ is **multiplicatively perturbed** to $\tilde{B}$ if $\tilde{B} = D_L^* B D_R$ for some $D_L \in \mathbb{C}^{m \times m}$ and $D_R \in \mathbb{C}^{n \times n}$.

Denote the singular values of B and $\tilde{B}$ as

$$\text{sv}(B) = \{\sigma_1, \sigma_2, \dots, \sigma_{\min\{m,n\}}\}, \quad \text{sv}(\tilde{B}) = \{\tilde{\sigma}_1, \tilde{\sigma}_2, \dots, \tilde{\sigma}_{\min\{m,n\}}\}.$$

B is said to be (highly) **graded** if it can be scaled as $B = GS$ such that G is “well-behaved” (i.e., $\kappa_2(G)$ is of modest magnitude), where S is a scaling matrix, often diagonal but not required so for the facts below. Interesting cases are when $\kappa_2(G) \ll \kappa_2(B)$.

Facts:

1. Let $B, \tilde{B} = D_L^* B D_R \in \mathbb{C}^{m \times n}$, where D_L and D_R are nonsingular.
 - (a) [EI95] For $1 \leq j \leq n$, $\frac{\sigma_j}{\|D_L^{-1}\|_2 \|D_R^{-1}\|_2} \leq \tilde{\sigma}_j \leq \sigma_j \|D_L\|_2 \|D_R\|_2$.
 - (b) [Li98], [LM99]

$$\begin{aligned} & \|\text{diag}(\zeta(\sigma_1, \tilde{\sigma}_1), \dots, \zeta(\sigma_n, \tilde{\sigma}_n))\|_{\text{UI}} \\ & \leq \frac{1}{2} \|D_L^* - D_L^{-1}\|_{\text{UI}} + \frac{1}{2} \|D_R^* - D_R^{-1}\|_{\text{UI}}. \end{aligned}$$

2. [Li99a] Let $B, \tilde{B} = D_L^* B D_R \in \mathbb{C}^{m \times n}$ ($m \geq n$) have decompositions

$$\begin{bmatrix} U_1^* \\ U_2^* \end{bmatrix} B \begin{bmatrix} V_1 & V_2 \end{bmatrix} = \begin{bmatrix} B_1 & 0 \\ 0 & B_2 \end{bmatrix}, \quad \begin{bmatrix} \tilde{U}_1^* \\ \tilde{U}_2^* \end{bmatrix} \tilde{B} \begin{bmatrix} \tilde{V}_1 & \tilde{V}_2 \end{bmatrix} = \begin{bmatrix} \tilde{B}_1 & 0 \\ 0 & \tilde{B}_2 \end{bmatrix},$$

where $[U_1 \ U_2], [V_1 \ V_2], [\tilde{U}_1 \ \tilde{U}_2]$, and $[\tilde{V}_1 \ \tilde{V}_2]$ are unitary, and $U_1, \tilde{U}_1 \in \mathbb{C}^{m \times k}$, $V_1, \tilde{V}_1 \in \mathbb{C}^{n \times k}$. Set $\Theta_U = \Theta(U_1, \tilde{U}_1)$, $\Theta_V = \Theta(V_1, \tilde{V}_1)$. If $\text{sv}(B_1) \cap \text{sv}_{\text{ext}}(\tilde{B}_2) = \emptyset$, then

$$\begin{aligned} & \sqrt{\|\sin \Theta_U\|_F^2 + \|\sin \Theta_V\|_F^2} \\ & \leq \frac{1}{\eta_2} [\|(I - D_L^*)U_1\|_F^2 + \|(I - D_L^{-1})U_1\|_F^2 \\ & \quad + \|(I - D_R^*)V_1\|_F^2 + \|(I - D_R^{-1})V_1\|_F^2]^{1/2}, \end{aligned}$$

where $\eta_2 = \min \varrho_2(\mu, \tilde{\mu})$ over all $\mu \in \text{sv}(B_1)$ and $\tilde{\mu} \in \text{sv}_{\text{ext}}(\tilde{B}_2)$.

3. [Li98], [Li99a], [LM99] Let $B = GS$ and $\tilde{B} = \tilde{G}\tilde{S}$ be two $m \times n$ matrices, where $\text{rank}(G) = n$, and let $\Delta G = \tilde{G} - G$. Then $\tilde{B} = DB$, where $D = I + (\Delta G)G^\dagger$. Fact 1 and Fact 2 apply with $D_L = D$ and $D_R = I$. Note that

$$\|D^* - D^{-1}\|_{\text{UI}} \leq \left(1 + \frac{1}{1 - \|(\Delta G)G^\dagger\|_2}\right) \frac{\|(\Delta G)G^\dagger\|_{\text{UI}}}{2}.$$

Examples:

1. [BD90], [DK90], [EI95] B is a real bidiagonal matrix with diagonal entries $a_1, a_2, \dots, a_n$ and off-diagonal (the one above the diagonal) entries are $b_1, b_2, \dots, b_{n-1}$. $\tilde{B}$ is the same as B , except for its diagonal entries, which change to $\alpha_1 a_1, \alpha_2 a_2, \dots, \alpha_n a_n$, and its off-diagonal entries, which change to $\beta_1 b_1, \beta_2 b_2, \dots, \beta_{n-1} b_{n-1}$. Then $\tilde{B} = D_L^* B D_R$ with

$$D_L = \text{diag} \left(\alpha_1, \frac{\alpha_1 \alpha_2}{\beta_1}, \frac{\alpha_1 \alpha_2 \alpha_3}{\beta_1 \beta_2}, \dots \right),$$

$$D_R = \text{diag} \left(1, \frac{\beta_1}{\alpha_1}, \frac{\beta_1 \beta_2}{\alpha_1 \alpha_2}, \dots \right).$$

Let $\alpha = \prod_{j=1}^n \max\{\alpha_j, 1/\alpha_j\}$ and $\beta = \prod_{j=1}^{n-1} \max\{\beta_j, 1/\beta_j\}$. Then

$$(\alpha\beta)^{-1} \leq (\|D_L^{-1}\|_2 \|D_R^{-1}\|_2)^{-1} \leq \|D_L\|_2 \|D_R\|_2 \leq \alpha\beta,$$

and Fact 1 and Fact 2 apply. Now if all $1 - \varepsilon \leq \alpha_i, \beta_j \leq 1 + \varepsilon$, then $(1 - \varepsilon)^{2n-1} \leq (\alpha\beta)^{-1} \leq (\alpha\beta) \leq (1 + \varepsilon)^{2n-1}$.

2. Consider block partitioned matrices

$$B = \begin{bmatrix} B_{11} & B_{12} \\ 0 & B_{22} \end{bmatrix},$$

$$\tilde{B} = \begin{bmatrix} B_{11} & 0 \\ 0 & B_{22} \end{bmatrix} = B \begin{bmatrix} I & -B_{11}^{-1} B_{12} \\ 0 & I \end{bmatrix} = B D_R.$$

By Fact 2, $\zeta(\sigma_j, \tilde{\sigma}_j) \leq \frac{1}{2} \|B_{11}^{-1} B_{12}\|_2$. Interesting cases are when $\|B_{11}^{-1} B_{12}\|_2$ is tiny enough to be treated as zero and so $\text{sv}(\tilde{B})$ approximates $\text{sv}(B)$ well. This situation occurs in computing the SVD of a bidiagonal matrix.

Author Note: Supported in part by the National Science Foundation under Grant No. DMS-0510664.

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16

Pseudospectra

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Eigenvalues often provide great insight into the behavior of matrices, precisely explaining, for example, the *asymptotic* character of functions of matrices like A^k and e^{tA} . Yet many important applications produce matrices whose behavior cannot be explained by eigenvalues alone. In such circumstances further information can be gleaned from broader sets in the complex plane, such as the numerical range (see Chapter 18), the polynomial numerical hull [Nev93], [Gre02], and the subject of this section, pseudospectra.

The ε -pseudospectrum is a subset of the complex plane that always includes the spectrum, but can potentially contain points far from any eigenvalue. Unlike the spectrum, pseudospectra vary with choice of norm and, thus, for a given application one must take care to work in a physically appropriate norm. Unless otherwise noted, throughout this chapter we assume that $A \in \mathbb{C}^{n \times n}$ is a square matrix with complex entries, and that $\|\cdot\|$ denotes a vector space norm and the matrix norm it induces. When speaking of a norm associated with an inner product, we presume that adjoints and normal and unitary matrices are defined with respect to that inner product. All computational examples given here use the 2-norm.

For further details about theoretical aspects of this subject and the application of pseudospectra to a variety of problems see [TE05]; for applications in control theory, see [HP05]; and for applications in perturbation theory see [CCF96].

16.1 Fundamentals of Pseudospectra

Definitions:

The ε -**pseudospectrum** of a matrix $A \in \mathbb{C}^{n \times n}$, $\varepsilon > 0$, is the set

$$\sigma_\varepsilon(A) = \{z \in \mathbb{C} : z \in \sigma(A + E) \text{ for some } E \in \mathbb{C}^{n \times n} \text{ with } \|E\| < \varepsilon\}.$$

(This definition is sometimes written with a weak inequality, $\|E\| \leq \varepsilon$; for matrices the difference has little significance, but the strict inequality proves to be convenient for infinite-dimensional operators.)

If $\|A\mathbf{v} - z\mathbf{v}\| < \varepsilon\|\mathbf{v}\|$ for some $\mathbf{v} \neq 0$, then z is an ε -**pseudoeigenvalue** of A with corresponding ε -**pseudoeigenvector** $\mathbf{v}$.

The **resolvent** of the matrix $A \in \mathbb{C}^{n \times n}$ at a point $z \notin \sigma(A)$ is the matrix $(zI - A)^{-1}$.

Facts: [TE05]

1. Equivalent definitions. The set $\sigma_\varepsilon(A)$ can be equivalently defined as:

- (a) The subset of the complex plane bounded within the $1/\varepsilon$ level set of the norm of the resolvent:

$$\sigma_\varepsilon(A) = \{z \in \mathbb{C} : \|(zI - A)^{-1}\| > \varepsilon^{-1}\}, \quad (16.1)$$

with the convention that $\|(zI - A)^{-1}\| = \infty$ when $zI - A$ is not invertible, i.e., when $z \in \sigma(A)$.

- (b) The set of all ε -pseudoeigenvalues of A :

$$\sigma_\varepsilon(A) = \{z \in \mathbb{C} : \|A\mathbf{v} - z\mathbf{v}\| < \varepsilon \text{ for some unit vector } \mathbf{v} \in \mathbb{C}^n\}.$$

2. For finite $\varepsilon > 0$, $\sigma_\varepsilon(A)$ is a bounded open set in $\mathbb{C}$ containing no more than n connected components, and $\sigma(A) \subset \sigma_\varepsilon(A)$. Each connected component must contain at least one eigenvalue of A .
3. Pseudospectral mapping theorems.

- (a) For any $\alpha, \gamma \in \mathbb{C}$ with $\gamma \neq 0$, $\sigma_\varepsilon(\alpha I + \gamma A) = \alpha + \sigma_{\varepsilon/|\gamma|}(A)$.

- (b) [Lui03] Suppose f is a function analytic on $\sigma_\varepsilon(A)$ for some $\varepsilon > 0$, and define $\gamma(\varepsilon) = \sup_{\|E\| \leq \varepsilon} \|f(A+E) - f(A)\|$. Then $f(\sigma_\varepsilon(A)) \subseteq \sigma_{\gamma(\varepsilon)}(f(A))$. See [Lui03] for several more inclusions of this type.

4. Stability of pseudospectra. For any $\varepsilon > 0$ and E such that $\|E\| < \varepsilon$,

$$\sigma_{\varepsilon - \|E\|}(A) \subseteq \sigma_\varepsilon(A + E) \subseteq \sigma_{\varepsilon + \|E\|}(A).$$

5. Properties of pseudospectra as $\varepsilon \rightarrow 0$.

- (a) If λ is a eigenvalue of A with index k , then there exist constants d and C such that $\|(zI - A)^{-1}\| \leq C|z - \lambda|^{-k}$ for all z such that $|z - \lambda| < d$.
- (b) Any two matrices with the same ε -pseudospectra for all $\varepsilon > 0$ have the same minimal polynomial.

6. Suppose $\|\cdot\|$ is the natural norm in an inner product space.

- (a) The matrix A is normal (see Section 7.2) if and only if $\sigma_\varepsilon(A)$ equals the union of open ε -balls about each eigenvalue for all $\varepsilon > 0$.
- (b) For any $A \in \mathbb{C}^{n \times n}$, $\sigma_\varepsilon(A^*) = \overline{\sigma_\varepsilon(A)}$.

7. [BLO03] Suppose $\|\cdot\|$ is the natural norm in an inner product space. The point $z = x + iy$, $x, y \in \mathbb{R}$, is on the boundary of $\sigma_\varepsilon(A)$ provided iy is an eigenvalue of the Hamiltonian matrix

$$\begin{bmatrix} xI - A^* & \varepsilon I \\ -\varepsilon I & A - xI \end{bmatrix}.$$

This fact implies that the boundary of $\sigma_\varepsilon(A)$ cannot contain a segment of any vertical line or, substituting $e^{i\theta}A$ for A , a segment of any straight line.

8. The following results provide lower and upper bounds on the ε -pseudospectrum; Δ_δ denotes the open unit ball of radius δ in $\mathbb{C}$, and $\kappa(X) = \|X\|\|X^{-1}\|$.

- (a) For all $\varepsilon > 0$, $\sigma(A) + \Delta_\varepsilon \subseteq \sigma_\varepsilon(A)$.

- (b) For any nonsingular $S \in \mathbb{C}^{n \times n}$ and all $\varepsilon > 0$,

$$\sigma_{\varepsilon/\kappa(S)}(SAS^{-1}) \subseteq \sigma_\varepsilon(A) \subseteq \sigma_{\varepsilon\kappa(S)}(SAS^{-1}).$$

- (c) (Bauer–Fike Theorems [BF60], [Dem97]) Let $\|\cdot\|$ denote a monotone norm. If A is diagonalizable, $A = V\Lambda V^{-1}$, then for all $\varepsilon > 0$,

$$\sigma_\varepsilon(A) \subseteq \sigma(A) + \Delta_{\varepsilon\kappa(V)}.$$

If $A \in \mathbb{C}^{n \times n}$ has n distinct eigenvalues $\lambda_1, \dots, \lambda_n$, then for all $\varepsilon > 0$,

$$\sigma_\varepsilon(A) \subseteq \bigcup_{j=1}^N (\lambda_j + \Delta_{\varepsilon n \kappa(\lambda_j)}),$$

where $\kappa(\lambda_j)$ here denotes the eigenvalue condition number of λ_j (i.e., $\kappa(\lambda_j) = 1/|\widehat{\mathbf{v}}_j^* \mathbf{v}_j|$, where $\widehat{\mathbf{v}}_j$ and $\mathbf{v}_j$ are unit-length left and right eigenvectors of A corresponding to the eigenvalue λ_j).

- (d) If $\|\cdot\|$ is the natural norm in an inner product space, then for any $\varepsilon > 0$, $\sigma_\varepsilon(A) \subseteq W(A) + \Delta_\varepsilon$, where $W(\cdot)$ denotes the numerical range (Chapter 18).
- (e) If $\|\cdot\|$ is the natural norm in an inner product space and U is a unitary matrix, then $\sigma_\varepsilon(U^*AU) = \sigma_\varepsilon(A)$ for all $\varepsilon > 0$.
- (f) If $\|\cdot\|$ is unitarily invariant, then $\sigma_\varepsilon(A) \subseteq \sigma(A) + \Delta_{\varepsilon + \text{dep}(A)}$, where $\text{dep}(\cdot)$ denotes Henrici's departure from normality (i.e., the norm of the off-diagonal part of the triangular factor in a Schur decomposition, minimized over all such decompositions).
- (g) (Geršgorin Theorem for pseudospectra [ET01]) Using the induced matrix 2-norm, for any $\varepsilon > 0$,

$$\sigma_\varepsilon(A) \subseteq \bigcup_{j=1}^n (a_{jj} + \Delta_{r_j + \varepsilon \sqrt{n}}),$$

where $r_j = \sum_{k=1, k \neq j}^n |a_{jk}|$.

9. The following results bound $\sigma_\varepsilon(A)$ by pseudospectra of smaller matrices. Here $\|\cdot\|$ is the natural norm in an inner product space.

- (a) [GL02] If A has the block-triangular form

$$A = \begin{bmatrix} B & C \\ D & E \end{bmatrix},$$

then

$$\sigma_\varepsilon(A) \subseteq \sigma_\delta(B) \cup \sigma_\delta(E),$$

where $\delta = (\varepsilon + \|D\|)\sqrt{1 + \|C\|/(\varepsilon + \|D\|)}$. Provided $\varepsilon > \|D\|$,

$$\sigma_\gamma(B) \cup \sigma_\gamma(E) \subseteq \sigma_\varepsilon(A),$$

where $\gamma = \varepsilon - \|D\|$.

- (b) If the columns of $V \in \mathbb{C}^{n \times m}$ form a basis for an invariant subspace of A , and $\widehat{V} \in \mathbb{C}^{n \times m}$ is such that $\widehat{V}^*V = I$, then $\sigma_\varepsilon(\widehat{V}^*AV) \subseteq \sigma_\varepsilon(A)$. In particular, if the columns of U form an orthonormal basis for an invariant subspace of A , then $\sigma_\varepsilon(U^*AU) \subseteq \sigma_\varepsilon(A)$.
- (c) [ET01] If $U \in \mathbb{C}^{n \times m}$ has orthonormal columns and $AU = UH + R$, then $\sigma(H) \subseteq \sigma_\varepsilon(A)$ for $\varepsilon = \|R\|$.
- (d) (Arnoldi factorization) If $AU = [U \mathbf{u}]H$, where $H \in \mathbb{C}^{(m+1) \times m}$ is an upper Hessenberg matrix ($h_{jk} = 0$ if $j > k + 1$) and the columns of $[U \mathbf{u}] \in \mathbb{C}^{n \times (m+1)}$ are orthonormal, then $\sigma_\varepsilon(H) \subseteq \sigma_\varepsilon(A)$. (The ε -pseudospectrum of a rectangular matrix is defined in section 16.5 below.)

Examples:

The plots of pseudospectra that follow show the boundary of $\sigma_\varepsilon(A)$ for various values of ε , with the smallest values of ε corresponding to those boundaries closest to the eigenvalues. In all cases, $\|\cdot\|$ is the 2-norm.

1. The following three matrices all have the same eigenvalues, $\sigma(A) = \{1, \pm i\}$, yet their pseudospectra, shown in Figure 16.1, differ considerably:

$$\begin{bmatrix} 0 & -1 & 10 \\ 1 & 0 & 5 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & -1 & 10 \\ 1 & 0 & 5i \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 2 & -5 & 10 \\ 1 & -2 & 5i \\ 0 & 0 & 1 \end{bmatrix}.$$

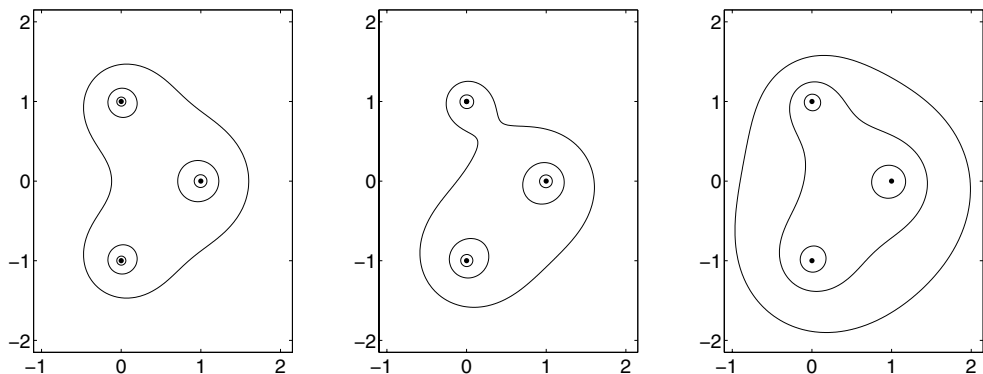


FIGURE 16.1 Spectra (solid dots) and ε -pseudospectra of the three matrices of Example 1, each with $\sigma(A) = \{1, i, -i\}$; $\varepsilon = 10^{-1}, 10^{-1.5}, 10^{-2}$.

2. [RT92] For any matrix that is zero everywhere except the first superdiagonal, $\sigma_\varepsilon(A)$ consists of an open disk centered at zero whose radius depends on ε for all $\varepsilon > 0$. Figure 16.2 shows pseudospectra for two such examples of dimension $n = 50$:

$$\begin{bmatrix} 0 & 1 & & & \\ & 0 & 1 & & \\ & & \ddots & \ddots & \\ & & & 0 & 1 \\ & & & & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 3 & & & \\ & 0 & 3/2 & & \\ & & \ddots & \ddots & \\ & & & 0 & 3/(n-1) \\ & & & & 0 \end{bmatrix}.$$

Though these matrices have the same minimal polynomial, the pseudospectra differ considerably.

3. It is evident from Figure 16.1 that the components of $\sigma_\varepsilon(A)$ need not be convex. In fact, they need not be simply connected; that is, $\sigma_\varepsilon(A)$ can have “holes.” This is illustrated in Figure 16.3 for the following examples, a circulant (hence, normal) matrix and a defective matrix constructed

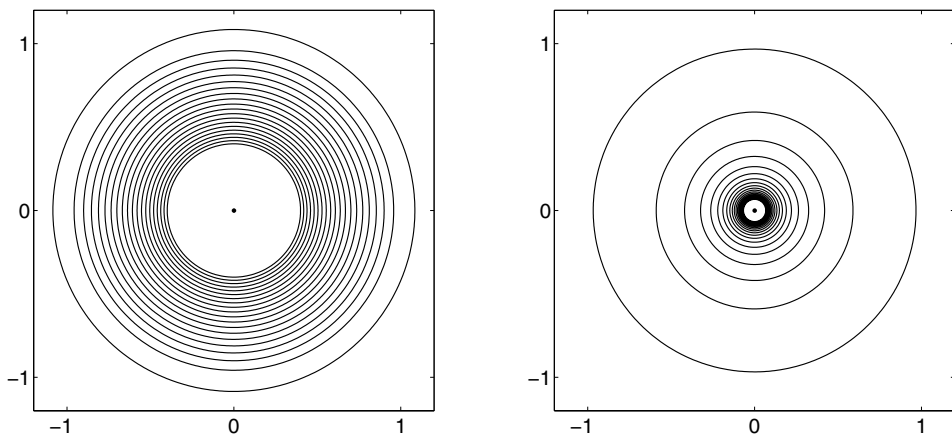


FIGURE 16.2 Spectra (solid dots) and ε -pseudospectra of the matrices in Example 2 for $\varepsilon = 10^{-1}, 10^{-2}, \dots, 10^{-20}$.

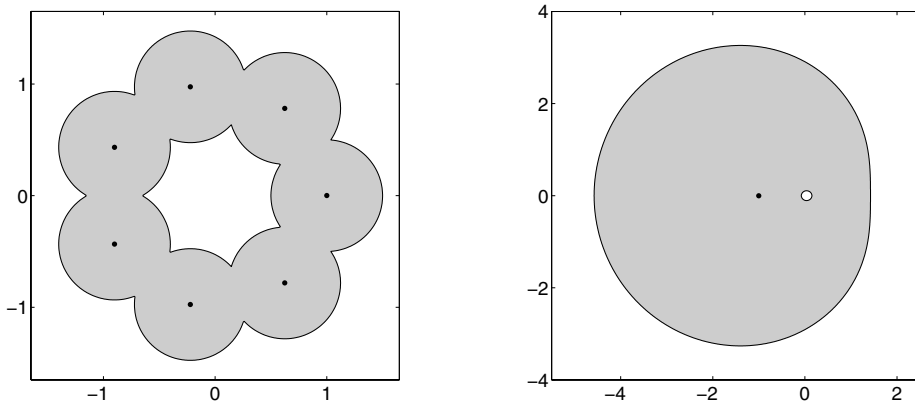


FIGURE 16.3 Spectra (solid dots) and ε -pseudospectra (gray regions) of the matrices in Example 3 for $\varepsilon = .5$ (left) and $\varepsilon = 10^{-3}$ (right). Both plotted pseudospectra are doubly connected.

by Demmel [Dem87]:

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} -1 & -100 & -10000 \\ 0 & -1 & -100 \\ 0 & 0 & -1 \end{bmatrix}.$$

16.2 Toeplitz Matrices

Given the rich variety of important applications in which Toeplitz matrices arise, we are fortunate that so much is now understood about their spectral properties. Nonnormal Toeplitz matrices are prominent examples of matrices whose eigenvalues provide only limited insight into system behavior. The spectra of infinite-dimensional Toeplitz matrices are easily characterized, and one would hope to use these results to approximate the spectra of more recalcitrant large, finite-dimensional examples. For generic problems, the spectra of finite-dimensional Toeplitz matrices do not converge to the spectrum of the corresponding infinite-dimensional Toeplitz operator. However, the ε -pseudospectra do converge in the $n \rightarrow \infty$ limit for all $\varepsilon > 0$, and, moreover, for banded Toeplitz matrices this convergence is especially striking as the resolvent grows exponentially with n in certain regions. Comprehensive references addressing the pseudospectra of Toeplitz matrices include the books [BS99] and [BG05]. For a generalization of these results to “twisted Toeplitz matrices,” where the entries on each diagonal are samples of a smoothly varying function, see [TC04].

Definitions:

A **Toeplitz operator** is a singly infinite matrix with constant entries on each diagonal:

$$T = \begin{bmatrix} a_0 & a_{-1} & a_{-2} & a_{-3} & \cdots \\ a_1 & a_0 & a_{-1} & a_{-2} & \ddots \\ a_2 & a_1 & a_0 & a_{-1} & \ddots \\ a_3 & a_2 & a_1 & a_0 & \ddots \\ \vdots & \ddots & \ddots & \ddots & \ddots \end{bmatrix}$$

for $a_0, a_{\pm 1}, a_{\pm 2}, \dots \in \mathbb{C}$.

Provided it is well defined for all z on the unit circle $\mathbb{T}$ in the complex plane, the function $a(z) = \sum_{k=-\infty}^{\infty} a_k z^k$ is called the **symbol** of T .

The set $a(\mathbb{T}) \subset \mathbb{C}$ is called the **symbol curve**.

Given a symbol a , the corresponding n -dimensional **Toeplitz matrix** takes the form

$$T_n = \begin{bmatrix} a_0 & a_{-1} & a_{-2} & \cdots & a_{1-n} \\ a_1 & a_0 & a_{-1} & \ddots & \vdots \\ a_2 & a_1 & a_0 & \ddots & a_{-2} \\ \vdots & \ddots & \ddots & \ddots & a_{-1} \\ a_{n-1} & \cdots & a_2 & a_1 & a_0 \end{bmatrix} \in \mathbb{C}^{n \times n}.$$

For a symbol $a(z) = \sum_{k=-\infty}^{\infty} a_k z^k$, the set $\{T_n\}_{n \geq 1}$ is called a **family of Toeplitz matrices**.

A family of Toeplitz matrices with symbol a is **banded** if there exists some $m \geq 0$ such that $a_{\pm k} = 0$ for all $k \geq m$.

Facts:

1. [Böt94] (Convergence of pseudospectra) Let $\|\cdot\|$ denote any p norm. If the symbol a is a continuous function on $\mathbb{T}$, then

$$\lim_{n \rightarrow \infty} \sigma_\varepsilon(T_n) \rightarrow \sigma_\varepsilon(T)$$

as $n \rightarrow \infty$, where T is the infinite-dimensional Toeplitz operator with symbol a acting on the space ℓ_p , and its ε -pseudospectrum is a natural generalization of the first definition in section 16.1. The convergence of sets is understood in the Hausdorff sense [Hau91, p. 167], i.e., the distance between bounded sets $\Sigma_1, \Sigma_2 \subseteq \mathbb{C}$ is given by

$$d(\Sigma_1, \Sigma_2) = \max \left\{ \sup_{s_1 \in \Sigma_1} \inf_{s_2 \in \Sigma_2} |s_1 - s_2|, \sup_{s_2 \in \Sigma_2} \inf_{s_1 \in \Sigma_1} |s_2 - s_1| \right\}.$$

2. [BS99] Provided the symbol a is a continuous function on $\mathbb{T}$, the spectrum $\sigma(T)$ of the infinite dimensional Toeplitz operator T on ℓ_p comprises $a(\mathbb{T})$ together with all points $z \in \mathbb{C} \setminus a(\mathbb{T})$ that $a(\mathbb{T})$ encloses with winding number

$$\frac{1}{2\pi i} \int_{a(\mathbb{T})} \frac{1}{\zeta - z} d\zeta$$

nonzero. From the previous fact, we deduce that $\|(zI - T_n)^{-1}\| \rightarrow \infty$ as $n \rightarrow \infty$ if $z \in \sigma(T)$ and that, for any fixed $\varepsilon > 0$, there exists some $N \geq 1$ such that $\sigma(T) \subseteq \sigma_\varepsilon(T_n)$ for all $n \geq N$.

3. [RT92] (Exponential growth of the resolvent) If the family of Toeplitz matrices T_n is banded, then for any fixed $z \in \mathbb{C}$ such that the winding number of $a(\mathbb{T})$ with respect to z is nonzero, there exists some $\gamma > 1$ and $N \geq 1$ such that $\|(zI - T_n)^{-1}\| \geq \gamma^n$ for all $n \geq N$.

Examples:

1. Consider the family of Toeplitz matrices with symbol

$$a(t) = t - \frac{1}{2} - \frac{1}{16}t^{-1}.$$

For any dimension n , the spectrum $\sigma(T_n(a))$ is contained in the line segment $[-\frac{1}{2} - \frac{1}{2}i, -\frac{1}{2} + \frac{1}{2}i]$ in the complex plane. This symbol was selected so that $\sigma(A)$ falls in both the left half-plane and the unit disk, while even for relatively small values of ε , $\sigma_\varepsilon(A)$ contains both points in the right half-plane and points outside the unit disk for all but the smallest values of n ; see Figure 16.4 for $n = 50$.

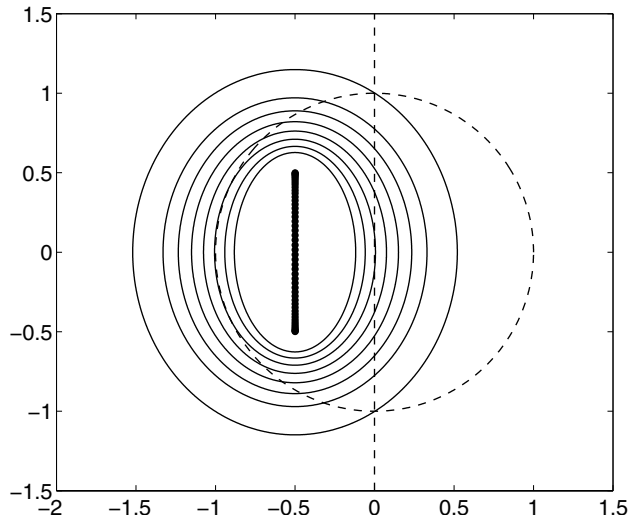


FIGURE 16.4 Spectrum (solid dots, so close they appear to be a thick line segment with real part $-1/2$) and ε -pseudospectra of the Toeplitz matrix T_{50} from Example 1; $\varepsilon = 10^{-1}, 10^{-3}, \dots, 10^{-15}$. The dashed lines show the unit circle and the imaginary axis.

2. Pseudospectra of matrices with the symbols

$$a(t) = it^4 + t^2 + 2t + 5t^{-2} + it^{-5}$$

and

$$a(t) = 3it^4 + t + t^{-1} + 3it^{-4}$$

are shown in Figure 16.5.

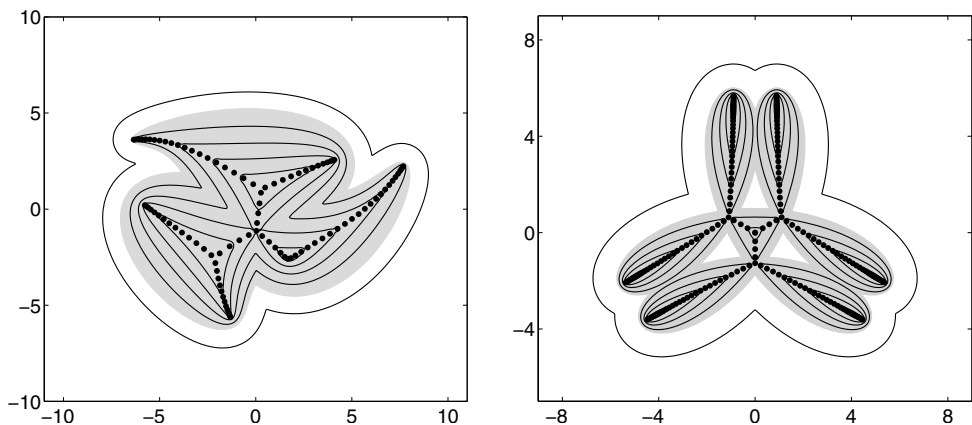


FIGURE 16.5 Spectra (solid dots) and ε -pseudospectra of Toeplitz matrices from Example 2 with the first symbol on the left ($n = 100$) and the second symbol on the right ($n = 200$), both with $\varepsilon = 10^0, 10^{-2}, \dots, 10^{-8}$. In each plot, the gray region is the spectrum of the underlying infinite dimensional Toeplitz operator.

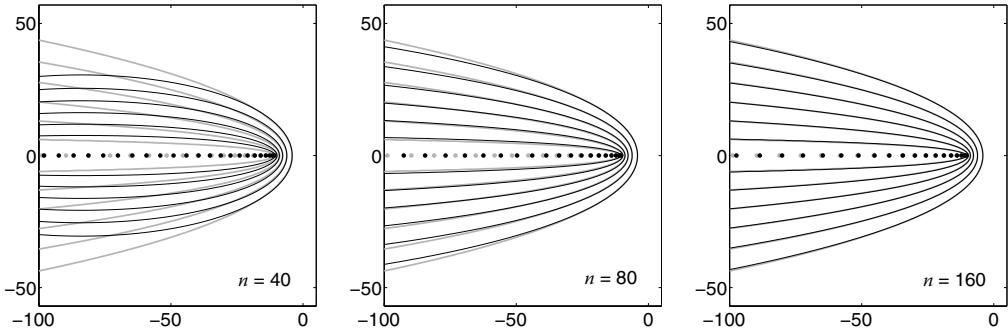


FIGURE 16.6 Spectra (solid dots) and ε -pseudospectra of Toeplitz matrices for the discretization of a convection–diffusion operator described in Application 1 with $\nu = 1/50$ and three values of n ; $\varepsilon = 10^{-1}, 10^{-2}, \dots, 10^{-6}$. The gray dots and lines in each plot show eigenvalues and pseudospectra of the differential operator to which the matrix spectra and pseudospectra converge.

Applications:

1. [RT94] Discretization of the one-dimensional convection-diffusion equation

$$\nu u''(x) + u'(x) = f(x), \quad u(0) = u(1) = 0$$

for $x \in [0, 1]$ with second-order centered finite differences on a uniform grid with spacing $h = 1/(n+1)$ between grid points results in an $n \times n$ Toeplitz matrix with symbol

$$a(t) = \left(\frac{\nu}{h^2} + \frac{1}{2h} \right) t - \left(\frac{2\nu}{h^2} \right) + \left(\frac{\nu}{h^2} - \frac{1}{2h} \right) t^{-1}.$$

On the right-most part of the spectrum, both the eigenvalues and pseudospectra of the discretization matrix converge to those of the underlying differential operator

$$\mathcal{L}u = \nu u'' + u'$$

whose domain is the space of functions that are square-integrable over $[0, 1]$ and satisfy the boundary conditions $u(0) = u(1) = 0$; see Figure 16.6.

16.3 Behavior of Functions of Matrices

In practice, pseudospectra are most often used to investigate the behavior of a function of a matrix. Does the solution $\mathbf{x}(t) = e^{tA}\mathbf{x}(0)$ or $\mathbf{x}_k = A^k\mathbf{x}_0$ of the linear dynamical system $\mathbf{x}'(t) = A\mathbf{x}(t)$ or $\mathbf{x}_{k+1} = A\mathbf{x}_k$ grow or decay as $t, k \rightarrow \infty$? Eigenvalues provide an answer: If $\sigma(A)$ lies in the open unit disk or left half-plane, the solution must eventually decay. However, the results described in this section show that if ε -pseudoeigenvalues of A extend well beyond the unit disk or left half-plane for small values of ε , then the system must exhibit transient growth for some initial states. While such growth is notable even for purely linear problems, it should spark special caution when observed for a dynamical system that arises from the linearization of a nonlinear system about a steady state based on the assumption that disturbances from that state are small in magnitude. This reasoning has been applied extensively in recent years in fluid dynamics; see, e.g., [TTRD93].

Definitions:

The ε -**pseudospectral abscissa** of A measures the rightmost extent of $\sigma_\varepsilon(A)$: $\alpha_\varepsilon(A) = \sup_{z \in \sigma_\varepsilon(A)} \operatorname{Re} z$.

The ε -**pseudospectral radius** of A measures the maximum magnitude in $\sigma_\varepsilon(A)$: $\rho_\varepsilon(A) = \sup_{z \in \sigma_\varepsilon(A)} |z|$.

Facts: [TE05, §§14–19]

1. For $\varepsilon > 0$, $\sup_{t \in \mathbb{R}, t \geq 0} \|e^{tA}\| \geq \alpha_\varepsilon(A)/\varepsilon$.
2. For $\varepsilon > 0$, $\sup_{k \in \mathbb{N}, k \geq 0} \|A^k\| \geq (\rho_\varepsilon(A) - 1)/\varepsilon$.
3. For any function f that is analytic on the spectrum of A ,

$$\|f(A)\| \geq \max_{\lambda \in \sigma(A)} |f(\lambda)|.$$

Equality holds when $\|\cdot\|$ is the natural norm in an inner product space in which A is normal.

4. In the special case of matrix exponentials e^{tA} and matrix powers A^k , this last fact implies that

$$\|e^{tA}\| \geq e^{t\alpha(A)}, \quad \|A^k\| \geq \rho(A)^k$$

for all $t \geq 0$ and integers $k \geq 0$, where $\alpha(A) = \max_{\lambda \in \sigma(A)} \operatorname{Re} \lambda$ is the spectral abscissa and $\rho(A) = \max_{\lambda \in \sigma(A)} |\lambda|$ is the spectral radius.

5. Let Γ_ε be a finite union of Jordan curves containing $\sigma_\varepsilon(A)$ in their collective interior for some $\varepsilon > 0$, and suppose f is a function analytic on Γ_ε and its interior. Then

$$\|f(A)\| \leq \frac{L_\varepsilon}{2\pi\varepsilon} \max_{z \in \Gamma_\varepsilon} |f(z)|,$$

where L_ε denotes the arc-length of Γ_ε .

6. In the special case of matrix exponentials e^{tA} and matrix powers A^k , this last fact implies that for all $t \geq 0$ and integers $k \geq 0$,

$$\|e^{tA}\| \leq \frac{L_\varepsilon}{2\pi\varepsilon} e^{t\alpha_\varepsilon(A)}, \quad \|A^k\| \leq \rho_\varepsilon(A)^{k+1}/\varepsilon.$$

In typical cases, larger values of ε give superior bounds for small t and k , while smaller values of ε yield more descriptive bounds for larger t and k ; see Figure 16.7.

7. Suppose $z \in \mathbb{C} \setminus \sigma(A)$ with $a \equiv \operatorname{Re} z$ and $\varepsilon \equiv 1/\|(zI - A)^{-1}\|$. Provided $a > \varepsilon$, then for any fixed $\tau > 0$,

$$\sup_{0 < t \leq \tau} \|e^{tA}\| \geq \frac{ae^{\tau a}}{a + \varepsilon(e^{\tau a} - 1)}.$$

8. Suppose $z \in \mathbb{C} \setminus \sigma(A)$ with $a \equiv \operatorname{Re} z$ and $\varepsilon \equiv 1/\|(zI - A)^{-1}\|$, and that $\|e^{\tau A}\| \leq M$ for all $\tau > 0$ with $M \geq a/\varepsilon$. Then for any $t > 0$,

$$\|e^{tA}\| \geq e^{ta}(1 - \varepsilon M/a) + \varepsilon M/a.$$

9. Suppose $z \in \mathbb{C} \setminus \sigma(A)$ with $r \equiv |z|$ and $\varepsilon \equiv 1/\|(zI - A)^{-1}\|$. Provided $r > 1 + \varepsilon$, then for any fixed integer $\kappa \geq 1$,

$$\sup_{0 \leq k \leq \kappa} \|A^k\| \geq \frac{r^\kappa(r - 1 - \varepsilon) + \varepsilon r^{\kappa-1}}{(r - 1 - \varepsilon) + \varepsilon r^{\kappa-1}}.$$

10. Suppose $z \in \mathbb{C} \setminus \sigma(A)$ with $r \equiv |z|$ and $\varepsilon \equiv 1/\|(zI - A)^{-1}\|$, and that $\|A^\kappa\| \leq M$ for all integers $\kappa \geq 0$ with $M \geq (r - 1)/\varepsilon$. Then for any integer $k \geq 0$,

$$\|A^k\| \geq r^k(r - 1 - \varepsilon M) + \varepsilon M.$$

11. For any $A \in \mathbb{C}^{n \times n}$,

$$\|A^k\| < e(k+1) \sup_{\varepsilon > 0} \frac{\rho_\varepsilon(A) - 1}{\varepsilon}.$$

12. (Kreiss Matrix Theorem) For any $A \in \mathbb{C}^{n \times n}$,

$$\sup_{\varepsilon > 0} \frac{\rho_\varepsilon(A) - 1}{\varepsilon} \leq \sup_{k \geq 0} \|A^k\| \leq e n \sup_{\varepsilon > 0} \frac{\rho_\varepsilon(A) - 1}{\varepsilon}.$$

13. [GT93] There exist matrices A and B such that, in the induced 2-norm, $\sigma_\varepsilon(A) = \sigma_\varepsilon(B)$ for all $\varepsilon > 0$, yet $\|f(A)\|_2 \neq \|f(B)\|_2$ for some polynomial f ; see Example 2. That is, even if the 2-norm of the resolvents of A and B are identical for all $z \in \mathbb{C}$, the norms of other matrix functions in A and B need not agree. (Curiously, if the *Frobenius* norm of the resolvents of A and B agree for all $z \in \mathbb{C}$, then $\|f(A)\|_F = \|f(B)\|_F$ for all polynomials f .)

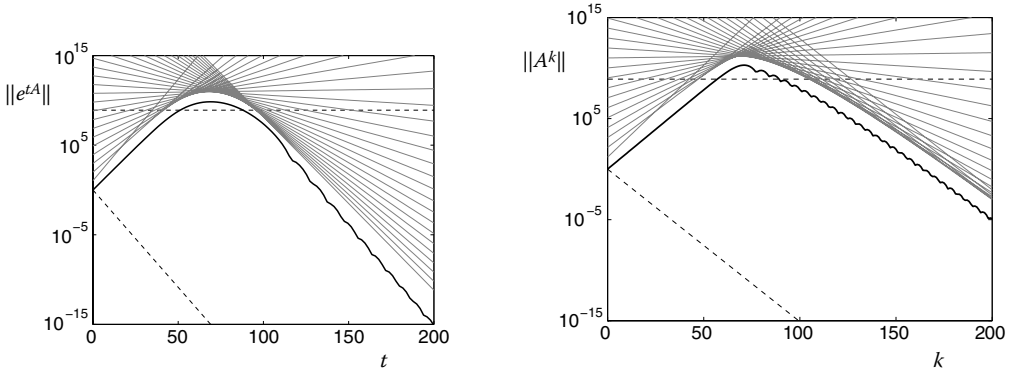


FIGURE 16.7 The functions $\|e^{tA}\|$ and $\|A^k\|$ exhibit transient growth before exponential decay for the Toeplitz matrix of dimension $n = 50$, whose pseudospectra were illustrated in Figure 16.4. The horizontal dashed lines show the lower bounds on maximal growth given in Facts 1 and 2, while the lower dashed lines show the lower bounds of Fact 4. The gray lines show the upper bounds in Fact 6 for $\varepsilon = 10^{-1}, 10^{-2}, \dots, 10^{-28}$ (ordered by decreasing slope).

Examples:

- Consider the tridiagonal Toeplitz matrix of dimension $n = 50$ from Example 1 of the last section, whose pseudospectra were illustrated in Figure 16.4. Since all the eigenvalues of this matrix are contained in both the left half-plane and the unit disk, $e^{tA} \rightarrow 0$ as $t \rightarrow \infty$ and $A^k \rightarrow 0$ as $k \rightarrow \infty$. However, $\sigma_\varepsilon(A)$ extends far into the right half-plane and beyond the unit disk even for ε as small as 10^{-7} . Consequently, the lower bounds in Facts 1 and 2 guarantee that $\|e^{tA}\|$ and $\|A^k\|$ exhibit transient growth before their eventual decay; results such as Fact 6 limit the extent of the transient growth. These bounds are illustrated in Figure 16.7. (For a similar example involving a different matrix, see the “Transient” demonstration in [Wri02b].)
- [GT93] The matrices

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

have the same 2-norm ε -pseudospectra for all $\varepsilon > 0$. However, $\|A\|_2 = \sqrt{2} > 1 = \|B\|_2$.

Applications:

1. Fact 5 leads to a convergence result for the GMRES algorithm (Chapter 41), which constructs estimates $\mathbf{x}_k$ to the solution $\mathbf{x}$ of the linear system $A\mathbf{x} = \mathbf{b}$. The k th residual $\mathbf{r}_k = \mathbf{b} - A\mathbf{x}_k$ is bounded by

$$\frac{\|\mathbf{r}_k\|_2}{\|\mathbf{r}_0\|_2} \leq \frac{L_\varepsilon}{2\pi\varepsilon} \min_{\substack{p \in \mathbb{C}[z;k] \\ p(0)=1}} \max_{z \in \Gamma_\varepsilon} |p(z)|,$$

where $\mathbb{C}[z;k]$ denotes the set of polynomials of degree k or less, Γ_ε is a finite union of Jordan curves containing $\sigma_\varepsilon(A)$ in their collective interior for some $\varepsilon > 0$, and L_ε is the arc-length of Γ_ε .

2. For further examples of the use of pseudospectra to analyze matrix iterations and the stability of discretizations of differential equations, see [TE05, §§24–34].

16.4 Computation of Pseudospectra

This section describes techniques for computing and approximating pseudospectra, focusing primarily on the induced matrix 2-norm, the case most studied in the literature and for which very satisfactory algorithms exist. For further details, see [Tre99], [TE05, §§39–44], or [Wri02a].

Facts: [TE05]

1. There are two general approaches to computing pseudospectra, both based on the expression for $\sigma_\varepsilon(A)$ in Fact 1(a) of Section 16.1. The most widely-used method computes the resolvent norm, $\|(zI - A)^{-1}\|_2$, on a grid of points in the complex plane and submits the results to a contour-plotting program; the second approach uses a curve-tracing algorithm to track the ε^{-1} -level curve of the resolvent norm ([Brü96]). Both approaches exploit the fact that the 2-norm of the resolvent is the reciprocal of the minimum singular value of $zI - A$. A third approach, based on the characterization of $\sigma_\varepsilon(A)$ as the set of all ε -pseudoeigenvalues, *approximates* $\sigma_\varepsilon(A)$ by the union of the eigenvalues of $A + E$ for randomly generated $E \in \mathbb{C}^{n \times n}$ with $\|E\| < \varepsilon$.
2. For dense matrices A , the computation of the minimum singular value of $zI - A$ requires $\mathcal{O}(n^3)$ floating point operations for each distinct value of z . Hence, the contour-plotting approach to computing pseudospectra based on a grid of $m \times m$ points in the complex plane, implemented via the most naive method, requires $\mathcal{O}(m^2 n^3)$ operations.
3. [Lui97] Improved efficiency is obtained through the use of iterative methods for computing the minimum singular value of the resolvent. The most effective methods (inverse iteration or the inverse Lanczos method) require matrix-vector products of the form $(zI - A)^{-1}\mathbf{x}$ at each iteration. For dense A , this approach requires $\mathcal{O}(n^3)$ operations per grid point. One can decrease this labor to $\mathcal{O}(n^2)$ by first reducing A to Schur form, $A = UTU^*$, and then noting that $\|(zI - A)^{-1}\|_2 = \|(zI - T)^{-1}\|_2$. Vectors of the form $(zI - T)^{-1}\mathbf{x}$ can be computed in $\mathcal{O}(n^2)$ operations since T is triangular. As the inverse iteration and inverse Lanczos methods typically converge to the minimum singular value in a small number of iterations at each grid point, the total complexity of the contour-plotting approach is $\mathcal{O}(n^3 + m^2 n^2)$.
4. For large-scale problems (say, $n > 1000$), the cost of preliminary triangularization can be prohibitive. Several alternatives are available: Use sparse direct or iterative methods to compute $(zI - A)^{-1}\mathbf{x}$ at each grid point, or reduce the dimension of the problem by replacing A with a smaller matrix, such as the $(m+1) \times m$ upper Hessenberg matrix in an Arnoldi decomposition, or U^*AU , where the columns of $U \in \mathbb{C}^{n \times m}$ form an orthonormal basis for an invariant subspace corresponding to physically relevant eigenvalues, with $m \ll n$. As per results stated in Fact 9 of Section 16.1, the pseudospectra of these smaller matrices provide a lower bounds on the pseudospectra of A .

5. [Wri02b] EigTool is a freely available MATLAB package based on a highly-efficient, robust implementation of the grid-based method with preliminary triangularization and inverse Lanczos iteration. For large-scale problems, EigTool uses ARPACK (Chapter 76), to compute a subspace that includes an invariant subspace associated with eigenvalues in a given region of the complex plane. The EigTool software, which was used to compute the pseudospectra shown throughout this section, can be downloaded from <http://www.comlab.ox.ac.uk/pseudospectra/eigtool>.
6. Curve-tracing algorithms can also benefit from iterative computation of the resolvent norm, though the standard implementation requires both left and right singular vectors associated with the minimal singular value ([Brü96]). Robust implementations require measures to ensure that all components of $\sigma_\varepsilon(A)$ have been located and to handle cusps in the boundary; see, e.g., [BG01].
7. Software for computing 2-norm pseudospectra can be used to compute pseudospectra in any norm induced by an inner product. Suppose the inner product of $\mathbf{x}$ and $\mathbf{y}$ is given by $(\mathbf{x}, \mathbf{y})_W = (W\mathbf{x}, \mathbf{y})$, where $(\cdot, \cdot)$ denotes the Euclidean inner product and $W = LL^*$, where L^* denotes the conjugate transpose of L . Then the W -norm pseudospectra of A are equal to the 2-norm pseudospectra of L^*AL^{-*} .
8. For norms not associated with inner products, all known grid-based algorithms require $\mathcal{O}(n^3)$ operations per grid point, typically involving the construction of the resolvent $(zI - A)^{-1}$. Higham and Tisseur ([HT00]) have proposed an efficient approach for approximating 1-norm pseudospectra using a norm estimator.
9. [BLO03],[MO05] There exist efficient algorithms, based on Fact 7 of section 16.1, for computing the 2-norm pseudospectral radius and abscissa without first determining the entire pseudospectrum.

16.5 Extensions

The previous sections address the standard formulation of the ε -pseudospectrum, the union of all eigenvalues of $A + E$ for a square matrix A and general complex perturbations E , with $\|E\| < \varepsilon$. Natural modifications restrict the structure of E or adapt the definition to more general eigenvalue problems. The former topic has attracted considerable attention in the control theory literature and is presented in detail in [HP05].

Definitions:

The **spectral value set**, or **structured ε -pseudospectrum**, of the matrix triplet (A, B, C) , $A \in \mathbb{C}^{n \times n}$, $B \in \mathbb{C}^{n \times m}$, $C \in \mathbb{C}^{p \times n}$, for $\varepsilon > 0$ is the set

$$\sigma_\varepsilon(A; B, C) = \{z \in \mathbb{C} : z \in \sigma(A + BEC) \text{ for some } E \in \mathbb{C}^{m \times p} \text{ with } \|E\| < \varepsilon\}.$$

The **real structured ε -pseudospectrum** of $A \in \mathbb{R}^{n \times n}$ is the set

$$\sigma_\varepsilon^R(A) = \{z \in \sigma(A + E) : E \in \mathbb{R}^{n \times n}, \|E\| < \varepsilon\}.$$

The **ε -pseudospectrum of a rectangular matrix** $A \in \mathbb{C}^{n \times m}$ ($n \geq m$) for $\varepsilon > 0$ is the set

$$\sigma_\varepsilon(A) = \{z \in \mathbb{C} : (A + E)\mathbf{x} = z\mathbf{x} \text{ for some } \mathbf{x} \neq 0 \text{ and } \|E\| < \varepsilon\}.$$

[Ruh95] For $A \in \mathbb{C}^{n \times n}$ and invertible $B \in \mathbb{C}^{n \times n}$, the **ε -pseudospectrum of the matrix pencil** $A - \lambda B$ (or generalized eigenvalue problem $A\mathbf{x} = \lambda B\mathbf{x}$) for $\varepsilon > 0$ is the set

$$\sigma_\varepsilon(A, B) = \sigma_\varepsilon(B^{-1}A).$$

[TH01] The ε -**pseudospectrum of the matrix polynomial** $P(\lambda)$ (or polynomial eigenvalue problem $P(\lambda)\mathbf{x} = \mathbf{0}$), where $P(\lambda) = \lambda^p A_p + \lambda^{p-1} A_{p-1} + \cdots + A_0$ and $\varepsilon > 0$, is the set

$$\sigma_\varepsilon(P) = \{z \in \mathbb{C} : z \in \sigma(P + E) \text{ for some } E(\lambda) = \lambda^p E_p + \cdots + E_0, \|E_j\| \leq \varepsilon \alpha_j, \quad j = 0, \dots, p\},$$

for values $\alpha_0, \dots, \alpha_p$. For most applications, one would either take $\alpha_j = 1$ for all j , or $\alpha_j = \|A_j\|$. (This definition differs considerably from the one given for the pseudospectrum of a matrix pencil. In particular, when $p = 1$ the present definition *does not* reduce to the above definition for the pencil; see Fact 6 below.)

Facts:

1. [HP92, HK93] The above definition of the spectral value set $\sigma_\varepsilon(A; B, C)$ is equivalent to

$$\sigma_\varepsilon(A; B, C) = \{z \in \mathbb{C} : \|C(zI - A)^{-1}B\| > \varepsilon^{-1}\}.$$

2. [Kar03] The above definition of the real structured ε -pseudospectrum $\sigma_\varepsilon^R(A)$ is equivalent to

$$\sigma_\varepsilon^R(A) = \{z \in \mathbb{C} : r(A, z) < \varepsilon\},$$

where

$$r(A, z)^{-1} = \inf_{\gamma \in (0,1)} \sigma_2 \left(\begin{bmatrix} \operatorname{Re}(zI - A)^{-1} & -\gamma \operatorname{Im}(zI - A)^{-1} \\ \gamma^{-1} \operatorname{Im}(zI - A)^{-1} & \operatorname{Re}(zI - A)^{-1} \end{bmatrix} \right)$$

and $\sigma_2(\cdot)$ denotes the second largest singular value. From this formulation, one can derive algorithms for computing $\sigma_\varepsilon^R(A)$ akin to those used for computing $\sigma_\varepsilon(A)$.

3. The definition of $\sigma_\varepsilon^R(A)$ suggests similar formulations that impose different restrictions upon E , such as a sparsity pattern, Toeplitz structure, nonnegativity or stochasticity of $A + E$, etc. Such structured pseudospectra are often difficult to compute or approximate.
4. [WT02] The above definition of the ε -pseudospectrum $\sigma_\varepsilon(A)$ of a rectangular matrix $A \in \mathbb{C}^{n \times m}$, $n \geq m$, is equivalent to

$$\sigma_\varepsilon(A) = \{z \in \mathbb{C} : \|(z\tilde{I} - A)^\dagger\| > \varepsilon^{-1}\},$$

where $(\cdot)^\dagger$ denotes the Moore–Penrose pseudoinverse and $\tilde{I}$ denotes the $n \times m$ matrix that has the $m \times m$ identity in the first m rows and is zero elsewhere.

5. The following facts apply to the ε -pseudospectrum of a rectangular matrix $A \in \mathbb{C}^{m \times n}$, $m \geq n$.

- (a) [WT02] It is possible that $\sigma_\varepsilon(A) = \emptyset$.
- (b) [BLO04] For $A \in \mathbb{C}^{m \times n}$, $m \geq n$, and any $\varepsilon > 0$, the set $\sigma_\varepsilon(A)$ contains no more than $2n^2 - n + 1$ connected components.

6. [TE05] Alternative definitions have been proposed for the pseudospectrum of the matrix pencil $A - \lambda B$. The definition presented above has the advantage that the pseudospectrum is invariant to premultiplication of the pencil by a nonsingular matrix, which is consistent with the fact that premultiplication of the differential equation $B\mathbf{x}' = A\mathbf{x}$ does not affect the solution $\mathbf{x}$. Here are two alternative definitions, *neither of which are equivalent to the previous definition*.

- (a) [Rie94] If B is Hermitian positive definite with Cholesky factorization $B = LL^*$, then the pseudospectrum of the pencil can be defined in terms of the standard pseudospectrum of a transformed problem:

$$\sigma_\varepsilon(A, B) = \sigma_\varepsilon(L^{-1}AL^{-*}).$$

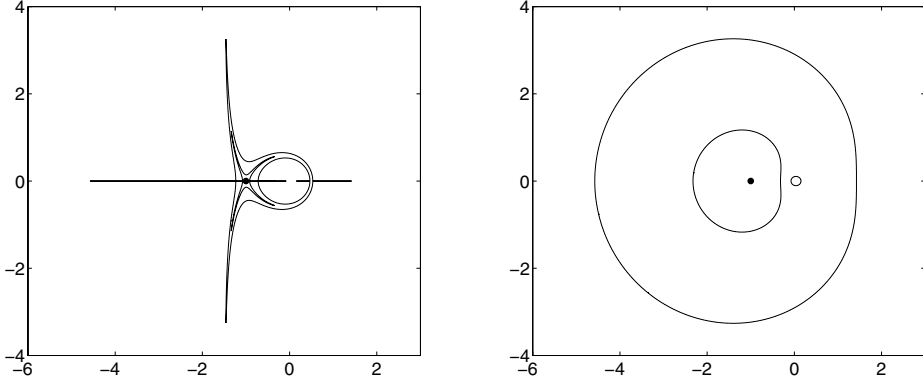


FIGURE 16.8 Spectrum (solid dot) and real structured ε -pseudospectra $\sigma_\varepsilon^R(A)$ (left) and unstructured ε -pseudospectra $\sigma_\varepsilon(A)$ of the second matrix of Example 3 in section 16.1 for $\varepsilon = 10^{-3}, 10^{-4}$.

- (b) [FGNT96, TH01] The following definition is more appropriate for the study of eigenvalue perturbations:

$$\sigma_\varepsilon(A, B) = \{z \in \mathbb{C} : (A + E_0)\mathbf{x} = z(B + E_1)\mathbf{x} \text{ for some } \mathbf{x} \neq \mathbf{0} \text{ and } E_0, E_1 \text{ with } \|E_0\| < \varepsilon\alpha_0, \|E_1\| < \varepsilon\alpha_1\},$$

where generally either $\alpha_j = 1$ for $j = 0, 1$, or $\alpha_0 = \|A\|$ and $\alpha_1 = \|B\|$. This is a special case of the definition given above for the pseudospectrum of a matrix polynomial.

7. [TH01] The above definition of the ε -pseudospectrum of a matrix polynomial, $\sigma_\varepsilon(P)$, is equivalent to

$$\sigma_\varepsilon(P) = \{z \in \mathbb{C} : \|P(z)^{-1}\| > 1/(\varepsilon\phi(|z|))\},$$

where $\phi(z) = \sum_{j=0}^p \alpha_j z^j$ for the same values of $\alpha_0, \dots, \alpha_p$ used in the earlier definition.

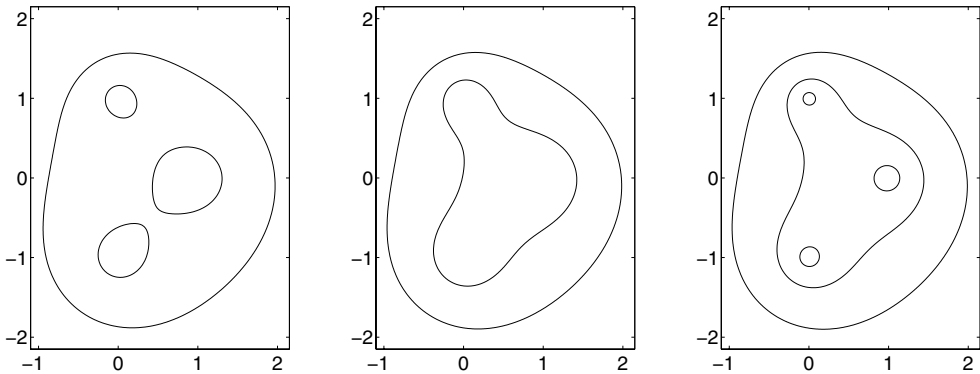


FIGURE 16.9 ε -pseudospectra of the rectangular matrix in Example 2 with $\delta = 0.02$ (left), $\delta = 0.01$ (middle), $\delta = 0.005$ (right), and $\varepsilon = 10^{-1}, 10^{-1.5}$, and 10^{-2} . Note that in the first two plots, $\sigma_\varepsilon(A) = \emptyset$ for $\varepsilon = 10^{-2}$.

Examples:

1. Figure 16.8 compares real structured ε -pseudospectra $\sigma_\varepsilon^R(A)$ to the (unstructured) pseudospectra $\sigma_\varepsilon(A)$ for the second matrix in Example 3 of Section 16.1; cf. [TE05, Fig. 50.3].
2. Figure 16.9 shows pseudospectra of the rectangular matrix

$$A = \begin{bmatrix} 2 & -5 & 10 \\ 1 & -2 & 5i \\ 0 & 0 & 1 \\ \delta & \delta & \delta \end{bmatrix},$$

which is the third matrix in Example 1 of Section 16.1, but with an extra row appended.

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17

Singular Values and Singular Value Inequalities

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17.1 Definitions and Characterizations

Singular values and the singular value decomposition are defined in Chapter 5.6. Additional information on computation of the singular value decomposition can be found in Chapter 45. A brief history of the singular value decomposition and early references can be found in [H]91, Chap. 3].

Throughout this chapter, $q = \min\{m, n\}$, and if $A \in \mathbb{C}^{n \times n}$ has real eigenvalues, then they are ordered $\lambda_1(A) \geq \cdots \geq \lambda_n(A)$.

Definitions:

For $A \in \mathbb{C}^{m \times n}$, define the **singular value vector** $\text{sv}(A) = (\sigma_1(A), \dots, \sigma_q(A))$.

For $A \in \mathbb{C}^{m \times n}$, define $r_1(A) \geq \cdots \geq r_m(A)$ and $c_1(A) \geq \cdots \geq c_n(A)$ to be the ordered Euclidean row and column lengths of A , that is, the square roots of the ordered diagonal entries of AA^* and A^*A .

For $A \in \mathbb{C}^{m \times n}$ define $|A|_{pd} = (A^*A)^{1/2}$. This is called the **spectral absolute value** of A . (This is also called the **absolute value**, but the latter term will not be used in this chapter due to potential confusion with the entry-wise absolute value of A , denoted $|A|$.)

A **polar decomposition** or **polar form** of the matrix $A \in \mathbb{C}^{m \times n}$ with $m \geq n$ is a factorization $A = UP$, where $P \in \mathbb{C}^{n \times n}$ is positive semidefinite and $U \in \mathbb{C}^{m \times n}$ satisfies $U^*U = I_n$.

Facts:

The following facts can be found in most books on matrix theory, for example [HJ91, Chap. 3] or [Bha97].

1. Take $A \in \mathbb{C}^{m \times n}$, and set

$$B = \begin{bmatrix} A & 0 \\ 0 & 0 \end{bmatrix}.$$

Then $\sigma_i(A) = \sigma_i(B)$ for $i = 1, \dots, q$ and $\sigma_i(B) = 0$ for $i > q$. We may choose the zero blocks in B to ensure that B is square. In this way we can often generalize results on the singular values of square matrices to rectangular matrices. For simplicity of exposition, in this chapter we will sometimes state a result for square matrices rather than the more general result for rectangular matrices.

2. (Unitary invariance) Take $A \in \mathbb{C}^{m \times n}$. Then for any unitary $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$,

$$\sigma_i(A) = \sigma_i(UAV), \quad i = 1, 2, \dots, q.$$

3. Take $A, B \in \mathbb{C}^{m \times n}$. There are unitary matrices $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$ such that $A = UBV$ if and only if $\sigma_i(A) = \sigma_i(B)$, $i = 1, 2, \dots, q$.
4. Let $A \in \mathbb{C}^{m \times n}$. Then $\sigma_i^2(A) = \lambda_i(AA^*) = \lambda_i(A^*A)$ for $i = 1, 2, \dots, q$.
5. Let $A \in \mathbb{C}^{m \times n}$. Let $\mathcal{S}_i$ denote the set of subspaces of $\mathbb{C}^n$ of dimension i . Then for $i = 1, 2, \dots, q$,

$$\begin{aligned} \sigma_i(A) &= \min_{\mathcal{X} \in \mathcal{S}_{n-i+1}} \max_{\mathbf{x} \in \mathcal{X}, \|\mathbf{x}\|_2=1} \|\mathbf{A}\mathbf{x}\|_2 = \min_{\mathcal{Y} \in \mathcal{S}_{i-1}} \max_{\mathbf{x} \perp \mathcal{Y}, \|\mathbf{x}\|_2=1} \|\mathbf{A}\mathbf{x}\|_2, \\ \sigma_i(A) &= \max_{\mathcal{X} \in \mathcal{S}_i} \min_{\mathbf{x} \in \mathcal{X}, \|\mathbf{x}\|_2=1} \|\mathbf{A}\mathbf{x}\|_2 = \max_{\mathcal{Y} \in \mathcal{S}_{n-i}} \min_{\mathbf{x} \perp \mathcal{Y}, \|\mathbf{x}\|_2=1} \|\mathbf{A}\mathbf{x}\|_2. \end{aligned}$$

6. Let $A \in \mathbb{C}^{m \times n}$ and define the Hermitian matrix

$$J = \begin{bmatrix} 0 & A \\ A^* & 0 \end{bmatrix} \in \mathbb{C}^{m+n, m+n}.$$

The eigenvalues of J are $\pm\sigma_1(A), \dots, \pm\sigma_q(A)$ together with $|m - n|$ zeros. The matrix J is called the Jordan–Wielandt matrix. Its use allows one to deduce singular value results from results for eigenvalues of Hermitian matrices.

7. Take $m \geq n$ and $A \in \mathbb{C}^{m \times n}$. Let $A = UP$ be a polar decomposition of A . Then $\sigma_i(A) = \lambda_i(P)$, $i = 1, 2, \dots, q$.
8. Let $A \in \mathbb{C}^{m \times n}$ and $1 \leq k \leq q$. Then

$$\begin{aligned} \sum_{i=1}^k \sigma_i(A) &= \max\{\operatorname{Re} \operatorname{tr} U^*AV : U \in \mathbb{C}^{m \times k}, V \in \mathbb{C}^{n \times k}, U^*U = V^*V = I_k\}, \\ \prod_{i=1}^k \sigma_i(A) &= \max\{|\det U^*AV| : U \in \mathbb{C}^{m \times k}, V \in \mathbb{C}^{n \times k}, U^*U = V^*V = I_k\}. \end{aligned}$$

If $m = n$, then

$$\sum_{i=1}^n \sigma_i(A) = \max \left\{ \sum_{i=1}^n |(U^*AU)_{ii}| : U \in \mathbb{C}^{n \times n}, U^*U = I_n \right\}.$$

We cannot replace the n by a general $k \in \{1, \dots, n\}$.

9. Let $A \in \mathbb{C}^{m \times n}$. A yields

$$(a) \sigma_i(A^T) = \sigma_i(A^*) = \sigma_i(\bar{A}) = \sigma_i(A), \text{ for } i = 1, 2, \dots, q.$$

(b) Let $k = \text{rank}(A)$. Then $\sigma_i(A^\dagger) = \sigma_{k-i+1}^{-1}(A)$ for $i = 1, \dots, k$, and $\sigma_i(A^\dagger) = 0$ for $i = k + 1, \dots, q$. In particular, if $m = n$ and A is invertible, then

$$\sigma_i(A^{-1}) = \sigma_{n-i+1}^{-1}(A), \quad i = 1, \dots, n.$$

(c) For any $j \in \mathbb{N}$

$$\sigma_i((A^*A)^j) = \sigma_i^{2j}(A), \quad i = 1, \dots, q;$$

$$\sigma_i((A^*A)^j A^*) = \sigma_i(A(A^*A)^j) = \sigma_i^{2j+1}(A) \quad i = 1, \dots, q.$$

10. Let UP be a polar decomposition of $A \in \mathbb{C}^{m \times n}$ ($m \geq n$). The positive semidefinite factor P is uniquely determined and is equal to $|A|_{pd}$. The factor U is uniquely determined if A has rank n . If A has singular value decomposition $A = U_1 \Sigma U_2^*$ ($U_1 \in \mathbb{C}^{m \times n}$, $U_2 \in \mathbb{C}^{n \times n}$), then $P = U_2 \Sigma U_2^*$, and U may be taken to be $U_1 U_2^*$.

11. Take $A, U \in \mathbb{C}^{n \times n}$ with U unitary. Then $A = U|A|_{pd}$ if and only if $A = |A^*|_{pd}U$.

Examples:

1. Take

$$A = \begin{bmatrix} 11 & -3 & -5 & 1 \\ 1 & -5 & -3 & 11 \\ -5 & 1 & 11 & -3 \\ -3 & 11 & 1 & -5 \end{bmatrix}.$$

The singular value decomposition of A is $A = U \Sigma V^*$, where $\Sigma = \text{diag}(20, 12, 8, 4)$, and

$$U = \frac{1}{2} \begin{bmatrix} -1 & 1 & -1 & 1 \\ -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad \text{and} \quad V = \frac{1}{2} \begin{bmatrix} -1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ -1 & -1 & 1 & 1 \end{bmatrix}.$$

The singular values of A are 20, 12, 8, 4. Let Q denote the permutation matrix that takes (x_1, x_2, x_3, x_4) to (x_1, x_4, x_3, x_2) . Let $P = |A|_{pd} = QA$. The polar decomposition of A is $A = QP$. (To see this, note that a permutation matrix is unitary and that P is positive definite by Geršgorin's theorem.) Note also that $|A|_{pd} \neq |A^*|_{pd} = AQ$.

17.2 Singular Values of Special Matrices

In this section, we present some matrices where the singular values (or some of the singular values) are known, and facts about the singular values of certain structured matrices.

Facts:

The following results can be obtained by straightforward computations if no specific reference is given.

1. Let $D = \text{diag}(\alpha_1, \dots, \alpha_n)$, where the α_i are integers, and let H_1 and H_2 be Hadamard matrices. (See Chapter 32.2.) Then the matrix $H_1 D H_2$ has integer entries and has integer singular values $n|\alpha_1|, \dots, n|\alpha_n|$.

2. (2×2 matrix) Take $A \in \mathbb{C}^{2 \times 2}$. Set $D = |\det(A)|^2$, $N = \|A\|_F^2$. The singular values of A are

$$\sqrt{\frac{N \pm \sqrt{N^2 - 4D}}{2}}.$$

3. Let $X \in \mathbb{C}^{m \times n}$ have singular values $\sigma_1 \geq \cdots \geq \sigma_q$ ($q = \min\{m, n\}$). Set

$$A = \begin{bmatrix} I & 2X \\ 0 & I \end{bmatrix} \in \mathbb{C}^{m+n, m+n}.$$

The $m + n$ singular values of A are

$$\sigma_1 + \sqrt{\sigma_1^2 + 1}, \dots, \sigma_q + \sqrt{\sigma_q^2 + 1}, 1, \dots, 1, \sqrt{\sigma_q^2 + 1} - \sigma_q, \dots, \sqrt{\sigma_1^2 + 1} - \sigma_1.$$

4. [HJ91, Theorem 4.2.15] Let $A \in \mathbb{C}^{m_1 \times n_1}$ and $B \in \mathbb{C}^{m_2 \times n_2}$ have rank m and n . The nonzero singular values of $A \otimes B$ are $\sigma_i(A)\sigma_j(B)$, $i = 1, \dots, m$, $j = 1, \dots, n$.
5. Let $A \in \mathbb{C}^{n \times n}$ be normal with eigenvalues $\lambda_1, \dots, \lambda_n$, and let p be a polynomial. Then the singular values of $p(A)$ are $|p(\lambda_k)|$, $k = 1, \dots, n$. In particular, if A is a circulant with first row $a_0, \dots, a_{n-1}$, then A has singular values

$$\left| \sum_{j=0}^{n-1} a_j e^{-2\pi i j k / n} \right|, \quad k = 1, \dots, n.$$

6. Take $A \in \mathbb{C}^{n \times n}$ and nonzero $\mathbf{x} \in \mathbb{C}^n$. If $A\mathbf{x} = \lambda\mathbf{x}$ and $\mathbf{x}^*A = \lambda\mathbf{x}^*$, then $|\lambda|$ is a singular value of A . In particular, if A is doubly stochastic, then $\sigma_1(A) = 1$.
7. [Kit95] Let A be the companion matrix corresponding to the monic polynomial $p(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_1t + a_0$. Set $N = 1 + \sum_{i=0}^{n-1} |a_i|^2$. The n singular values of A are

$$\sqrt{\frac{N + \sqrt{N^2 - 4|a_0|^2}}{2}}, 1, \dots, 1, \sqrt{\frac{N - \sqrt{N^2 - 4|a_0|^2}}{2}}.$$

8. [Hig96, p. 167] Take $s, c \in \mathbb{R}$ such that $s^2 + c^2 = 1$. The matrix

$$A = \text{diag}(1, s, \dots, s^{n-1}) \begin{bmatrix} 1 & -c & -c & \cdots & -c \\ & 1 & -c & \cdots & -c \\ & & \ddots & \ddots & \vdots \\ & & & \ddots & -c \\ & & & & 1 \end{bmatrix}$$

is called a Kahan matrix. If c and s are positive, then $\sigma_{n-1}(A) = s^{n-2}\sqrt{1+c}$.

9. [GE95, Lemma 3.1] Take $0 = d_1 < d_2 < \cdots < d_n$ and $0 \neq z_i \in \mathbb{C}$. Let

$$A = \begin{bmatrix} z_1 & & & \\ z_2 & d_2 & & \\ \vdots & & \ddots & \\ z_n & & & d_n \end{bmatrix}.$$

The singular values of A satisfy the equation

$$f(t) = 1 + \sum_{i=1}^n \frac{|z_i|^2}{d_i^2 - t^2} = 0$$

and exactly one lies in each of the intervals $(d_1, d_2), \dots, (d_{n-1}, d_n), (d_n, d_n + \|z\|_2)$. Let $\sigma_i = \sigma_i(A)$. The left and right i th singular vectors of A are $\mathbf{u}/\|\mathbf{u}\|_2$ and $\mathbf{v}/\|\mathbf{v}\|_2$ respectively, where

$$\mathbf{u} = \left[\frac{z_1}{d_1^2 - \sigma_i^2}, \dots, \frac{z_n}{d_n^2 - \sigma_i^2} \right]^T \quad \text{and} \quad \mathbf{v} = \left[-1, \frac{d_2 z_2}{d_2^2 - \sigma_i^2}, \dots, \frac{d_n z_n}{d_n^2 - \sigma_i^2} \right]^T.$$

10. (*Bidiagonal*) Take

$$B = \begin{bmatrix} \alpha_1 & \beta_1 & & & \\ & \alpha_2 & \ddots & & \\ & & \ddots & \beta_{n-1} & \\ & & & \alpha_n & \end{bmatrix} \in \mathbb{C}^{n \times n}.$$

If all the α_i and β_i are nonzero, then B is called an *unreduced bidiagonal matrix* and

- (a) The singular values of B are distinct.
 - (b) The singular values of B depend only on the moduli of $\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_{n-1}$.
 - (c) The largest singular value of B is a strictly increasing function of the modulus of each of the α_i and β_i .
 - (d) The smallest singular value of B is a strictly increasing function of the modulus of each of the α_i and a strictly decreasing function of the modulus of each of the β_i .
 - (e) (High relative accuracy) Take $\tau > 1$ and multiply one of the entries of B by τ to give $\hat{B}$. Then $\tau^{-1}\sigma_i(B) \leq \sigma_i(\hat{B}) \leq \tau\sigma_i(B)$.
11. [HJ85, Sec. 4.4, prob. 26] Let $A \in \mathbb{C}^{n \times n}$ be skew-symmetric (and possibly complex). The nonzero singular values of A occur in pairs.

17.3 Unitarily Invariant Norms

Throughout this section, $q = \min\{m, n\}$.

Definitions:

A vector norm $\|\cdot\|$ on $\mathbb{C}^{m \times n}$ is **unitarily invariant** (*u.i.*) if $\|A\| = \|UAV\|$ for any unitary $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$ and any $A \in \mathbb{C}^{m \times n}$.

$\|\cdot\|_{UI}$ is used to denote a **general unitarily invariant norm**.

A function $g: \mathbb{R}^n \rightarrow \mathbb{R}_0^+$ is a **permutation invariant absolute norm** if it is a norm, and in addition $g(x_1, \dots, x_n) = g(|x_1|, \dots, |x_n|)$ and $g(\mathbf{x}) = g(P\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^n$ and all permutation matrices $P \in \mathbb{R}^{n \times n}$. (Many authors call a permutation invariant absolute norm a symmetric gauge function.)

The **Ky Fan k norms** of $A \in \mathbb{C}^{m \times n}$ are

$$\|A\|_{K,k} = \sum_{i=1}^k \sigma_i(A), \quad k = 1, 2, \dots, q.$$

The **Schatten- p norms** of $A \in \mathbb{C}^{m \times n}$ are

$$\|A\|_{S,p} = \left(\sum_{i=1}^q \sigma_i^p(A) \right)^{1/p} = (\text{tr } |A|_{pd}^p)^{1/p} \quad 0 \leq p < \infty$$

$$\|A\|_{S,\infty} = \sigma_1(A).$$

The **trace norm** of $A \in \mathbb{C}^{m \times n}$ is

$$\|A\|_{\text{tr}} = \sum_{i=1}^q \sigma_i(A) = \|A\|_{K,q} = \|A\|_{S,1} = \text{tr } |A|_p.$$

Other norms discussed in this section, such as the **spectral norm** $\|\cdot\|_2$ ($\|A\|_2 = \sigma_1(A) = \max_{\mathbf{x} \neq 0} \frac{\|A\mathbf{x}\|_2}{\|\mathbf{x}\|_2}$) and the **Frobenius norm** $\|\cdot\|_F$ ($\|A\|_F = (\sum_{i=1}^q \sigma_i^2(A))^{1/2} = (\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2)^{1/2}$), are defined in Section 7.1. and discussed extensively in Chapter 37.

Warning: There is potential for considerable confusion. For example, $\|A\|_2 = \|A\|_{K,1} = \|A\|_{S,\infty}$, while $\|\cdot\|_\infty \neq \|\cdot\|_{S,\infty}$ (unless $m = 1$), and generally $\|A\|_2$, $\|A\|_{S,2}$ and $\|A\|_{K,2}$ are all different, as are $\|A\|_1$, $\|A\|_{S,1}$ and $\|A\|_{K,1}$. Nevertheless, many authors use $\|\cdot\|_k$ for $\|\cdot\|_{K,k}$ and $\|\cdot\|_p$ for $\|\cdot\|_{S,p}$.

Facts:

The following standard facts can be found in many texts, e.g., [HJ91, §3.5] and [Bha97, Chap. IV].

1. Let $\|\cdot\|$ be a norm on $\mathbb{C}^{m \times n}$. It is unitarily invariant if and only if there is a permutation invariant absolute norm g on $\mathbb{R}^q$ such that $\|A\| = g(\sigma_1(A), \dots, \sigma_q(A))$ for all $A \in \mathbb{C}^{m \times n}$.
2. Let $\|\cdot\|$ be a unitarily invariant norm on $\mathbb{C}^{m \times n}$, and let g be the corresponding permutation invariant absolute norm g . Then the dual norms (see Chapter 37) satisfy $\|A\|^D = g^D(\sigma_1(A), \dots, \sigma_q(A))$.
3. [HJ91, Prob. 3.5.18] The spectral norm and trace norm are duals, while the Frobenius norm is self dual. The dual of $\|\cdot\|_{S,p}$ is $\|\cdot\|_{S,\tilde{p}}$, where $1/p + 1/\tilde{p} = 1$ and

$$\|A\|_{K,k}^D = \max \left\{ \|A\|_2, \frac{\|A\|_{\text{tr}}}{k} \right\}, \quad k = 1, \dots, q.$$

4. For any $A \in \mathbb{C}^{m \times n}$, $q^{-1/2} \|A\|_F \leq \|A\|_2 \leq \|A\|_F$.
5. If $\|\cdot\|$ is a u.i. norm on $\mathbb{C}^{m \times n}$, then $N(A) = \|A^*A\|^{1/2}$ is a u.i. norm on $\mathbb{C}^{n \times n}$. A norm that arises in this way is called a *Q-norm*.
6. Let $A, B \in \mathbb{C}^{m \times n}$ be given. The following are equivalent

- (a) $\|A\|_{UI} \leq \|B\|_{UI}$ for all unitarily invariant norms $\|\cdot\|_{UI}$.
- (b) $\|A\|_{K,k} \leq \|B\|_{K,k}$ for $k = 1, 2, \dots, q$.
- (c) $(\sigma_1(A), \dots, \sigma_q(A)) \leq_w (\sigma_1(B), \dots, \sigma_q(B))$. ($\leq_w$ is defined in Preliminaries)

The equivalence of the first two conditions is Fan's Dominance Theorem.

7. The Ky–Fan- k norms can be represented in terms of an extremal problem involving the spectral norm and the trace norm. Take $A \in \mathbb{C}^{m \times n}$. Then

$$\|A\|_{K,k} = \min\{\|X\|_{\text{tr}} + k\|Y\|_2 : X + Y = A\} \quad k = 1, \dots, q.$$

8. [HJ91, Theorem 3.3.14] Take $A, B \in \mathbb{C}^{m \times n}$. Then

$$|\text{tr } AB^*| \leq \sum_{i=1}^q \sigma_i(A) \sigma_i(B).$$

This is an important result in developing the theory of unitarily invariant norms.

Examples:

1. The matrix A in Example 1 of Section 17.1 has singular values 20, 12, 8, and 4. So

$$\begin{aligned}\|A\|_2 &= 20, & \|A\|_F &= \sqrt{624}, & \|A\|_{tr} &= 44; \\ \|A\|_{K,1} &= 20, & \|A\|_{K,2} &= 32, & \|A\|_{K,3} &= 40, & \|A\|_{K,4} &= 44; \\ \|A\|_{S,1} &= 44, & \|A\|_{S,2} &= \sqrt{624}, & \|A\|_{S,3} &= \sqrt[3]{10304} = 21.7605, & \|A\|_{S,\infty} &= 20.\end{aligned}$$

17.4 Inequalities

Throughout this section, $q = \min\{m, n\}$ and if $A \in \mathbb{C}^{m \times n}$ has real eigenvalues, then they are ordered $\lambda_1(A) \geq \cdots \geq \lambda_n(A)$.

Definitions:

Pinching is defined recursively. If

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \in \mathbb{C}^{m \times n}, \quad B = \begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix} \in \mathbb{C}^{m \times n},$$

then B is a pinching of A . (Note that we do not require the A_{ii} to be square.) Furthermore, any pinching of B is a pinching of A .

For positive α, β , define the **measure of relative separation** $\chi(\alpha, \beta) = |\sqrt{\alpha/\beta} - \sqrt{\beta/\alpha}|$.

Facts:

The following facts can be found in standard references, for example [HJ91, Chap. 3], unless another reference is given.

1. (*Submatrices*) Take $A \in \mathbb{C}^{m \times n}$ and let B denote A with one of its rows or columns deleted. Then $\sigma_{i+1}(A) \leq \sigma_i(B) \leq \sigma_i(A)$, $i = 1, \dots, q-1$.
2. Take $A \in \mathbb{C}^{m \times n}$ and let B be A with a row *and* a column deleted. Then

$$\sigma_{i+2}(A) \leq \sigma_i(B) \leq \sigma_i(A), \quad i = 1, \dots, q-2.$$

The $i+2$ cannot be replaced by $i+1$. (Example 2)

3. Take $A \in \mathbb{C}^{m \times n}$ and let B be an $(m-k) \times (n-l)$ submatrix of A . Then

$$\sigma_{i+k+l}(A) \leq \sigma_i(B) \leq \sigma_i(A), \quad i = 1, \dots, q-(k+l).$$

4. Take $A \in \mathbb{C}^{m \times n}$ and let B be A with some of its rows and/or columns set to zero. Then $\sigma_i(B) \leq \sigma_i(A)$, $i = 1, \dots, q$.
5. Let B be a pinching of A . Then $\text{sv}(B) \leq_w \text{sv}(A)$. The inequalities $\prod_{i=1}^k \sigma_i(B) \leq \prod_{i=1}^k \sigma_i(A)$ and $\sigma_k(B) \leq \sigma_k(A)$ are not necessarily true for $k > 1$. (Example 1)
6. (*Singular values of $A+B$*) Let $A, B \in \mathbb{C}^{m \times n}$.

- (a) $\text{sv}(A+B) \leq_w \text{sv}(A) + \text{sv}(B)$, or equivalently

$$\sum_{i=1}^k \sigma_i(A+B) \leq \sum_{i=1}^k \sigma_i(A) + \sum_{i=1}^k \sigma_i(B), \quad i = 1, \dots, q.$$

- (b) If $i+j-1 \leq q$ and $i, j \in \mathbb{N}$, then $\sigma_{i+j-1}(A+B) \leq \sigma_i(A) + \sigma_j(B)$.

- (c) We have the weak majorization $|\text{sv}(A + B) - \text{sv}(A)| \leq_w \text{sv}(B)$ or, equivalently, if $1 \leq i_1 < \dots < i_k \leq q$, then

$$\sum_{j=1}^k |\sigma_{i_j}(A + B) - \sigma_{i_j}(A)| \leq \sum_{j=1}^k \sigma_{j_j}(B),$$

$$\sum_{i=1}^k \sigma_{i_j}(A) - \sum_{j=1}^k \sigma_{j_j}(B) \leq \sum_{j=1}^k \sigma_{i_j}(A + B) \leq \sum_{i=1}^k \sigma_{i_j}(A) + \sum_{j=1}^k \sigma_{j_j}(B).$$

- (d) [Tho75] (Thompson's Standard Additive Inequalities) If $1 \leq i_1 < \dots < i_k \leq q$, $1 \leq j_1 < \dots < j_k \leq q$ and $i_k + j_k \leq q + k$, then

$$\sum_{s=1}^k \sigma_{i_s + j_s - s}(A + B) \leq \sum_{s=1}^k \sigma_{i_s}(A) + \sum_{s=1}^k \sigma_{j_s}(B).$$

7. (Singular values of AB) Take $A, B \in \mathbb{C}^{n \times n}$.

- (a) For all $k = 1, 2, \dots, n$ and all $p > 0$, we have

$$\prod_{i=n-k+1}^{i=n} \sigma_i(A) \sigma_i(B) \leq \prod_{i=n-k+1}^{i=n} \sigma_i(AB),$$

$$\prod_{i=1}^k \sigma_i(AB) \leq \prod_{i=1}^k \sigma_i(A) \sigma_i(B),$$

$$\sum_{i=1}^k \sigma_i^p(AB) \leq \sum_{i=1}^k \sigma_i^p(A) \sigma_i^p(B).$$

- (b) If $i, j \in \mathbb{N}$ and $i + j - 1 \leq n$, then $\sigma_{i+j-1}(AB) \leq \sigma_i(A) \sigma_j(B)$.

- (c) $\sigma_n(A) \sigma_i(B) \leq \sigma_i(AB) \leq \sigma_1(A) \sigma_i(B)$, $i = 1, 2, \dots, n$.

- (d) [LM99] Take $1 \leq j_1 < \dots < j_k \leq n$. If A is invertible and $\sigma_{j_i}(B) > 0$, then $\sigma_{j_i}(AB) > 0$ and

$$\prod_{i=n-k+1}^n \sigma_i(A) \leq \prod_{i=1}^k \max \left\{ \frac{\sigma_{j_i}(AB)}{\sigma_{j_i}(B)}, \frac{\sigma_{j_i}(B)}{\sigma_{j_i}(AB)} \right\} \leq \prod_{i=1}^k \sigma_i(A).$$

- (e) [LM99] Take invertible $S, T \in \mathbb{C}^{n \times n}$. Set $\tilde{A} = SAT$. Let the singular values of A and $\tilde{A}$ be $\sigma_1 \geq \dots \geq \sigma_n$ and $\tilde{\sigma}_1 \geq \dots \geq \tilde{\sigma}_n$. Then

$$\|\text{diag}(\chi(\sigma_1, \tilde{\sigma}_1), \dots, \chi(\sigma_n, \tilde{\sigma}_n))\|_{UI} \leq \frac{1}{2} (\|S^* - S^{-1}\|_{UI} + \|T^* - T^{-1}\|_{UI}).$$

- (f) [TT73] (Thompson's Standard Multiplicative Inequalities) Take $1 \leq i_1 < \dots < i_m \leq n$ and $1 \leq j_1 < \dots < j_m \leq n$. If $i_m + j_m \leq m + n$, then

$$\prod_{s=1}^m \sigma_{i_s + j_s - s}(AB) \leq \prod_{s=1}^m \sigma_{i_s}(A) \prod_{s=1}^m \sigma_{j_s}(B).$$

8. [Bha97, §IX.1] Take $A, B \in \mathbb{C}^{n \times n}$.

- (a) If AB is normal, then

$$\prod_{i=1}^k \sigma_i(AB) \leq \prod_{i=1}^k \sigma_i(BA), \quad k = 1, \dots, q,$$

and, consequently, $\text{sv}(AB) \leq_w \text{sv}(BA)$, and $\|AB\|_{UI} \leq \|BA\|_{UI}$.

(b) If AB is Hermitian, then $\text{sv}(AB) \leq_w \text{sv}(H(BA))$ and $\|AB\|_{UI} \leq \|H(BA)\|_{UI}$, where $H(X) = (X + X^*)/2$.

9. (Term-wise singular value inequalities) [Zha02, p. 28] Take $A, B \in \mathbb{C}^{m \times n}$. Then

$$2\sigma_i(AB^*) \leq \sigma_i(A^*A + B^*B), \quad i = 1, \dots, q$$

and, more generally, if $p, \bar{p} > 0$ and $1/p + 1/\bar{p} = 1$, then

$$\sigma_i(AB^*) \leq \sigma_i \left(\frac{(A^*A)^{p/2}}{p} + \frac{(B^*B)^{\bar{p}/2}}{\bar{p}} \right) = \sigma_i \left(\frac{|A|_{pd}^p}{p} + \frac{|B|_{pd}^{\bar{p}}}{\bar{p}} \right).$$

The inequalities $2\sigma_1(A^*B) \leq \sigma_1(A^*A + B^*B)$ and $\sigma_1(A + B) \leq \sigma_1(|A|_{pd} + |B|_{pd})$ are not true in general (Example 3), but we do have

$$\|A^*B\|_{UI}^2 \leq \|A^*A\|_{UI} \|B^*B\|_{UI}.$$

10. [Bha97, Prop. III.5.1] Take $A \in \mathbb{C}^{n \times n}$. Then $\lambda_i(A + A^*) \leq 2\sigma_i(A)$, $i = 1, 2, \dots, n$.

11. [LM02] (Block triangular matrices) Let $A = \begin{bmatrix} R & 0 \\ S & T \end{bmatrix} \in \mathbb{C}^{n \times n}$ ($R \in \mathbb{C}^{p \times p}$) have singular values $\alpha_1 \geq \dots \geq \alpha_n$. Let $k = \min\{p, n - p\}$. Then

(a) If $\sigma_{\min}(R) \geq \sigma_{\max}(T)$, then

$$\begin{aligned} \sigma_i(R) &\leq \alpha_i, \quad i = 1, \dots, p \\ \alpha_i &\leq \sigma_{i-p}(T), \quad i = p + 1, \dots, n. \end{aligned}$$

(b) $(\sigma_1(S), \dots, \sigma_k(S)) \leq_w (\alpha_1 - \alpha_n, \dots, \alpha_k - \alpha_{n-k+1})$.

(c) If A is invertible, then

$$\begin{aligned} (\sigma_1(T^{-1}SR^{-1}), \dots, \sigma_k(T^{-1}SR^{-1})) &\leq_w (\alpha_n^{-1} - \alpha_1^{-1}, \dots, \alpha_{n-k+1}^{-1} - \alpha_k^{-1}), \\ (\sigma_1(T^{-1}S), \dots, \sigma_k(T^{-1}S)) &\leq_w \frac{1}{2} \left(\frac{\alpha_1}{\alpha_n} - \frac{\alpha_n}{\alpha_1}, \dots, \frac{\alpha_k}{\alpha_{n-k+1}} - \frac{\alpha_{n-k+1}}{\alpha_k} \right). \end{aligned}$$

12. [LM02] (Block positive semidefinite matrices) Let $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{12}^* & A_{22} \end{bmatrix} \in \mathbb{C}^{n \times n}$ be positive definite with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n$. Assume $A_{11} \in \mathbb{C}^{p \times p}$. Set $k = \min\{p, n - p\}$. Then

$$\begin{aligned} \prod_{i=1}^j \sigma_i^2(A_{12}) &\leq \prod_{i=1}^j \sigma_i(A_{11}) \sigma_i(A_{22}), \quad j = 1, \dots, k, \\ (\sigma_1(A_{11}^{-1/2}A_{12}), \dots, \sigma_k(A_{11}^{-1/2}A_{12})) &\leq_w (\sqrt{\lambda_1} - \sqrt{\lambda_n}, \dots, \sqrt{\lambda_k} - \sqrt{\lambda_{n-k+1}}), \\ (\sigma_1(A_{11}^{-1}A_{12}), \dots, \sigma_k(A_{11}^{-1}A_{12})) &\leq_w \frac{1}{2} (\chi(\lambda_1, \lambda_n), \dots, \chi(\lambda_k, \lambda_{n-k+1})). \end{aligned}$$

If $k = n/2$, then

$$\|A_{12}\|_{UI}^2 \leq \|A_{11}\|_{UI} \|A_{22}\|_{UI}.$$

13. (Singular values and eigenvalues) Let $A \in \mathbb{C}^{n \times n}$. Assume $|\lambda_1(A)| \geq \dots \geq |\lambda_n(A)|$. Then

(a) $\prod_{i=1}^k |\lambda_i(A)| \leq \prod_{i=k}^n \sigma_i(A)$, $k = 1, \dots, n$, with equality for $k = n$.

- (b) Fix $p > 0$. Then for $k = 1, 2, \dots, n$,

$$\sum_{i=1}^k |\lambda_i^p(A)| \leq \sum_{i=1}^k \sigma_i^p(A).$$

Equality holds with $k = n$ if and only if equality holds for all $k = 1, 2, \dots, n$, if and only if A is normal.

- (c) [HJ91, p. 180] (Yamamoto's theorem) $\lim_{k \rightarrow \infty} (\sigma_i(A^k))^{1/k} = |\lambda_i(A)|$, $i = 1, \dots, n$.
14. [LM01] Let $\lambda_i \in \mathbb{C}$ and $\sigma_i \in \mathbb{R}_0^+$, $i = 1, \dots, n$ be ordered in nonincreasing absolute value. There is a matrix A with eigenvalues $\lambda_1, \dots, \lambda_n$ and singular values $\sigma_1, \dots, \sigma_n$ if and only if

$$\prod_{i=1}^k |\lambda_i| \leq \prod_{i=1}^k \sigma_i, \quad k = 1, \dots, n, \quad \text{with equality for } k = n.$$

In addition:

- (a) The matrix A can be taken to be upper triangular with the eigenvalues on the diagonal in any order.
- (b) If the complex entries in $\lambda_1, \dots, \lambda_n$ occur in conjugate pairs, then A may be taken to be in real Schur form, with the 1×1 and 2×2 blocks on the diagonal in any order.
- (c) There is a finite construction of the upper triangular matrix in cases (a) and (b).
- (d) If $n > 2$, then A cannot always be taken to be bidiagonal. (Example 5)
15. [Zha02, Chap. 2] (*Singular values of $A \circ B$*) Take $A, B \in \mathbb{C}^{n \times n}$.
- (a) $\sigma_i(A \circ B) \leq \min\{r_i(A), c_i(B)\} \cdot \sigma_1(B)$, $i = 1, 2, \dots, n$.
- (b) We have the following weak majorizations:

$$\begin{aligned} \sum_{i=1}^k \sigma_i(A \circ B) &\leq \sum_{i=1}^k \min\{r_i(A), c_i(A)\} \sigma_i(B), \quad k = 1, \dots, n, \\ \sum_{i=1}^k \sigma_i(A \circ B) &\leq \sum_{i=1}^k \sigma_i(A) \sigma_i(B), \quad k = 1, \dots, n, \\ \prod_{i=1}^k \sigma_i^2(A \circ B) &\leq \prod_{i=1}^k \sigma_i((A^* A) \circ (B^* B)), \quad k = 1, \dots, n. \end{aligned}$$

- (c) Take $X, Y \in \mathbb{C}^{n \times n}$. If $A = X^* Y$, then we have the weak majorization

$$\sum_{i=1}^k \sigma_i(A \circ B) \leq \sum_{i=1}^k c_i(X) c_i(Y) \sigma_i(B), \quad k = 1, \dots, n.$$

- (d) If B is positive semidefinite with diagonal entries $b_{11} \geq \dots \geq b_{nn}$, then

$$\sum_{i=1}^k \sigma_i(A \circ B) \leq \sum_{i=1}^k b_{ii} \sigma_i(A), \quad k = 1, \dots, n.$$

- (e) If both A and B are positive definite, then so is $A \circ B$ (Schur product theorem). In this case the singular values of A, B and $A \circ B$ are their eigenvalues and BA has positive eigenvalues and we have the weak multiplicative majorizations

$$\prod_{i=k}^n \lambda_i(B) \lambda_i(A) \leq \prod_{i=k}^n b_{ii} \lambda_i(A) \leq \prod_{i=k}^n \lambda_i(BA) \leq \prod_{i=k}^n \lambda_i(A \circ B), \quad k = 1, 2, \dots, n.$$

The inequalities are still valid if we replace $A \circ B$ by $A \circ B^T$. (Note B^T is not necessarily the same as $B^* = B$.)

16. Let $A \in \mathbb{C}^{m \times n}$. The following are equivalent:

- (a) $\sigma_1(A \circ B) \leq \sigma_1(B)$ for all $B \in \mathbb{C}^{m \times n}$.
- (b) $\sum_{i=1}^k \sigma_i(A \circ B) \leq \sum_{i=1}^k \sigma_i(B)$ for all $B \in \mathbb{C}^{m \times n}$ and all $k = 1, \dots, q$.
- (c) There are positive semidefinite $P \in \mathbb{C}^{n \times n}$ and $Q \in \mathbb{C}^{m \times m}$ such that

$$\begin{bmatrix} P & A \\ A^* & Q \end{bmatrix}$$

is positive semidefinite, and has diagonal entries at most 1.

17. (*Singular values and matrix entries*) Take $A \in \mathbb{C}^{m \times n}$. Then

$$\begin{aligned} (|a_{11}|^2, |a_{12}|^2, \dots, |a_{mn}|^2) &\preceq (\sigma_1^2(A), \dots, \sigma_q^2(A), 0, \dots, 0), \\ \sum_{i=1}^q \sigma_i^p(A) &\leq \sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^p, \quad 0 \leq p \leq 2, \\ \sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^p &\leq \sum_{i=1}^q \sigma_i^p(A), \quad 2 \leq p < \infty. \end{aligned}$$

If $\sigma_1(A) = |a_{ij}|$, then all the other entries in row i and column j of A are 0.

18. Take $\sigma_1 \geq \dots \geq \sigma_n \geq 0$ and $\alpha_1 \geq \dots \geq \alpha_n \geq 0$. Then

$$\exists A \in \mathbb{R}^{n \times n} \text{ s.t. } \sigma_i(A) = \sigma_i \quad \text{and} \quad c_i(A) = \alpha_i \Leftrightarrow (\alpha_1^2, \dots, \alpha_n^2) \preceq (\sigma_1^2, \dots, \sigma_n^2).$$

This statement is still true if we replace $\mathbb{R}^{n \times n}$ by $\mathbb{C}^{n \times n}$ and/or $c_i(\cdot)$ by $r_i(\cdot)$.

19. Take $A \in \mathbb{C}^{n \times n}$. Then

$$\prod_{i=k}^n \sigma_i(A) \leq \prod_{i=k}^n c_i(A), \quad k = 1, 2, \dots, n.$$

The case $k = 1$ is Hadamard's Inequality: $|\det(A)| \leq \prod_{i=1}^n c_i(A)$.

20. [Tho77] Take $F = \mathbb{C}$ or $\mathbb{R}$ and $d_1, \dots, d_n \in F$ such that $|d_1| \geq \dots \geq |d_n|$, and $\sigma_1 \geq \dots \geq \sigma_n \geq 0$. There is a matrix $A \in F^{n \times n}$ with diagonal entries $d_1, \dots, d_n$ and singular values $\sigma_1, \dots, \sigma_n$ if and only if

$$(|d_1|, \dots, |d_n|) \preceq_w (\sigma_1(A), \dots, \sigma_n(A)) \quad \text{and} \quad \sum_{j=1}^{n-1} |d_j| - |d_n| \leq \sum_{j=1}^{n-1} \sigma_j(A) - \sigma_n(A).$$

21. (*Nonnegative matrices*) Take $A = [a_{ij}] \in \mathbb{C}^{m \times n}$.

- (a) If $B = [|a_{ij}|]$, then $\sigma_1(A) \leq \sigma_1(B)$.
- (b) If A and B are real and $0 \leq a_{ij} \leq b_{ij} \quad \forall \quad i, j$, then $\sigma_1(A) \leq \sigma_1(B)$. The condition $0 \leq a_{ij}$ is essential. (Example 4)
- (c) The condition $0 \leq b_{ij} \leq 1 \quad \forall \quad i, j$ does not imply $\sigma_1(A \circ B) \leq \sigma_1(A)$. (Example 4)

22. (*Bound on σ_1*) Let $A \in \mathbb{C}^{m \times n}$. Then $\|A\|_2 = \sigma_1(A) \leq \sqrt{\|A\|_1 \|A\|_\infty}$.

23. [Zha99] (*Cartesian decomposition*) Let $C = A + iB \in \mathbb{C}^{n \times n}$, where A and B are Hermitian. Let A, B, C have singular values $\alpha_j, \beta_j, \gamma_j, j = 1, \dots, n$. Then

$$(\gamma_1, \dots, \gamma_n) \preceq_w \sqrt{2}(|\alpha_1 + i\beta_1|, \dots, |\alpha_n + i\beta_n|) \preceq_w 2(\gamma_1, \dots, \gamma_n).$$

Examples:

1. Take

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Then B is a pinching of A , and C is a pinching of both A and B . The matrices A, B, C have singular values $\alpha = (3, 0, 0)$, $\beta = (2, 1, 0)$, and $\gamma = (1, 1, 1)$. As stated in Fact 5, $\gamma \preceq_w \beta \preceq_w \alpha$. In fact, since the matrices are all positive semidefinite, we may replace $\preceq_w$ by $\preceq$. However, it is not true that $\gamma_i \leq \alpha_i$ except for $i = 1$. Nor is it true that $|\det(C)| \leq |\det(A)|$.

2. The matrices

$$A = \begin{bmatrix} 11 & -3 & -5 & 1 \\ 1 & -5 & -3 & 11 \\ -5 & 1 & 11 & -3 \\ -3 & 11 & 1 & -5 \end{bmatrix}, \quad B = \begin{bmatrix} 11 & -3 & -5 & 1 \\ 1 & -5 & -3 & 11 \\ -5 & 1 & 11 & -3 \end{bmatrix}, \quad C = \begin{bmatrix} 11 & -3 & -5 \\ 1 & -5 & -3 \\ -5 & 1 & 11 \end{bmatrix}$$

have singular values $\alpha = (20, 12, 8, 4)$, $\beta = (17.9, 10.5, 6.0)$, and $\gamma = (16.7, 6.2, 4.5)$ (to 1 decimal place). The singular values of B interlace those of A ($\alpha_4 \leq \beta_3 \leq \alpha_3 \leq \beta_2 \leq \alpha_2 \leq \beta_1 \leq \alpha_1$), but those of C do not. In particular, $\alpha_3 \not\leq \gamma_2$. It is true that $\alpha_{i+2} \leq \gamma_i \leq \alpha_i$ ($i = 1, 2$).

3. Take

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}.$$

Then $\|A + B\|_2 = \sigma_1(A + B) = 2 \not\leq \sqrt{2} = \sigma_1(|A|_{pd} + |B|_{pd}) = \| |A|_{pd} + |B|_{pd} \|_2$. Also, $2\sigma_1(A^*B) = 4 \not\leq 2 = \sigma_1(A^*A + B^*B)$.

4. Setting entries of a matrix to zero can increase the largest singular value. Take

$$A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \quad \text{and} \quad B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Then $\sigma_1(A) = \sqrt{2} < (1 + \sqrt{5})/2 = \sigma_1(B)$.

5. A bidiagonal matrix B cannot have eigenvalues 1, 1, 1 and singular values 1/2, 1/2, 4. If B is unreduced bidiagonal, then it cannot have repeated singular values. (See Fact 10, section 17.2.) However, if B were reduced, then it would have a singular value equal to 1.

17.5 Matrix Approximation

Recall that $\|\cdot\|_{UI}$ denotes a general unitarily invariant norm, and that $q = \min\{m, n\}$.

Facts:

The following facts can be found in standard references, for example, [HJ91, Chap. 3], unless another reference is given.

1. (*Best rank k approximation.*) Let $A \in \mathbb{C}^{m \times n}$ and $1 \leq k \leq q - 1$. Let $A = U\Sigma V^*$ be a singular value decomposition of A . Let $\tilde{\Sigma}$ be equal to Σ except that $\tilde{\Sigma}_{ii} = 0$ for $i > k$, and let $\tilde{A} = U\tilde{\Sigma}V^*$. Then $\text{rank}(\tilde{A}) \leq k$, and

$$\|\Sigma - \tilde{\Sigma}\|_{UI} = \|A - \tilde{A}\|_{UI} = \min\{\|A - B\|_{UI} : \text{rank}(B) \leq k\}.$$

In particular, for the spectral norm and the Frobenius norm, we have

$$\begin{aligned}\sigma_{k+1}(A) &= \min\{\|A - B\|_2 : \text{rank}(B) \leq k\}, \\ \left(\sum_{i=k+1}^q \sigma_{k+1}^2(A)\right)^{1/2} &= \min\{\|A - B\|_F : \text{rank}(B) \leq k\}.\end{aligned}$$

2. [Bha97, p. 276] (*Best unitary approximation*) Take $A, W \in \mathbb{C}^{n \times n}$ with W unitary. Let $A = UP$ be a polar decomposition of A . Then

$$\|A - U\|_{UI} \leq \|A - W\|_{UI} \leq \|A + U\|_{UI}.$$

3. [GV96, §12.4.1] [HJ85, Ex. 7.4.8] (*Orthogonal Procrustes problem*) Let $A, B \in \mathbb{C}^{m \times n}$. Let B^*A have a polar decomposition $B^*A = UP$. Then

$$\|A - BU\|_F = \min\{\|A - BW\|_F : W \in \mathbb{C}^{n \times n}, W^*W = I\}.$$

This result is not true if $\|\cdot\|_F$ is replaced by $\|\cdot\|_{UI}$ ([Mat93, §4]).

4. [Hig89] (*Best PSD approximation*) Take $A \in \mathbb{C}^{n \times n}$. Set $A_H = (A + A^*)/2$, $B = (A_H + |A_H|)/2$. Then B is positive semidefinite and is the unique solution to

$$\min\{\|A - X\|_F : X \in \mathbb{C}^{n \times n}, X \in \text{PSD}\}.$$

There is also a formula for the best PSD approximation in the spectral norm.

5. Let $A, B \in \mathbb{C}^{m \times n}$ have singular value decompositions $A = U_A \Sigma_A V_A^*$ and $B = U_B \Sigma_B V_B^*$. Let $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$ be any unitary matrices. Then

$$\|\Sigma_A - \Sigma_B\|_{UI} \leq \|A - UB V^*\|_{UI}.$$

17.6 Characterization of the Eigenvalues of Sums of Hermitian Matrices and Singular Values of Sums and Products of General Matrices

There are necessary and sufficient conditions for three sets of numbers to be the eigenvalues of Hermitian $A, B, C = A + B \in \mathbb{C}^{n \times n}$, or the singular values of $A, B, C = A + B \in \mathbb{C}^{m \times n}$, or the singular values of nonsingular $A, B, C = AB \in \mathbb{C}^{n \times n}$. The key results in this section were first proved by Klyachko ([Kly98]) and Knutson and Tao ([KT99]). The results presented here are from a survey by Fulton [Ful00]. Bhatia has written an expository paper on the subject ([Bha01]).

Definitions:

The inequalities are in terms of the sets T_r^n of triples (I, J, K) of subsets of $\{1, \dots, n\}$ of the same cardinality r , defined by the following inductive procedure. Set

$$U_r^n = \left\{ (I, J, K) \mid \sum_{i \in I} i + \sum_{j \in J} j = \sum_{k \in K} k + r(r+1)/2 \right\}.$$

When $r = 1$, set $T_1^n = U_1^n$. In general,

$$T_r^n = \left\{ (I, J, K) \in U_r^n \mid \text{for all } p < r \text{ and all } \right.$$

$$\left. (F, G, H) \in T_p^n, \sum_{f \in F} i_f + \sum_{g \in G} j_g \leq \sum_{h \in H} k_h + p(p+1)/2 \right\}.$$

In this section, the vectors α, β, γ will have real entries ordered in nonincreasing order.

Facts:

The following facts are in [Ful00]:

1. A triple (α, β, γ) of real n -vectors occurs as eigenvalues of Hermitian $A, B, C = A + B \in \mathbb{C}^{n \times n}$ if and only if $\sum \gamma_i = \sum \alpha_i + \sum \beta_i$ and the inequalities

$$\sum_{k \in K} \gamma_k \leq \sum_{i \in I} \alpha_i + \sum_{j \in J} \beta_j$$

hold for every (I, J, K) in T_r^n , for all $r < n$. Furthermore, the statement is true if $\mathbb{C}^{n \times n}$ is replaced by $\mathbb{R}^{n \times n}$.

2. Take Hermitian $A, B \in \mathbb{C}^{n \times n}$ (not necessarily PSD). Let the vectors of eigenvalues of $A, B, C = A + B$ be α, β , and γ . Then we have the (nonlinear) inequality

$$\min_{\pi \in S_n} \prod_{i=1}^n (\alpha_i + \beta_{\pi(i)}) \leq \prod_{i=1}^n \gamma_i \leq \max_{\pi \in S_n} \prod_{i=1}^n (\alpha_i + \beta_{\pi(i)}).$$

3. Fix m, n and set $q = \min\{m, n\}$. For any subset X of $\{1, \dots, m+n\}$, define $X_q = \{i : i \in X, i \leq q\}$ and $X'_q = \{i : i \leq q, m+n+1-i \in X\}$. A triple (α, β, γ) occurs as the singular values of $A, B, C = A + B \in \mathbb{C}^{m \times n}$, if and only if the inequalities

$$\sum_{k \in K_q} \gamma_k - \sum_{k \in K'_q} \gamma_k \leq \sum_{i \in I} \alpha_i - \sum_{i \in I'_q} \alpha_i + \sum_{j \in J_q} \beta_j - \sum_{j \in J'_q} \beta_j$$

are satisfied for all (I, J, K) in T_r^{m+n} , for all $r < m+n$. This statement is not true if $\mathbb{C}^{m \times n}$ is replaced by $\mathbb{R}^{m \times n}$. (See Example 1.)

4. A triple of positive real n -vectors (α, β, γ) occurs as the singular values of n by n matrices $A, B, C = AB \in \mathbb{C}^{n \times n}$ if and only if $\gamma_1 \cdots \gamma_n = \alpha_1 \cdots \alpha_n \beta_1 \cdots \beta_n$ and

$$\prod_{k \in K} \gamma_k \leq \prod_{i \in I} \alpha_i \cdot \prod_{j \in J} \beta_j$$

for all (I, J, K) in T_r^n , and all $r < n$. This statement is still true if $\mathbb{C}^{n \times n}$ is replaced by $\mathbb{R}^{n \times n}$.

Example:

1. There are $A, B, C = A + B \in \mathbb{C}^{2 \times 2}$ with singular values $(1, 1), (1, 0)$, and $(1, 1)$, but there are no $A, B, C = A + B \in \mathbb{R}^{2 \times 2}$ with these singular values.

In the complex case, take $A = \text{diag}(1, 1/2 + (\sqrt{3}/2)i)$, $B = \text{diag}(0, -1)$.

Now suppose that A and B are real 2×2 matrices such that A and $C = A + B$ both have singular values $(1, 1)$. Then A and C are orthogonal. Consider $BC^T = AC^T - CC^T = AC^T - I$. Because AC^T is real, it has eigenvalues $\alpha, \bar{\alpha}$ and so BC^T has eigenvalues $\alpha - 1, \bar{\alpha} - 1$. Because AC^T is orthogonal, it is normal and, hence, so is BC^T , and so its singular values are $|\alpha - 1|$ and $|\bar{\alpha} - 1|$, which are equal and, in particular, cannot be $(1, 0)$.

17.7 Miscellaneous Results and Generalizations

Throughout this section F can be taken to be either $\mathbb{R}$ or $\mathbb{C}$.

Definitions:

Let $\mathcal{X}, \mathcal{Y}$ be subspaces of $\mathbb{C}^r$ of dimension m and n . The **principal angles** $0 \leq \theta_1 \leq \dots \leq \theta_q \leq \pi/2$ between $\mathcal{X}$ and $\mathcal{Y}$ and **principal vectors** $\mathbf{u}_1, \dots, \mathbf{u}_q$ and $\mathbf{v}_1, \dots, \mathbf{v}_q$ are defined inductively:

$$\cos(\theta_1) = \max\{|\mathbf{x}^* \mathbf{y}| : \mathbf{x} \in \mathcal{X}, \max_{\mathbf{y} \in \mathcal{Y}} \|\mathbf{x}\|_2 = \|\mathbf{y}\|_2 = 1\}.$$

Let $\mathbf{u}_1$ and $\mathbf{v}_1$ be a pair of maximizing vectors. For $k = 2, \dots, q$,

$$\cos(\theta_k) = \max\{|\mathbf{x}^* \mathbf{y}| : \mathbf{x} \in \mathcal{X}, \mathbf{y} \in \mathcal{Y}, \|\mathbf{x}\|_2 = \|\mathbf{y}\|_2 = 1, \quad \mathbf{x}^* \mathbf{u}_i = \mathbf{y}^* \mathbf{v}_i = 0, \quad i = 1, \dots, k-1\}.$$

Let $\mathbf{u}_k$ and $\mathbf{v}_k$ be a pair of maximizing vectors. (Principal angles are also called **canonical angles**, and the cosines of the principal angles are called **canonical correlations**.)

Facts:

1. (*Principal Angles*) Let $\mathcal{X}, \mathcal{Y}$ be subspaces of $\mathbb{C}^r$ of dimension m and n .
 - (a) [BG73] The principal vectors obtained by the process above are not necessarily unique, but the principal angles are unique (and, hence, independent of the chosen principal vectors).
 - (b) Let $m = n \leq r/2$ and X, Y be matrices whose columns form orthonormal bases for the subspaces $\mathcal{X}$ and $\mathcal{Y}$, respectively.
 - i. The singular values of $X^* Y$ are the cosines of the principal angles between the subspaces $\mathcal{X}$ and $\mathcal{Y}$.
 - ii. There are unitary matrices $U \in \mathbb{C}^{r \times r}$ and V_X and $V_Y \in \mathbb{C}^{n \times n}$ such that

$$U X V_X = \begin{bmatrix} I_n \\ 0_n \\ 0_{r-n,n} \end{bmatrix}, \quad U Y V_Y = \begin{bmatrix} \Gamma \\ \Sigma \\ 0_{r-n,n} \end{bmatrix},$$

where Γ and Σ are nonnegative diagonal matrices. Their diagonal entries are the cosines and sines respectively of the principal angles between $\mathcal{X}$ and $\mathcal{Y}$.

- (c) [QZL05] Take $m = n$. For any permutation invariant absolute norm g on $\mathbb{R}^m$,

$$g(\sin(\theta_1), \dots, \sin(\theta_m)), \quad g(2 \sin(\theta_1/2), \dots, 2 \sin(\theta_m/2)), \quad \text{and} \quad g(\theta_1, \dots, \theta_m)$$

are metrics on the set of subspaces of dimension n of $\mathbb{C}^{r \times r}$.

2. [GV96, Theorem 2.6.2] (*CS decomposition*) Let $W \in F^{n \times n}$ be unitary. Take a positive integer l such that $2l \leq n$. Then there are unitary matrices $U_{11}, V_{11} \in F^{l \times l}$ and $U_{22}, V_{22} \in F^{(n-l) \times (n-l)}$ such that

$$\begin{bmatrix} U_{11} & 0 \\ 0 & U_{22} \end{bmatrix} W \begin{bmatrix} V_{11} & 0 \\ 0 & V_{22} \end{bmatrix} = \begin{bmatrix} \Gamma & -\Sigma & 0 \\ \Sigma & \Gamma & 0 \\ 0 & 0 & I_{n-2l} \end{bmatrix},$$

where $\Gamma = \text{diag}(\gamma_1, \dots, \gamma_l)$ and $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_l)$ are nonnegative and $\Gamma^2 + \Sigma^2 = I$.

3. [GV96, Theorem 8.7.4] (*Generalized singular value decomposition*) Take $A \in F^{p \times n}$ and $B \in F^{m \times n}$ with $p \geq n$. Then there is an invertible $X \in F^{n \times n}$, unitary $U \in F^{p \times p}$ and $V \in F^{m \times m}$, and nonnegative diagonal matrices $\Sigma_A \in \mathbb{R}^{n \times n}$ and $\Sigma_B \in \mathbb{R}^{q \times q}$ ($q = \min\{m, n\}$) such that $A = U \Sigma_A X$ and $B = V \Sigma_B X$.

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18

Numerical Range

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The numerical range $W(A)$ of an $n \times n$ complex matrix A is the collection of complex numbers of the form $\mathbf{x}^* A \mathbf{x}$, where $\mathbf{x} \in \mathbb{C}^n$ is a unit vector. It can be viewed as a “picture” of A containing useful information of A . Even if the matrix A is not known explicitly, the “picture” $W(A)$ would allow one to “see” many properties of the matrix. For example, the numerical range can be used to locate eigenvalues, deduce algebraic and analytic properties, obtain norm bounds, help find dilations with simple structure, etc. Related to the numerical range are the numerical radius of A defined by $w(A) = \max_{\mu \in W(A)} |\mu|$ and the distance of $W(A)$ to the origin denoted by $\tilde{w}(A) = \min_{\mu \in W(A)} |\mu|$. The quantities $w(A)$ and $\tilde{w}(A)$ are useful in studying perturbation, convergence, stability, and approximation problems.

Note that the spectrum $\sigma(A)$ can be viewed as another useful “picture” of the matrix $A \in M_n$. There are interesting relations between $\sigma(A)$ and $W(A)$.

18.1 Basic Properties and Examples

Definitions and Notation:

Let $A \in \mathbb{C}^{n \times n}$. The **numerical range** (also known as **the field of values**) of A is defined by

$$W(A) = \{\mathbf{x}^* A \mathbf{x} : \mathbf{x} \in \mathbb{C}^n, \mathbf{x}^* \mathbf{x} = 1\}.$$

The **numerical radius** of A and the **distance** of $W(A)$ to the origin are the quantities

$$w(A) = \max\{|\mu| : \mu \in W(A)\} \quad \text{and} \quad \tilde{w}(A) = \min\{|\mu| : \mu \in W(A)\}.$$

Furthermore, let

$$\overline{W(A)} = \{\bar{a} : a \in W(A)\}.$$

Facts:

The following basic facts can be found in most references on numerical ranges such as [GR96], [Hal82], and [HJ91].

1. Let $A \in \mathbb{C}^{n \times n}$, $a, b \in \mathbb{C}$. Then $W(aA + bI) = aW(A) + b$.
2. Let $A \in \mathbb{C}^{n \times n}$. Then $W(U^*AU) = W(A)$ for any unitary $U \in \mathbb{C}^{n \times n}$.
3. Let $A \in \mathbb{C}^{n \times n}$. Suppose $k \in \{1, \dots, n-1\}$ and $X \in \mathbb{C}^{n \times k}$ satisfies $X^*X = I_k$. Then

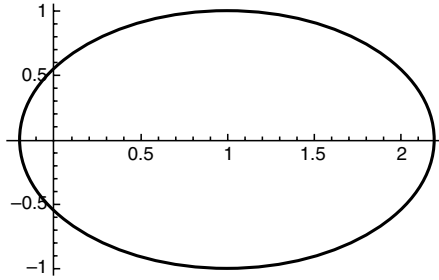
$$W(X^*AX) \subseteq W(A).$$

In particular, for any $k \times k$ principal submatrix B of A , we have $W(B) \subseteq W(A)$.

4. Let $A \in \mathbb{C}^{n \times n}$. Then $W(A)$ is a compact convex set in $\mathbb{C}$.
5. If $A_1 \oplus A_2 \in M_n$, then $W(A) = \text{conv} \{W(A_1) \cup W(A_2)\}$.
6. Let $A \in M_n$. Then $W(A) = W(A^T)$ and $W(A^*) = \overline{W(A)}$.
7. If $A \in \mathbb{C}^{2 \times 2}$ has eigenvalues λ_1, λ_2 , then $W(A)$ is an elliptical disk with foci λ_1, λ_2 , and minor axis with length $\{\text{tr}(A^*A) - |\lambda_1|^2 - |\lambda_2|^2\}^{1/2}$. Consequently, if $A = \begin{bmatrix} \lambda_1 & b \\ 0 & \lambda_2 \end{bmatrix}$, then the minor axis of the elliptical disk $W(A)$ has length $|b|$.
8. Let $A \in \mathbb{C}^{n \times n}$. Then $W(A)$ is a subset of a straight line if and only if there are $a, b \in \mathbb{C}$ with $a \neq 0$ such that $aA + bI$ is Hermitian. In particular, we have the following:
 - (a) $A = aI$ if and only if $W(A) = \{a\}$.
 - (b) $A = A^*$ if and only if $W(A) \subseteq \mathbb{R}$.
 - (c) $A = A^*$ is positive definite if and only if $W(A) \subseteq (0, \infty)$.
 - (d) $A = A^*$ is positive semidefinite if and only if $W(A) \subseteq [0, \infty)$.
9. If $A \in \mathbb{C}^{n \times n}$ is normal, then $W(A) = \text{conv} \sigma(A)$ is a convex polygon. The converse is true if $n \leq 4$.
10. Let $A \in \mathbb{C}^{n \times n}$. The following conditions are equivalent.
 - (a) $W(A) = \text{conv} \sigma(A)$.
 - (b) $W(A)$ is a convex polygon with vertices $\mu_1, \dots, \mu_k$.
 - (c) A is unitarily similar to $\text{diag}(\mu_1, \dots, \mu_k) \oplus B$ such that $W(B) \subseteq \text{conv} \{\mu_1, \dots, \mu_k\}$.
11. Let $A \in \mathbb{C}^{n \times n}$. Then A is unitary if and only if all eigenvalues of A have modulus one and $W(A) = \text{conv} \sigma(A)$.
12. Suppose $A = (A_{ij})_{1 \leq i, j \leq m} \in M_n$ is a block matrix such that $A_{11}, \dots, A_{mm}$ are square matrices and $A_{ij} = 0$ whenever $(i, j) \notin \{(1, 2), \dots, (m-1, m), (m, 1)\}$. Then $W(A) = cW(A)$ for any $c \in \mathbb{C}$ satisfying $c^m = 1$. If $A_{m,1}$ is also zero, then $W(A)$ is a circular disk centered at 0 with the radius equal to the largest eigenvalue of $(A + A^*)/2$.

Examples:

1. Let $A = \text{diag}(1, 0)$. Then $W(A) = [0, 1]$.
2. Let $A = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$. Then $W(A)$ is the closed unit disk $\mathbf{D} = \{a \in \mathbb{C} : |a| \leq 1\}$.
3. Let $A = \begin{bmatrix} 2 & 2 \\ 0 & 1 \end{bmatrix}$. Then by Fact 7 above, $W(A)$ is the convex set whose boundary is the ellipse with foci 1 and 2 and minor axis 2, as shown in Figure 18.1.
4. Let $A = \text{diag}(1, i, -1, -i) \oplus \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. By Facts 5 and 7, the boundary of $W(A)$ is the square with vertices $1, i, -1, -i$.

FIGURE 18.1 Numerical range of the matrix A in Example 3.**Applications:**

1. By Fact 6, if A is real, then $W(A)$ is symmetric about the real axis, i.e., $W(A) = \overline{W(A)}$.
2. Suppose $A \in \mathbb{C}^{n \times n}$, and there are $a, b \in \mathbb{C}$ such that $(A - aI)(A - bI) = 0_n$. Then A is unitarily similar to a matrix of the form

$$aI_r \oplus bI_s \oplus \begin{bmatrix} a & d_1 \\ 0 & b \end{bmatrix} \oplus \cdots \oplus \begin{bmatrix} a & d_t \\ 0 & b \end{bmatrix}$$

with $d_1 \geq \cdots \geq d_t > 0$, where $r + s + 2t = n$. By Facts 1, 5, and 7, the set $W(A)$ is the elliptical disk with foci a, b and minor axis of length d , where

$$d = d_1 = \{(\|A\|_2^2 - |a|^2)(\|A\|_2^2 - |b|^2)\}^{1/2} / \|A\|_2$$

if $t \geq 1$, and $d = 0$ otherwise.

3. By Fact 12, if $A \in \mathbb{C}^{n \times n}$ is the basic circulant matrix $E_{12} + E_{23} + \cdots + E_{n-1,n} + E_{n1}$, then $W(A) = \text{conv}\{c \in \mathbb{C} : c^n = 1\}$; if $A \in M_n$ is the Jordan block of zero $J_n(0)$, then $W(A) = \{c \in \mathbb{C} : |c| \leq \cos(\pi/(n+1))\}$.
4. Suppose $A \in \mathbb{C}^{n \times n}$ is a primitive nonnegative matrix. Then A is permutationally similar to a block matrix (A_{ij}) as described in Fact 12 and, thus, $W(A) = cW(A)$ for any $c \in \mathbb{C}$ satisfying $c^m = 1$.

18.2 The Spectrum and Special Boundary Points

Definitions and Notation:

Let $\partial\mathcal{S}$ and $\text{int}(\mathcal{S})$ be the boundary and the interior of a convex compact subset $\mathcal{S}$ of $\mathbb{C}$.

A **support line** ℓ of $\mathcal{S}$ is a line that intersects $\partial\mathcal{S}$ such that $\mathcal{S}$ lies entirely within one of the closed half-planes determined by ℓ .

A boundary point μ of $\mathcal{S}$ is **nondifferentiable** if there is more than one support line of $\mathcal{S}$ passing through μ .

An eigenvalue λ of $A \in \mathbb{C}^{n \times n}$ is a **reducing eigenvalue** if A is unitarily similar to $[\lambda] \oplus A_2$.

Facts:

The following facts can be found in [GR96], [Hal82], and [HJ91].

1. Let $A \in \mathbb{C}^{n \times n}$. Then

$$\sigma(A) \subseteq W(A) \subseteq \{a \in \mathbb{C} : |a| \leq \|A\|_2\}.$$